

Chapter 4. Localization and Mapping

Part 1. Kalman filter



All models are wrong. Some models are useful.
– George Box

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4.1 Mathematical background

Mean

- For example, given five different coins – two 5-cent coins and three 10-cent coins, we can easily calculate the mean value by averaging coins' values.



$$V_{mean} = \frac{1}{N}\sum_{n=1}^{N} V_n = \frac{1}{5}(5+5+10+10+10) = 8cent$$

- The above outcome cannot be defined as the expected value because the system states (the coin values) are not hidden, and we've used the entire population (all 5 coins) for the mean value calculation.

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Expected value

- Now assume five different weight measurements of the same person: 79.8kg, 80kg, 80.1kg, 79.8kg, and 80.2kg.
- The person is a system, and the person's weight is a system state.
- The measurements are different due to the random measurement error of the scales. We do not know the true value of the weight since it is a Hidden State. However, we can estimate the weight by averaging the scales' measurements.

$$W = \frac{1}{N}\sum_{n=1}^{N} W_n = \frac{1}{5}(79.8 + 80 + 80.1 + 79.8 + 80.2) = 79.98kg$$

- The outcome of the estimate is the expected value of the weight.
- The expected value is the value you would expect your hidden variable to have over a long time or many trials.

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Variance

The Variance is a measure of the spreading of the data set from its mean.

$$\sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2$$

Standard deviation

The standard deviation is a square root of the variance

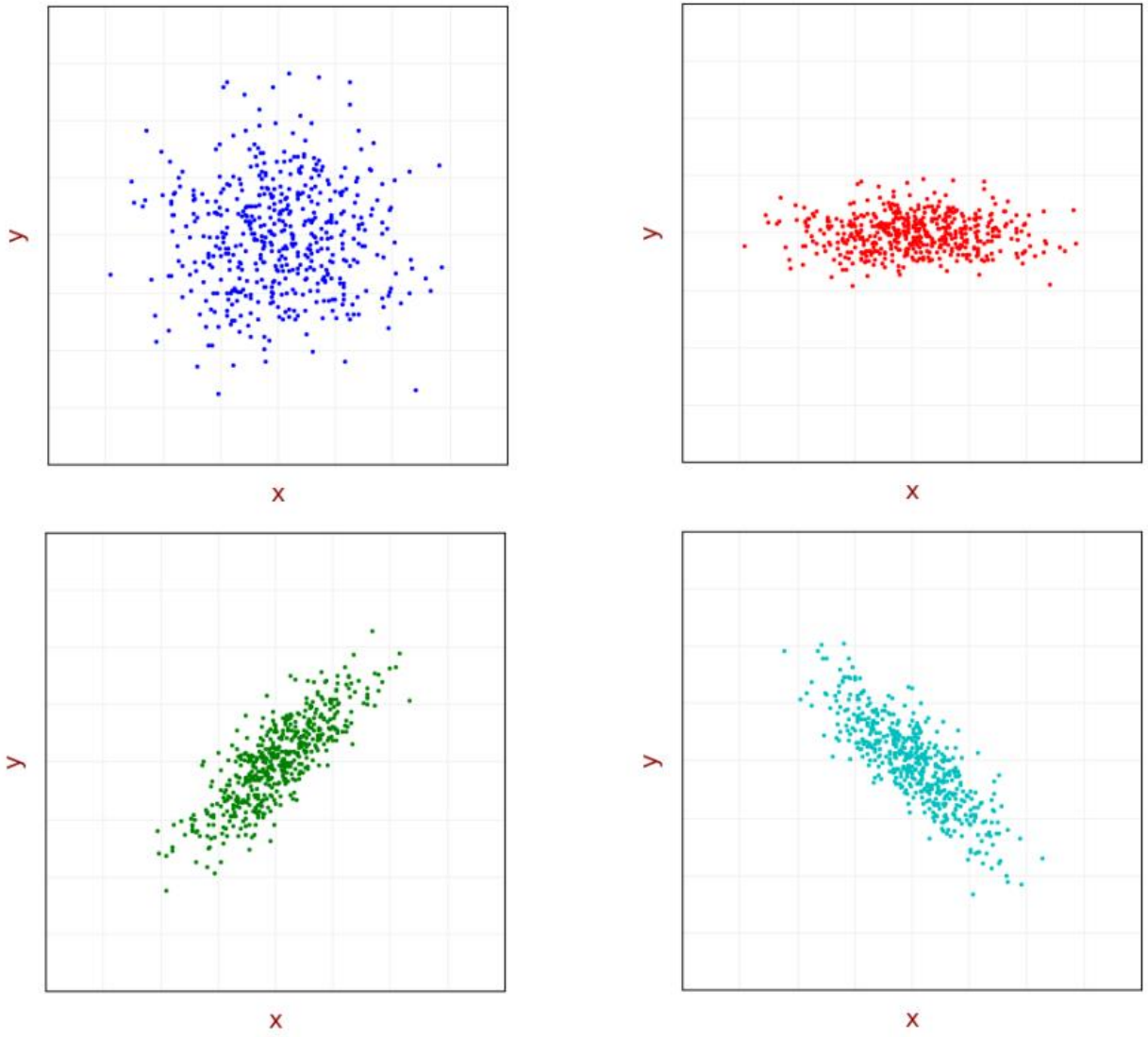
$$\sigma = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2}$$

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Covariance

Covariance is a measure of the strength of the correlation between two or more sets of random variates.

$$COV(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$$



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Covariance matrix

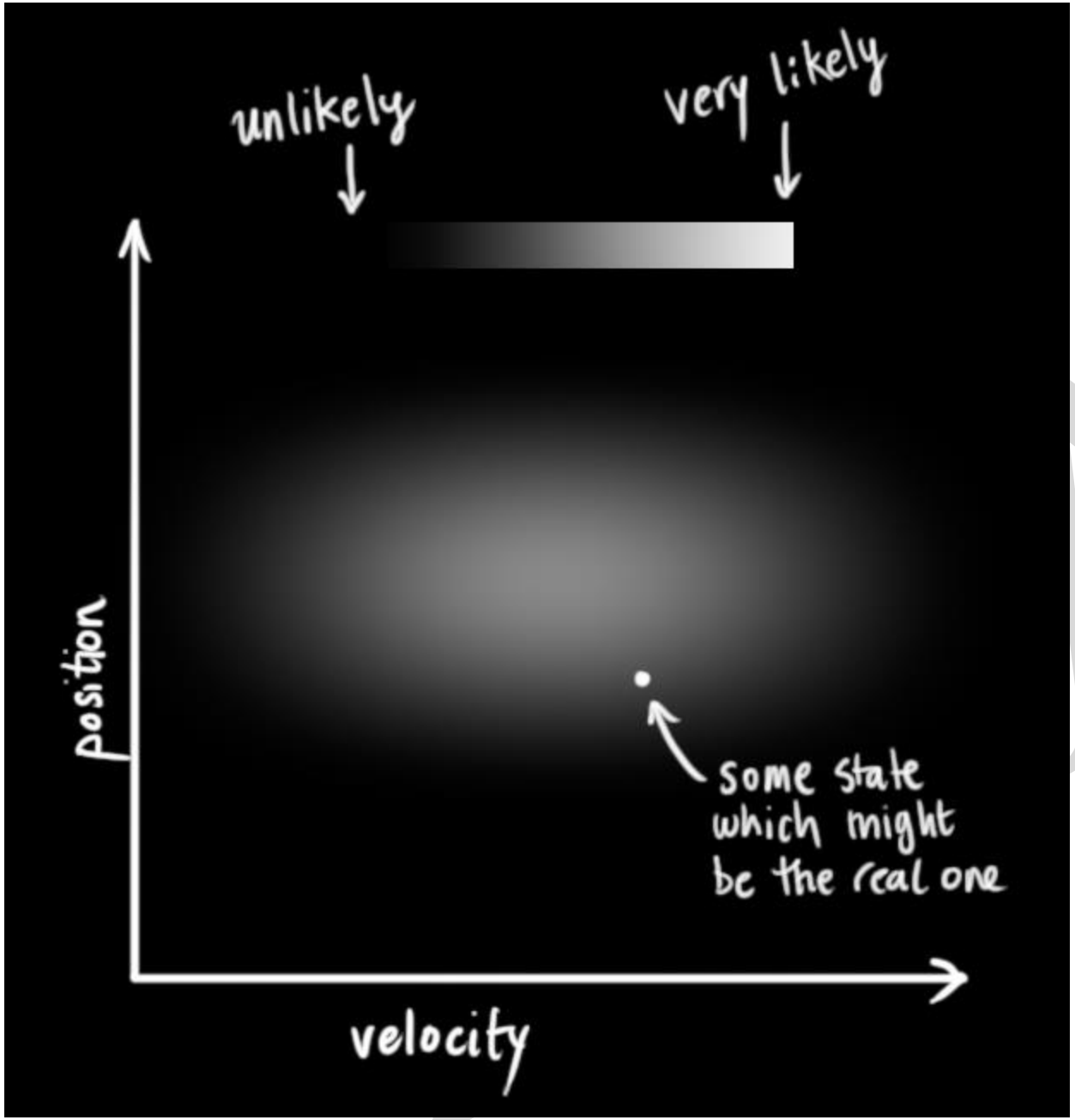
- A covariance matrix is a square matrix that represents the covariance between each pair of elements in a given multivariate random variable.
- For a two-dimensional random variable, the covariance matrix is:

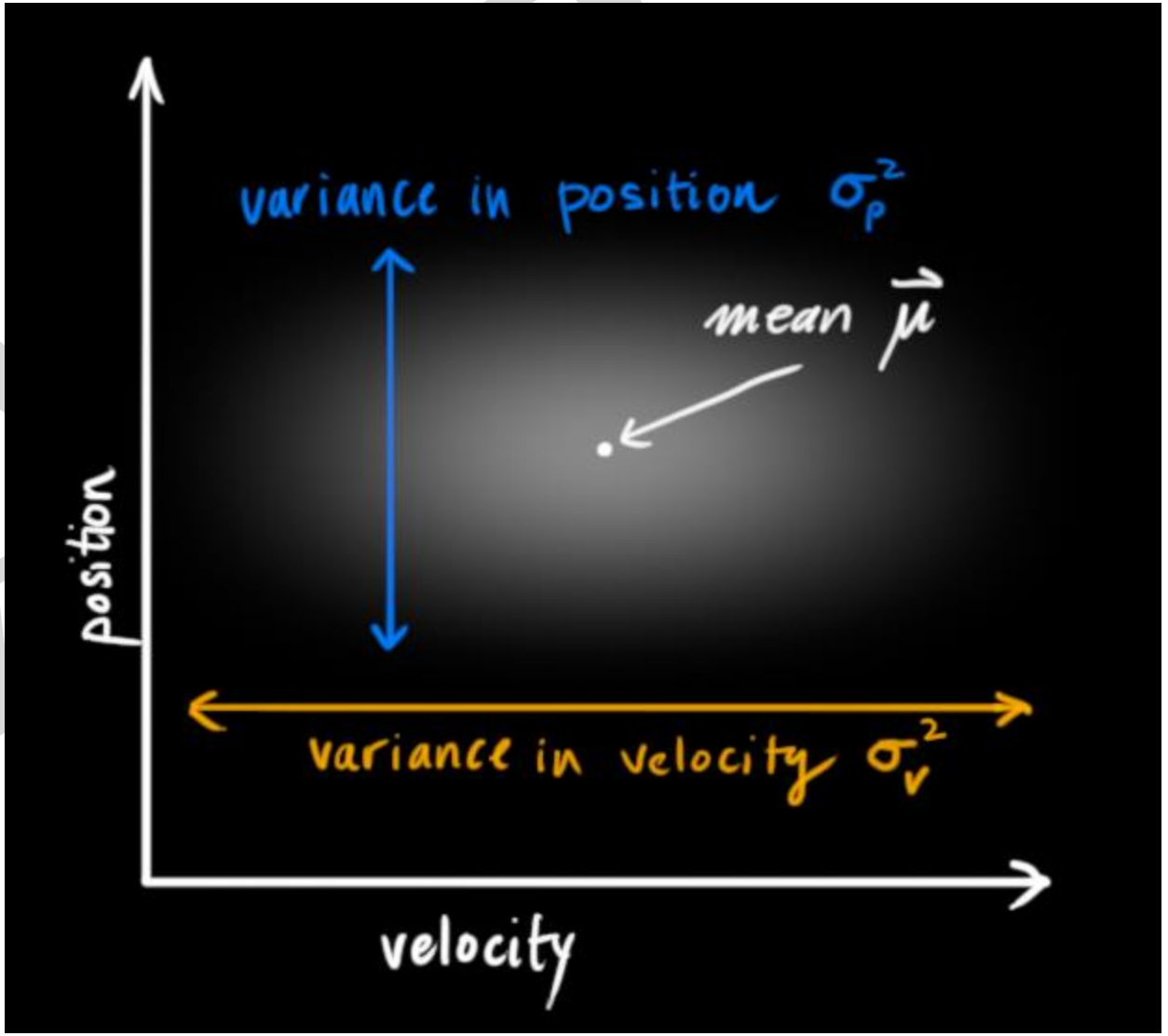
$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{yx} & \sigma_y^2 \end{bmatrix} = \begin{bmatrix} VAR(\boldsymbol{x}) & COV(\boldsymbol{x}, \boldsymbol{y}) \\ COV(\boldsymbol{y}, \boldsymbol{x}) & VAR(\boldsymbol{y}) \end{bmatrix}$$

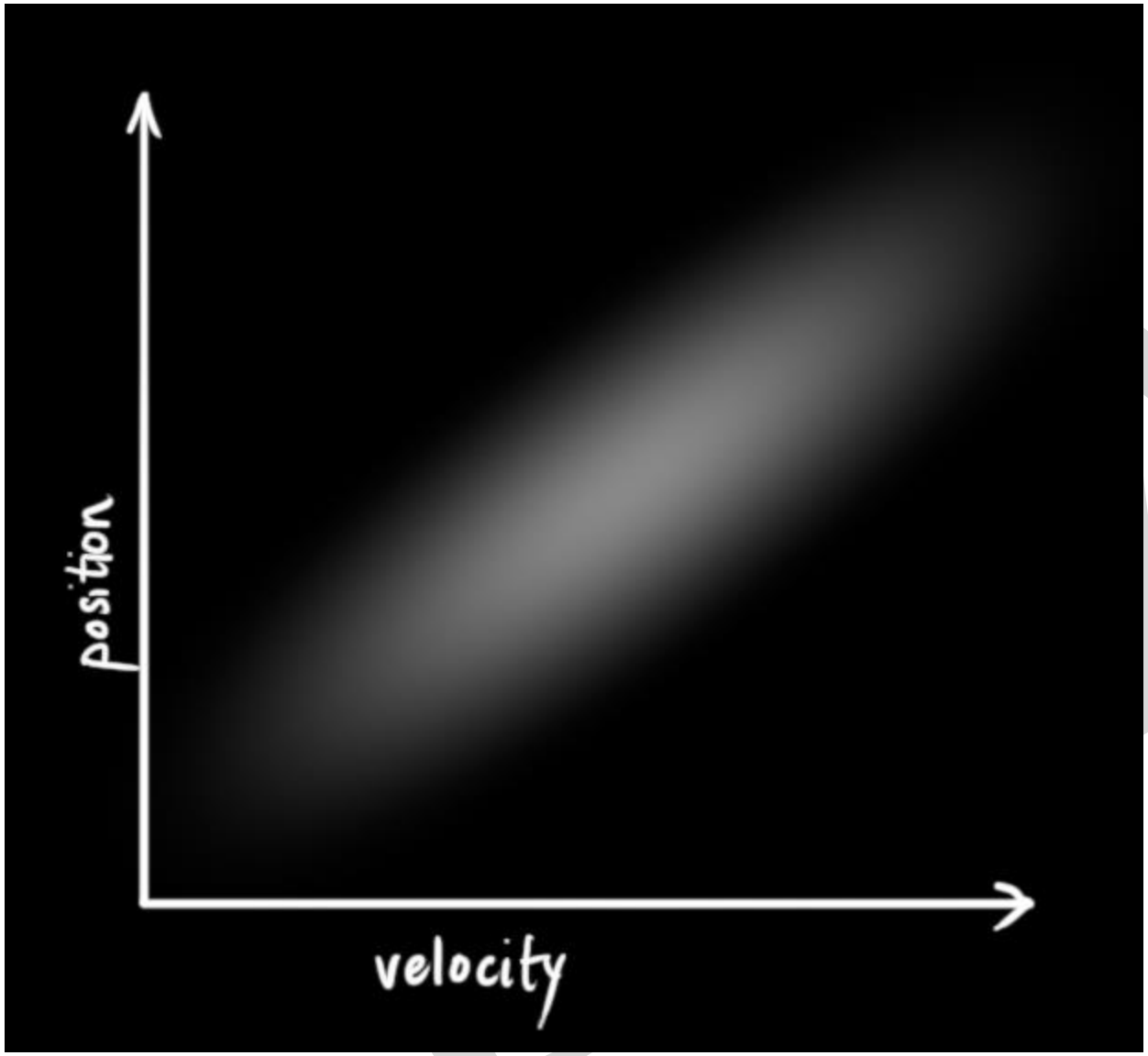
- For a n-dimensional random variable, the covariance matrix is:

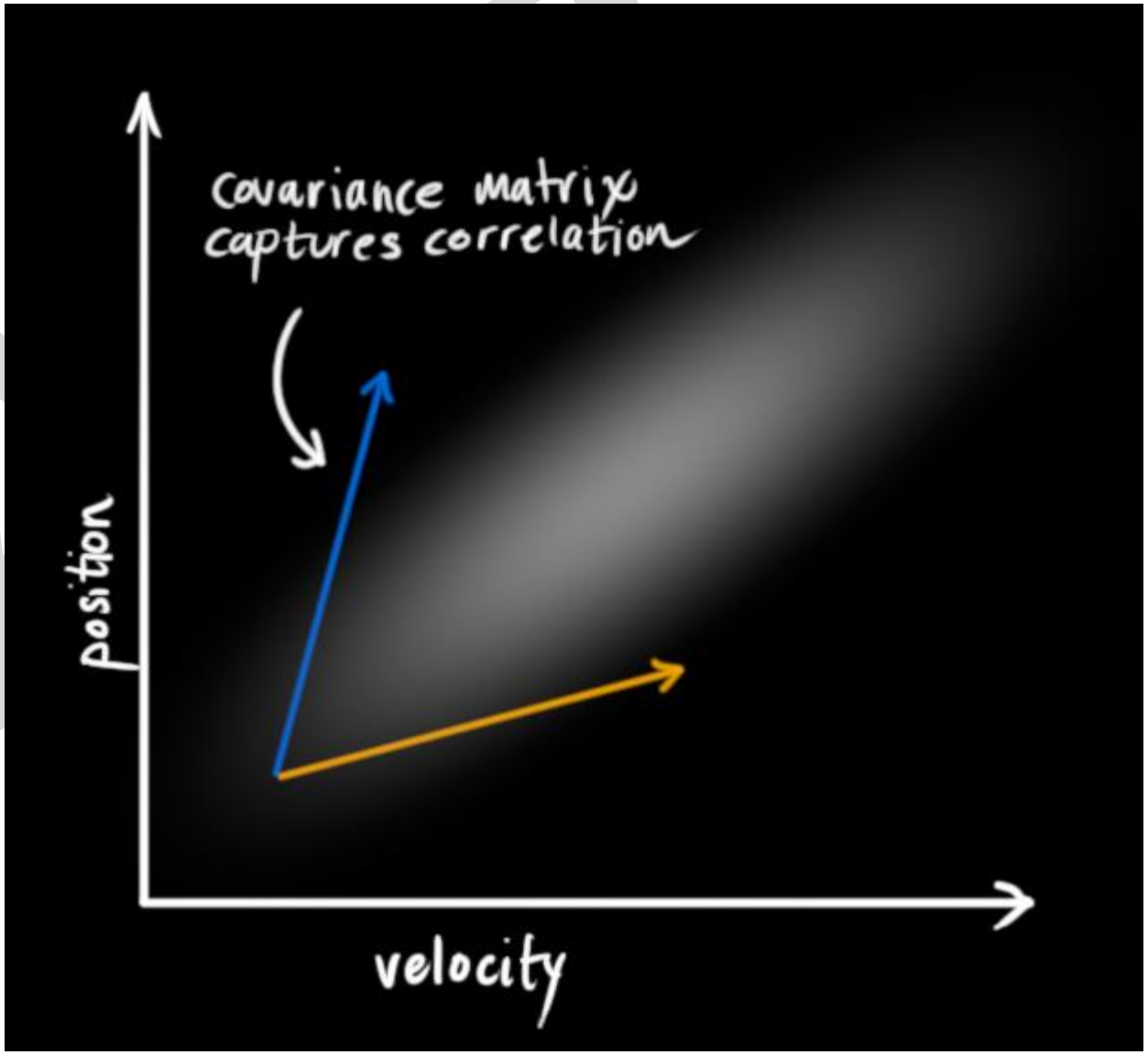
$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{bmatrix}$$

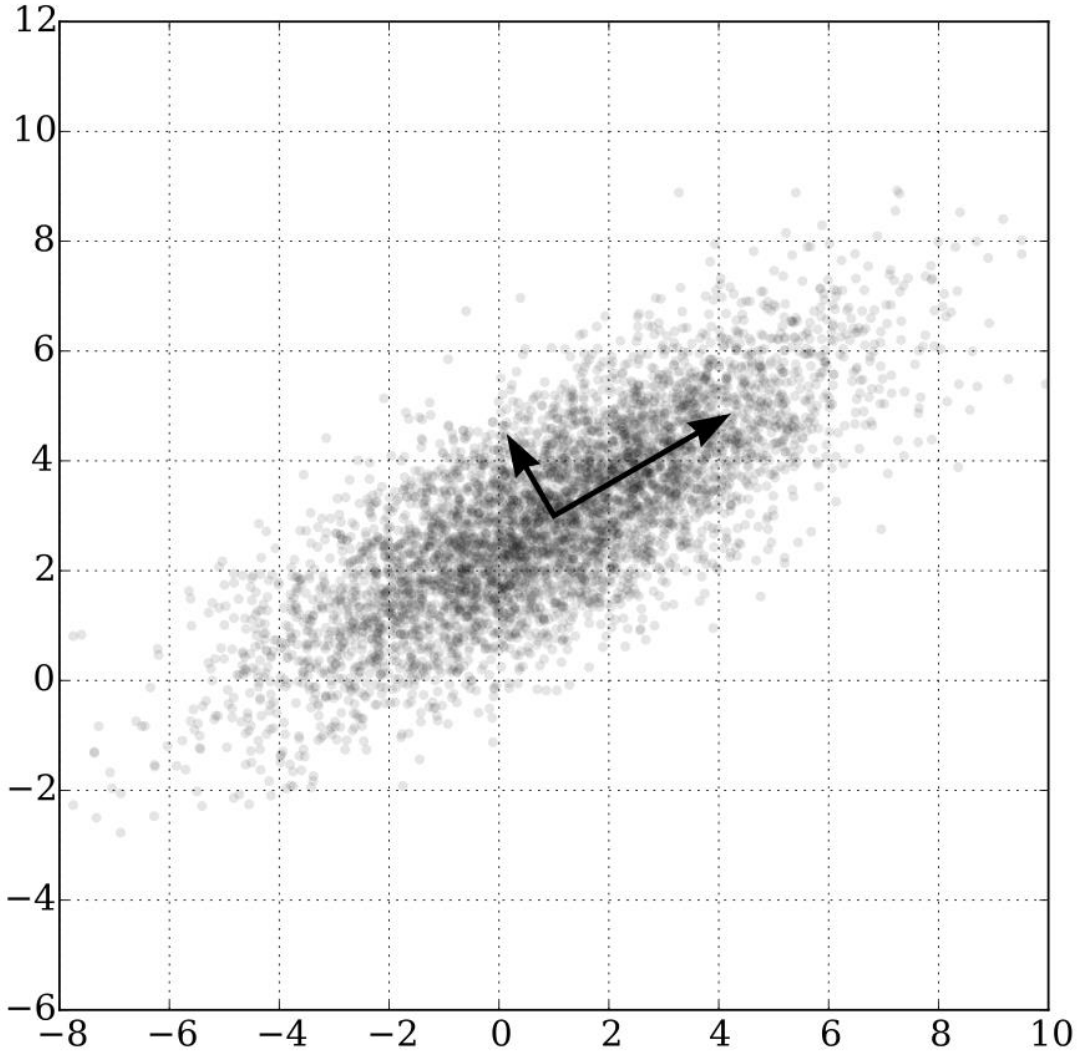
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unlikely
very likely
position
velocity
some state which might be the real one


variance in position σ_p^2
mean $\vec{\mu}$
position
variance in velocity σ_v^2
velocity


position
velocity


Covariance matrix
captures correlation
position
velocity



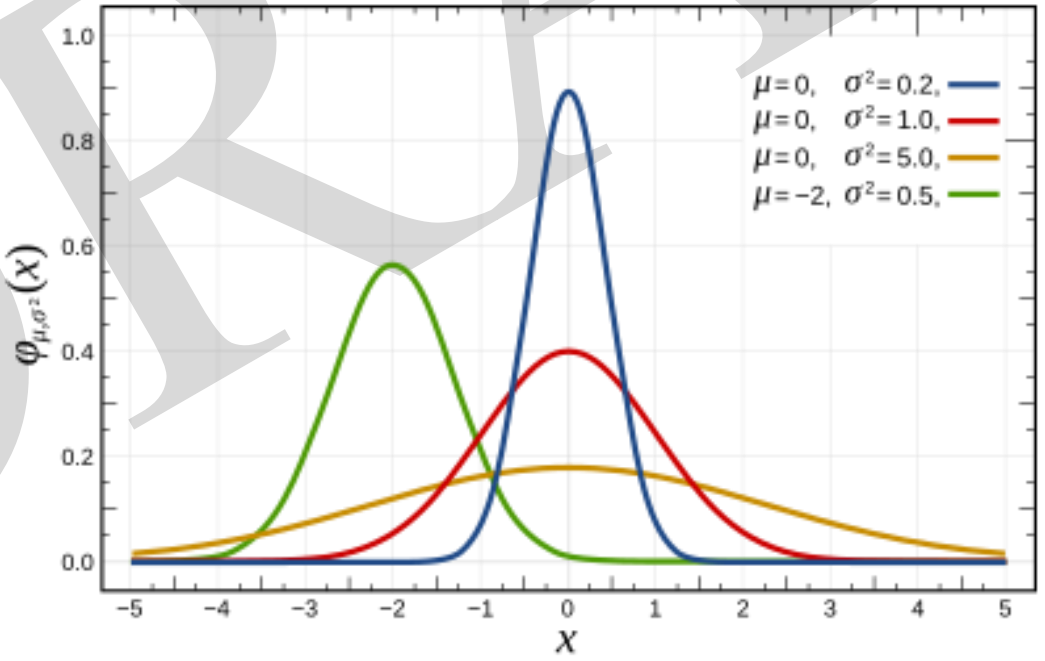
Sample points from a bivariate Gaussian distribution with a standard deviation of 3 in roughly the lower left–upper right direction and of 1 in the orthogonal direction. Because the x and y components co-vary, the variances of and do not fully describe the distribution. A covariance matrix is needed; the directions of the arrows correspond to the eigenvectors of this covariance matrix and their lengths to the square roots of the eigenvalues.

Normal distribution:

The normal distribution, also known as the Gaussian (named after the mathematician Carl Friedrich Gauss), is described by the following equation:

$$f\left(x;\mu,\sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

The Gaussian curve is also called the Probability Density Function (PDF) for the normal distribution.



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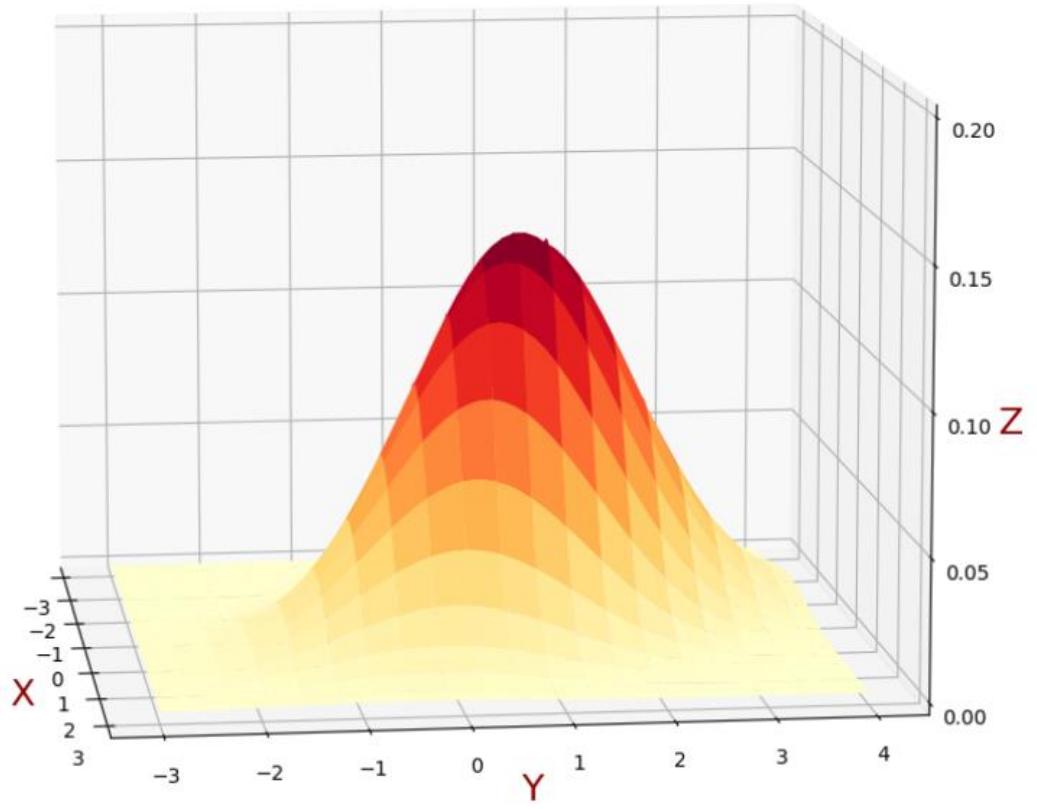
Multivariate normal distribution

The n - dimensional multivariate normal distribution is described by:

$$p(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^n |\boldsymbol{\Sigma}|}} exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

Where:

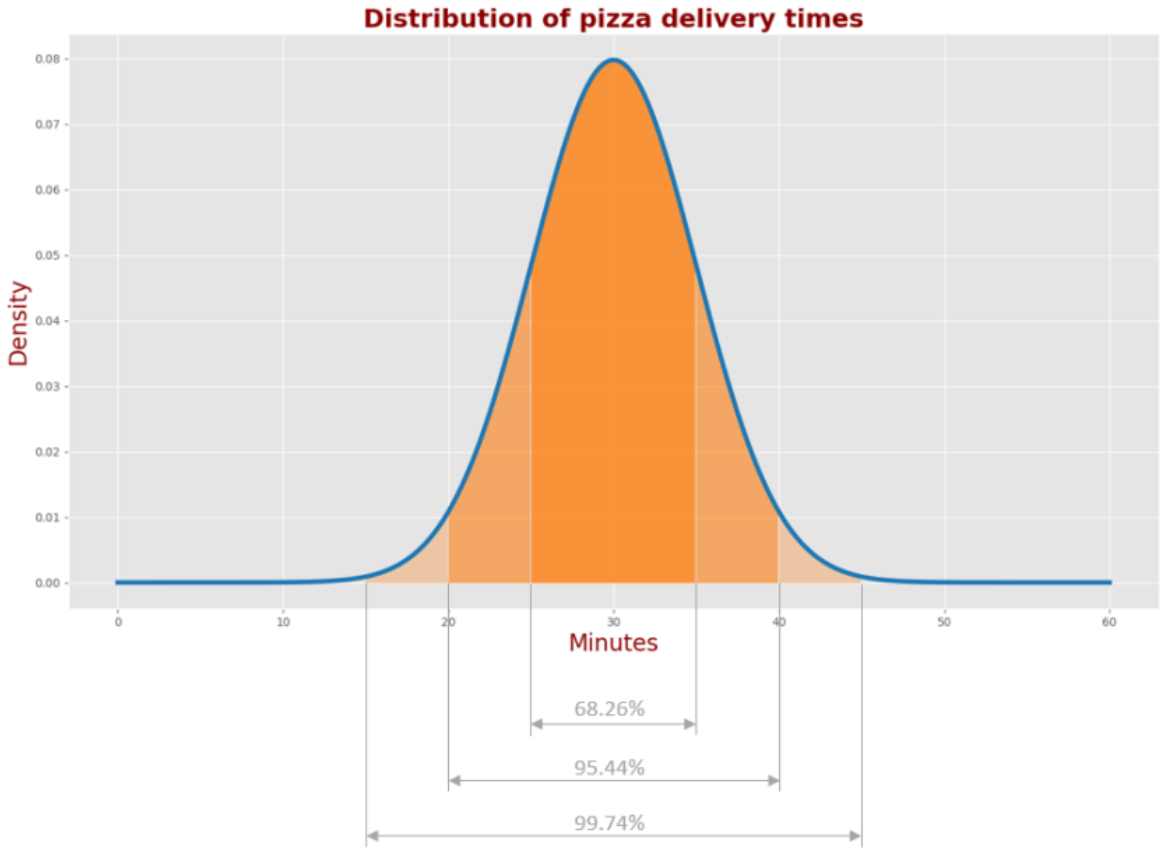
- $\boldsymbol{x}$ is an n - dimensional random vector
- $\boldsymbol{\mu}$ is an n - dimensional mean vector
- $\boldsymbol{\Sigma}$ is a square $n \times n$ covariance matrix

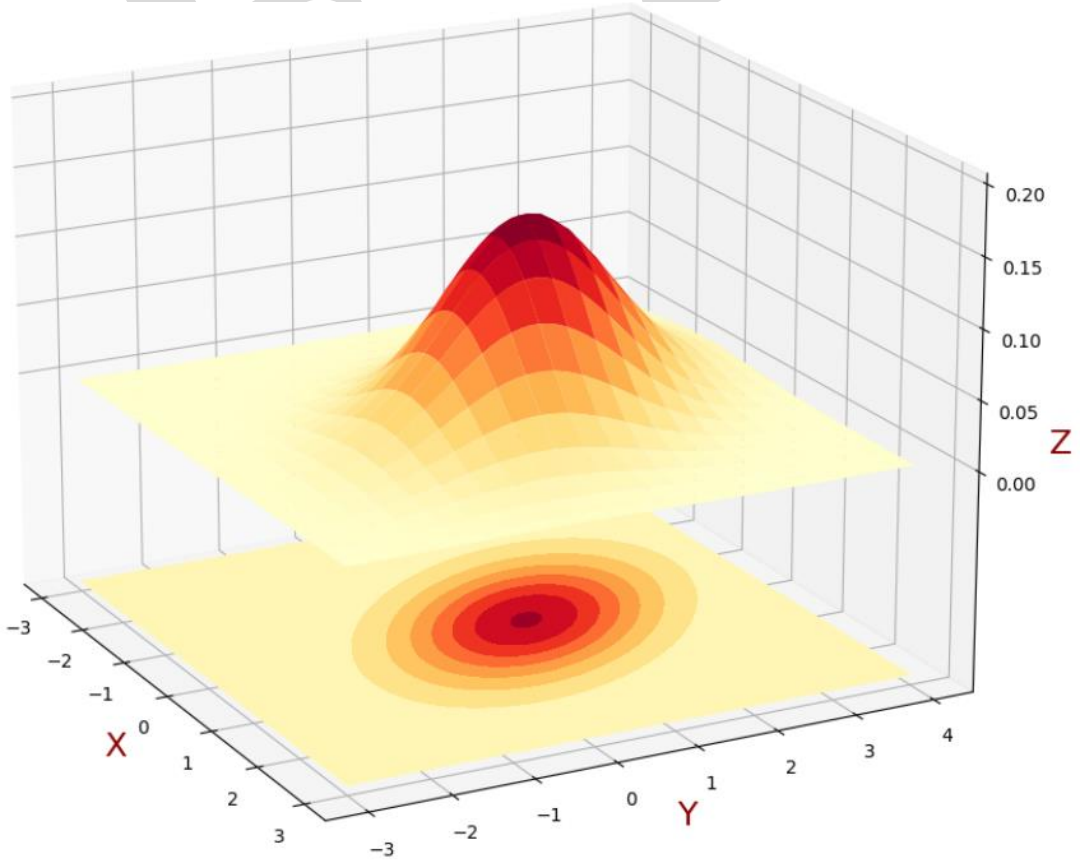


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Confidence intervals

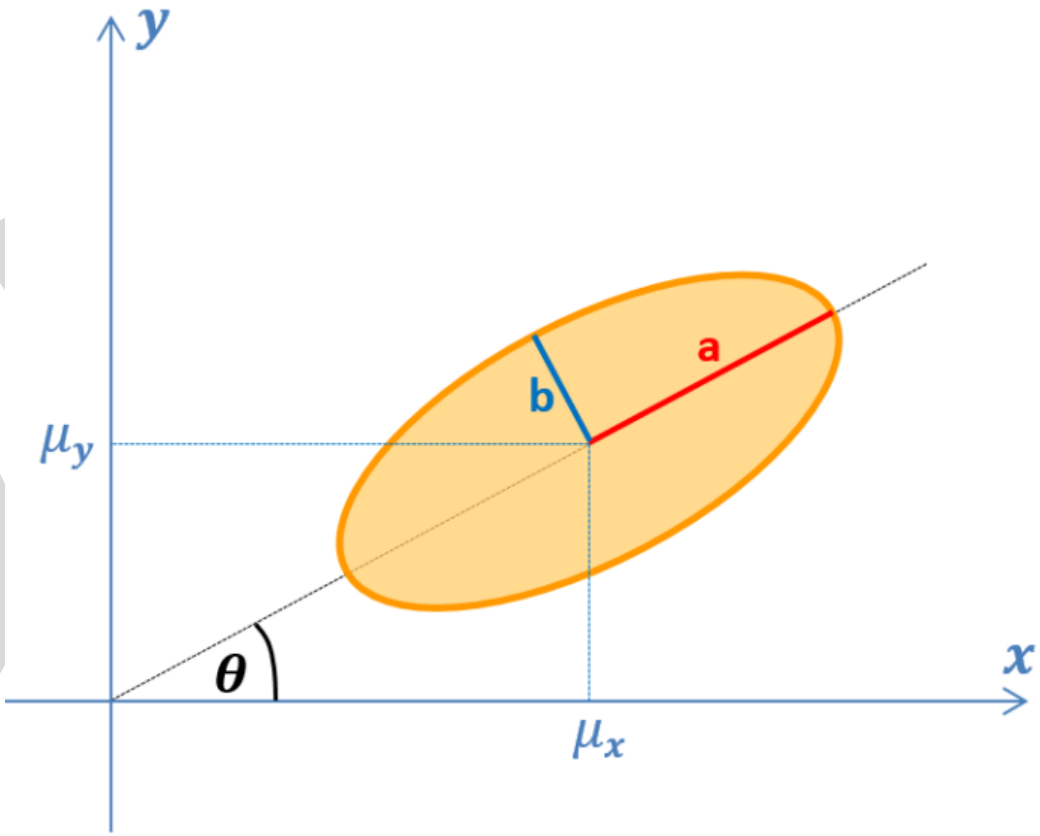
A confidence interval specifes the probability that a parameter falls between a pair of values around the mean for the univariate normal distribution.





Covariance ellipse

The covariance ellipse represents an iso-contour of the Gaussian distribution and allows visualization of a 1σ confidence interval in two dimensions. The covariance ellipse provides a geometric interpretation of the covariance matrix.



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Random variables

- A random variable describes the hidden state of the system. A random variable is a set of possible values from a random experiment.
- The random variable is described by the probability density function. The probability density function is characterized by moments.
- In this lecture, the random variables are characterized by:

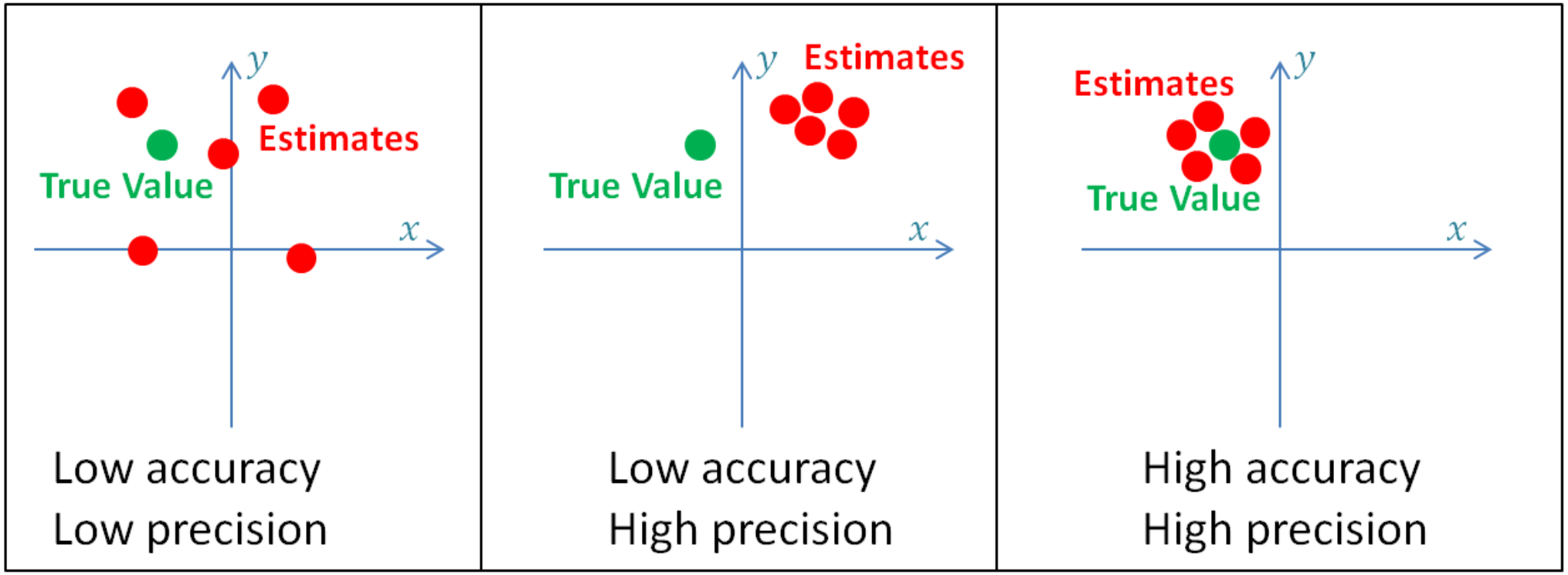
- The first raw moment $E(X)$ – the mean of the sequence of measurements;

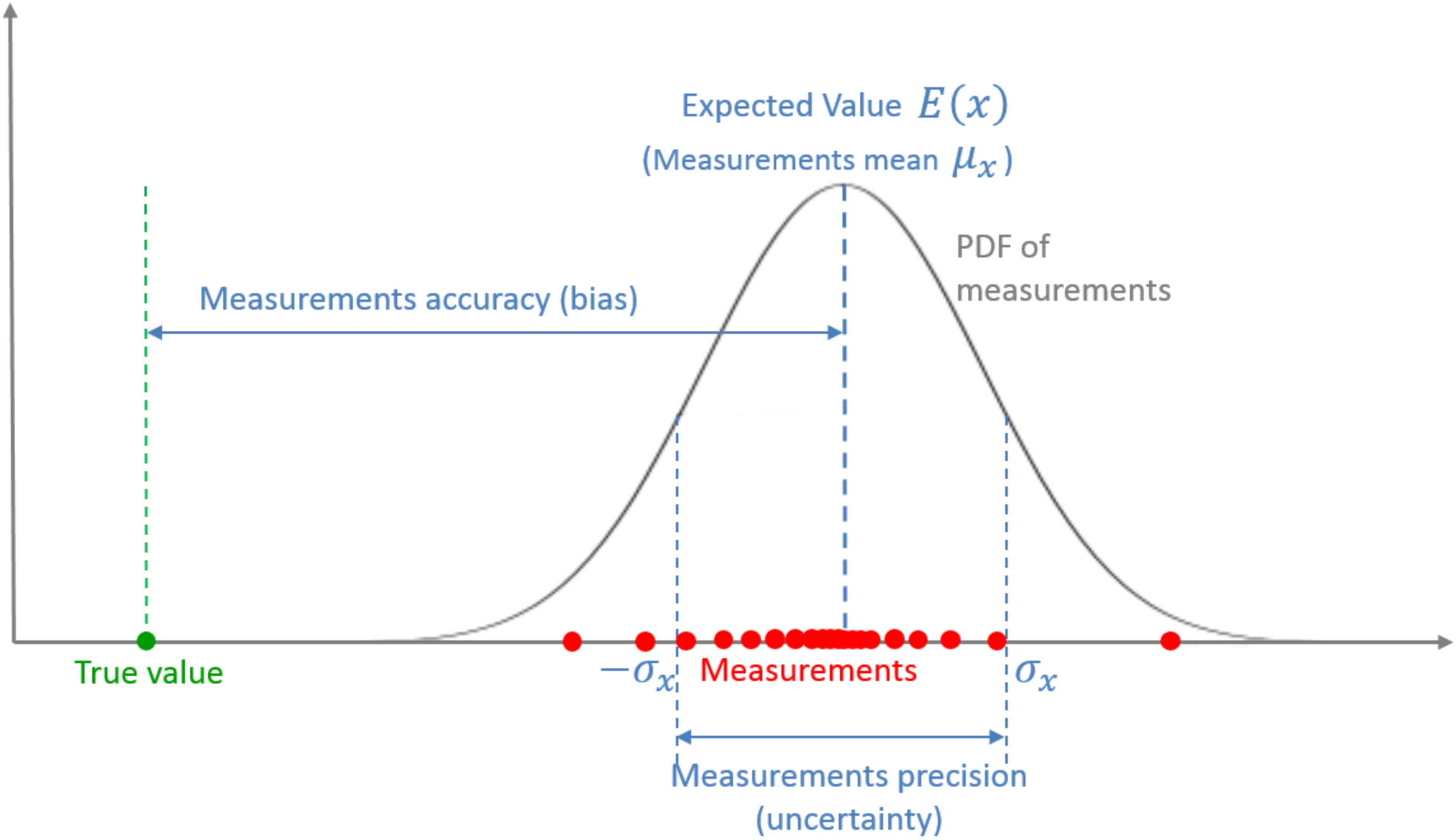
- The second central moment $E\left((X-\mu_X)^2\right)$ – the variance of the sequence of measurements.

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Estimate, Accuracy and Precision

- An Estimate is about evaluating the hidden state of the system.
- Accuracy indicates how close the measurement is to the true value.
- Precision describes the variability in a series of measurements of the same parameter. Accuracy and precision form the basis of the estimate.




Expected Value $E(x)$
(Measurements mean μ_x)
PDF of measurements
Measurements accuracy (bias)
True value
$-\sigma_x$
Measurements
σ_x
Measurements precision
(uncertainty)

4.2 Kalman filter in one dimension

- In this section, we derive the Kalman Filter in one dimension.
- The main goal of this section is to explain the Kalman Filter concept simply and intuitively without using math tools that may seem complex and confusing.
- We are going to advance toward the Kalman Filter equations step by step.

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Example

Consider the problem of estimating the distance to an object. Let z_1 and z_2 be the measurement results. Knowing that the measurement is affected by noise. Give a method to estimate the distance to the object based on the two measurement results above.

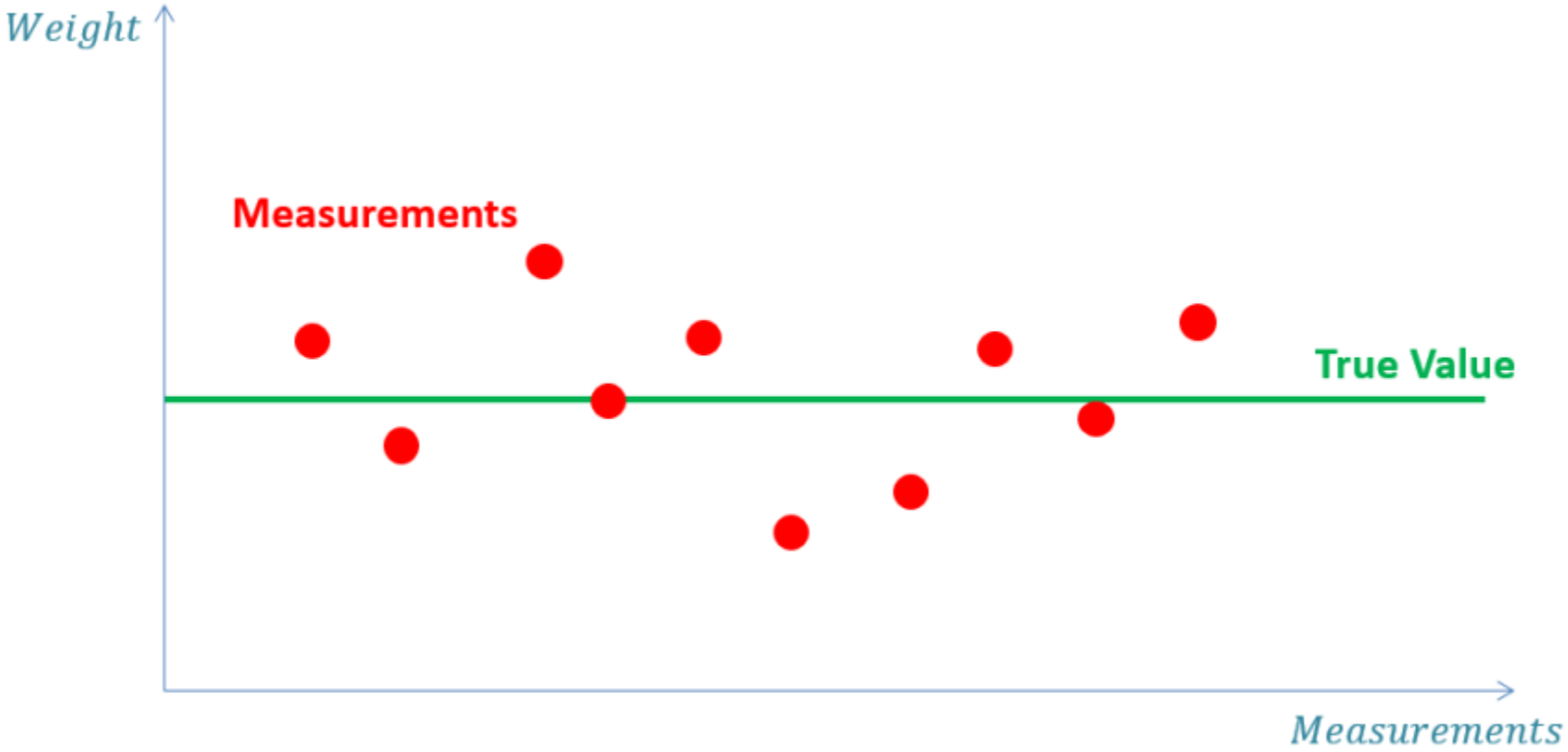
$$\hat{x} = w_1 z_1 + w_2 z_2$$
$$w_1 + w_2 = 1$$

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4.2.1 α filter

- We estimate a static system's state. A static system is a system that doesn't change its state over a reasonable period.
- In this example, we estimate a gold bar's weight. We have unbiased scales, i.e., the scales' measurements don't have a systematic error, but the measurements do include random noise.
- In this example, the system is the gold bar, and the system's state is the weight of the gold bar.
- The system's dynamic model is constant since we assume that the weight doesn't change over short periods.
- In order to estimate the system's state (i.e., the weight value), we can make multiple measurements and average them.

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Weight
Measurements
True Value
Measurements

- At the time n, the estimate $\hat{x}_{n,n}$ would be the average of all previous measurements:

$$\hat{x}_{n,n} = \frac{1}{n}(z_1 + z_2 + \ldots + z_{n-1} + z_n) = \frac{1}{n}\sum_{i=1}^{n}(z_i)$$

- Notation:

x	is the true value of the weight
z_n	is the measured value of the weight at time n
$\hat{x}_{n,n}$	is the estimate of x at time n (the estimate is made after taking the measurement z_n)
$\hat{x}_{n+1,n}$	is the estimate of the future state $(n+1)$ of x. The estimate is made at the time n. In other words, $\hat{x}_{n+1,n}$ is a predicted state or extrapolated state
$\hat{x}_{n-1,n-1}$	is the estimate of x at time $n-1$ (the estimate is made after taking the measurement z_{n-1})
$\hat{x}_{n,n-1}$	is a prior prediction - the estimate of the state at time n. The prediction is made at the time $n-1$

- The dynamic model in this example is constant, therefore $\hat{x}_{n+1,n} = \hat{x}_{n,n}$:

$$\hat{x}_{n,n} = \frac{1}{n}(z_1 + z_2 + \ldots + z_{n-1} + z_n) = \frac{1}{n}\sum_{i=1}^{n}(z_i)$$

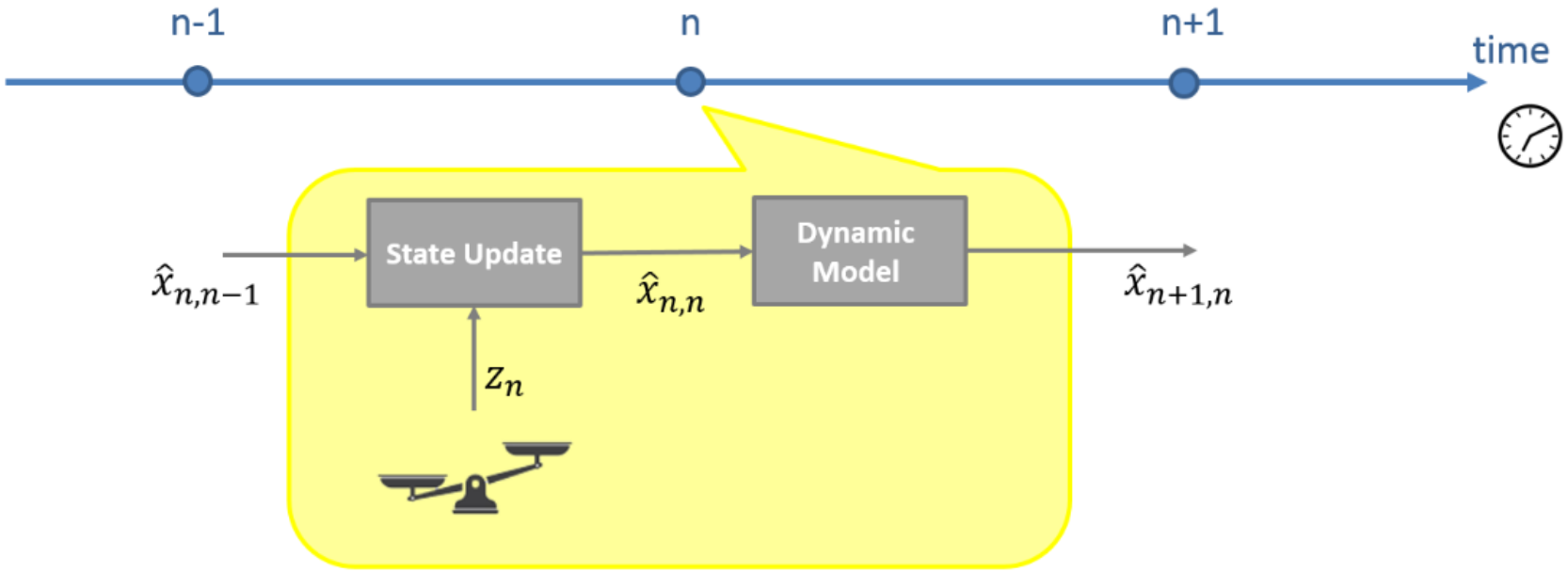
- Although the above equation is mathematically correct, it is not practical for implementation.
- In order to estimate $\hat{x}_{n,n}$ we need to remember all historical measurements; therefore, we need a large memory.
- We also need to recalculate the average repeatedly if we want to update the estimated value after every new measurement.
- Thus, we need a more powerful CPU.

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- It would be more practical to keep the last estimate only ($\hat{x}_{n-1,n-1}$) and update it after every new measurement.
- The following figure exemplifies the required algorithm:

- Estimate the current state based on the measurement and prior prediction.

- Predict the next state based on the current state estimate using the Dynamic Model.



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- We can modify the averaging equation for our needs using a small mathematical trick:

- $\hat{x}_{n-1,n-1}$ is the estimated state of x at the time n − 1, based on the measurement at the time n − 1.

- Let's fnd $\hat{x}_{n,n-1}$ (the predicted state of x at the time n), based on $\hat{x}_{n-1,n-1}$ (the estimation at the time n − 1). In other words, we would like to extrapolate $\hat{x}_{n-1,n-1}$ to the time n.

- Since the dynamic model in this example is static, the predicted state of x equals the estimated state of x: $\hat{x}_{n,n-1} = \hat{x}_{n-1,n-1}$.

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Equation	Notes
$\hat{x}_{n,n} = \frac{1}{n}\sum_{i=1}^{n}(z_i)$	Average formula: sum of n measurements divided by n
$= \frac{1}{n}\left(\sum_{i=1}^{n-1}(z_i) + z_n\right)$	Sum of the $n-1$ measurements plus the last measurement divided by n
$= \frac{1}{n}\sum_{i=1}^{n-1}(z_i) + \frac{1}{n}z_n$	Expand
$= \frac{1}{n}\frac{n-1}{n-1}\sum_{i=1}^{n-1}(z_i) + \frac{1}{n}z_n$	Multiply and divide by term $n-1$
$= \frac{n-1}{n}\frac{1}{n-1}\sum_{i=1}^{n-1}(z_i) + \frac{1}{n}z_n$	Reorder. The 'orange' term is the prior estimate
$= \frac{n-1}{n}\hat{x}_{n-1,n-1} + \frac{1}{n}z_n$	Rewriting the sum
$= \hat{x}_{n-1,n-1} - \frac{1}{n}\hat{x}_{n-1,n-1} + \frac{1}{n}z_n$	Distribute the term $\frac{n-1}{n}$
$= \hat{x}_{n-1,n-1} + \frac{1}{n}(z_n - \hat{x}_{n-1,n-1})$	Reorder

- Based on the above, we can write the **State Update Equation**:

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \frac{1}{n}\left(z_n - \hat{x}_{n,n-1}\right)$$

- The State Update Equation is one of the fve Kalman flter equations. It means the following:

The estimate of the current state = **Predicted value of the current state** + **Factor** X (**Measurement** − **Predicted value of the current state**)

- The factor 1/n is specifc to our example.

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- In the Kalman Filter, the factor is called the Kalman Gain. It is denoted by K_n. The subscript n indicates that the Kalman Gain can change with every iteration.
- The discovery of K_n was one of Rudolf Kalman's significant contributions.
- Before we get into the guts of the Kalman Filter, we use the Greek letter α_n instead of K_n.
- So, the State Update Equation looks as follows:

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \alpha_n \left(z_n - \hat{x}_{n,n-1} \right)$$

- The term $(z_n - x_{n,n-1})$ is the "measurement residual", also called the innovation. The innovation contains new information.

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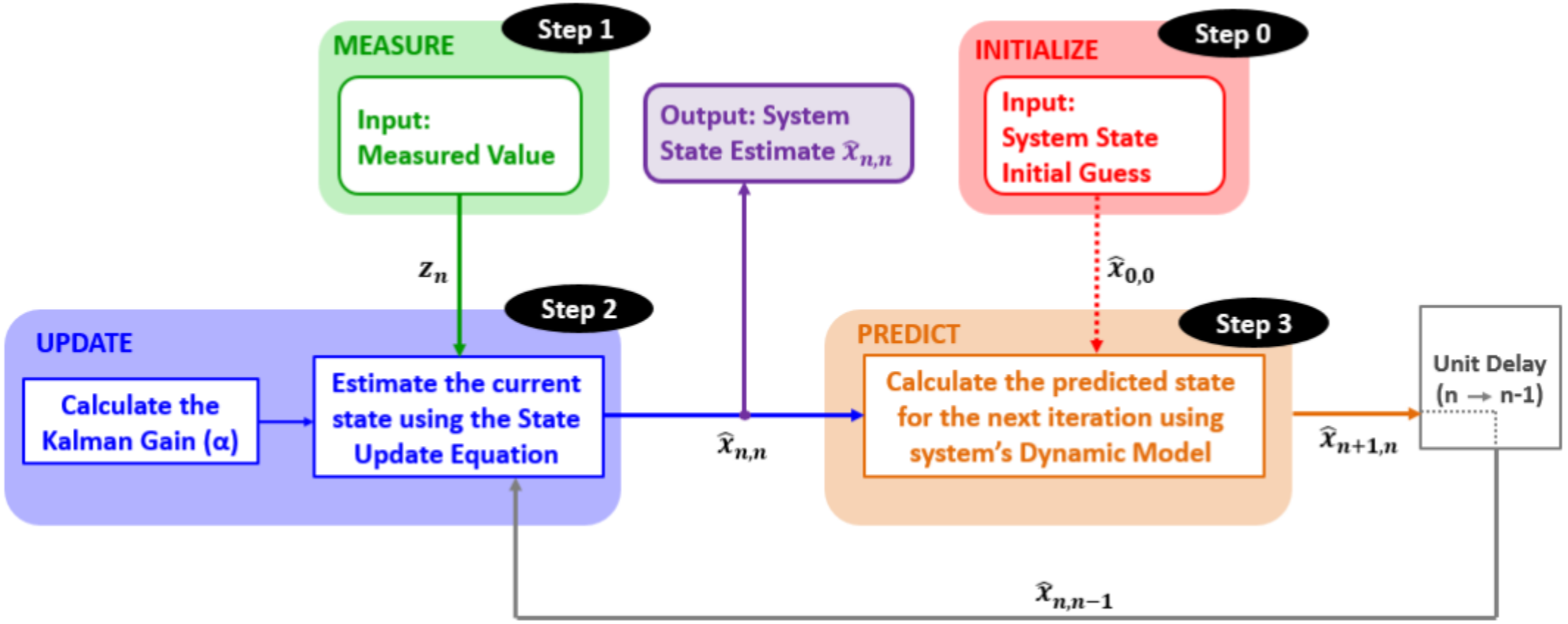
- In this example, $1/n$ decreases as n increases.
- In the beginning, we don't have enough information about the current state; thus, the first estimation is based on the first measurement $\frac{1}{n}\Big|_{n=1} = 1$.
- As we continue, each successive measurement has less weight in the estimation process, since $1/n$ decreases. At some point, the contribution of the new measurements would become negligible.
- Before we make the first measurement, we can guess (or rough estimate) the gold bar weight simply by reading the stamp on the gold bar. It is called the **Initial Guess**, and it is our first estimate.

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- The following chart depicts the estimation algorithm that is used in this example.

$$\hat{x}_{n,n-1} = \hat{x}_{n-1,n-1}$$

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \alpha_n \left(z_n - \hat{x}_{n,n-1} \right)$$



Example

Our initial guess of the gold bar weight is 1000 grams. The weight of the gold bar is not supposed to change. The measurements are given as follows:

n	1	2	3	4	5	6	7	8	9	10
z_n	996	994	1021	1000	1002	1010	983	971	993	1023

Find $\hat{x}_{n,n}$.

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \frac{1}{n}\left(z_n - \hat{x}_{n,n-1}\right)$$

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \alpha_n\left(z_n - \hat{x}_{n,n-1}\right)$$

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Iteration Zero

Initialization

Our initial guess of the gold bar weight is 1000 grams. The initial guess is used only once for the filter initiation. Thus, it won't be required for successive iterations.

$\hat{x}_{0,0} = 1000g$

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First Iteration

Step 1

Making the weight measurement with the scales:

$z_1 = 996g$

Step 2

Calculating the gain. In our example $\alpha_n = {}^1/_n$, thus:

$$\alpha_1 = \frac{1}{1} = 1$$

Calculating the current estimate using the State Update Equation:

$$\hat{x}_{1,1} = \hat{x}_{1,0} + \alpha_1 \left(z_1 - \hat{x}_{1,0}\right) = 1000 + 1\left(996 - 1000\right) = 996g$$

R The initial guess could be any number in this specific example. Since $\alpha_1 = 1$, the initial guess is eliminated in the first iteration.

Step 3

The dynamic model of the system is static; thus, the weight of the gold bar is not supposed to change. Our next state estimate (prediction) equals to current state estimate:

$\hat{x}_{2,1} = \hat{x}_{1,1} = 996g$

Second Iteration

After a unit time delay, the **predicted estimate** from the previous iteration becomes the **prior estimate** in the current iteration:

$\hat{x}_{2,1} = 996g$

Step 1

Making the second measurement of the weight:

$z_2 = 994g$

Step 2

Calculating the gain:

$$\alpha_2 = \frac{1}{2}$$

Calculating the current estimate:

$$\hat{x}_{2,2} = \hat{x}_{2,1} + \alpha_2 \left(z_2 - \hat{x}_{2,1}\right) = 996 + \frac{1}{2}(994 - 996) = 995g$$

Step 3

$$\hat{x}_{3,2} = \hat{x}_{2,2} = 995g$$

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Third Iteration

$$z_3 = 1021g$$

$$\alpha_3 = \frac{1}{3}$$

$$\hat{x}_{3,3} = 995 + \frac{1}{3}(1021 - 995) = 1003.67g$$

$$\hat{x}_{4,3} = 1003.67g$$

Fourth Iteration

$$z_4 = 1000g$$

$$\alpha_4 = \frac{1}{4}$$

$$\hat{x}_{4,4} = 1003.67 + \frac{1}{4}(1000 - 1003.67) = 1002.75g$$

$$\hat{x}_{5,4} = 1002.75g$$

Fifth Iteration

$$z_5 = 1002g$$

$$\alpha_5 = \frac{1}{5}$$

$$\hat{x}_{5,5} = 1002.75 + \frac{1}{5}(1002 - 1002.75) = 1002.6$$

$$\hat{x}_{6,5} = 1002.6g$$

Sixth Iteration

$$z_6 = 1010g$$

$$\alpha_6 = \frac{1}{6}$$

$$\hat{x}_{6,6} = 1002.6 + \frac{1}{6}(1010 - 1002.6) = 1003.83$$

$$\hat{x}_{7,6} = 1003.83g$$

Seventh Iteration

$$z_7 = 983g$$

$$\alpha_7 = \frac{1}{7}$$

$$\hat{x}_{7,7} = 1003.83 + \frac{1}{7}(983 - 1003.83) = 1000.86g$$

$$\hat{x}_{8,7} = 1000.86g$$

Eighth Iteration

$$z_8 = 971g$$

$$\alpha_8 = \frac{1}{8}$$

$$\hat{x}_{8,8} = 1000.86 + \frac{1}{8}(971 - 1000.86) = 997.125g$$

$$\hat{x}_{9,8} = 997.125g$$

Ninth Iteration

$$z_9 = 993g$$

$$\alpha_9 = \frac{1}{9}$$

$$\hat{x}_{9,9} = 997.125 + \frac{1}{9}(993 - 997.125) = 996.67g$$

$$\hat{x}_{10,9} = 996.67g$$

Tenth Iteration

$$z_{10} = 1023g$$

$$\alpha_{10} = \frac{1}{10}$$

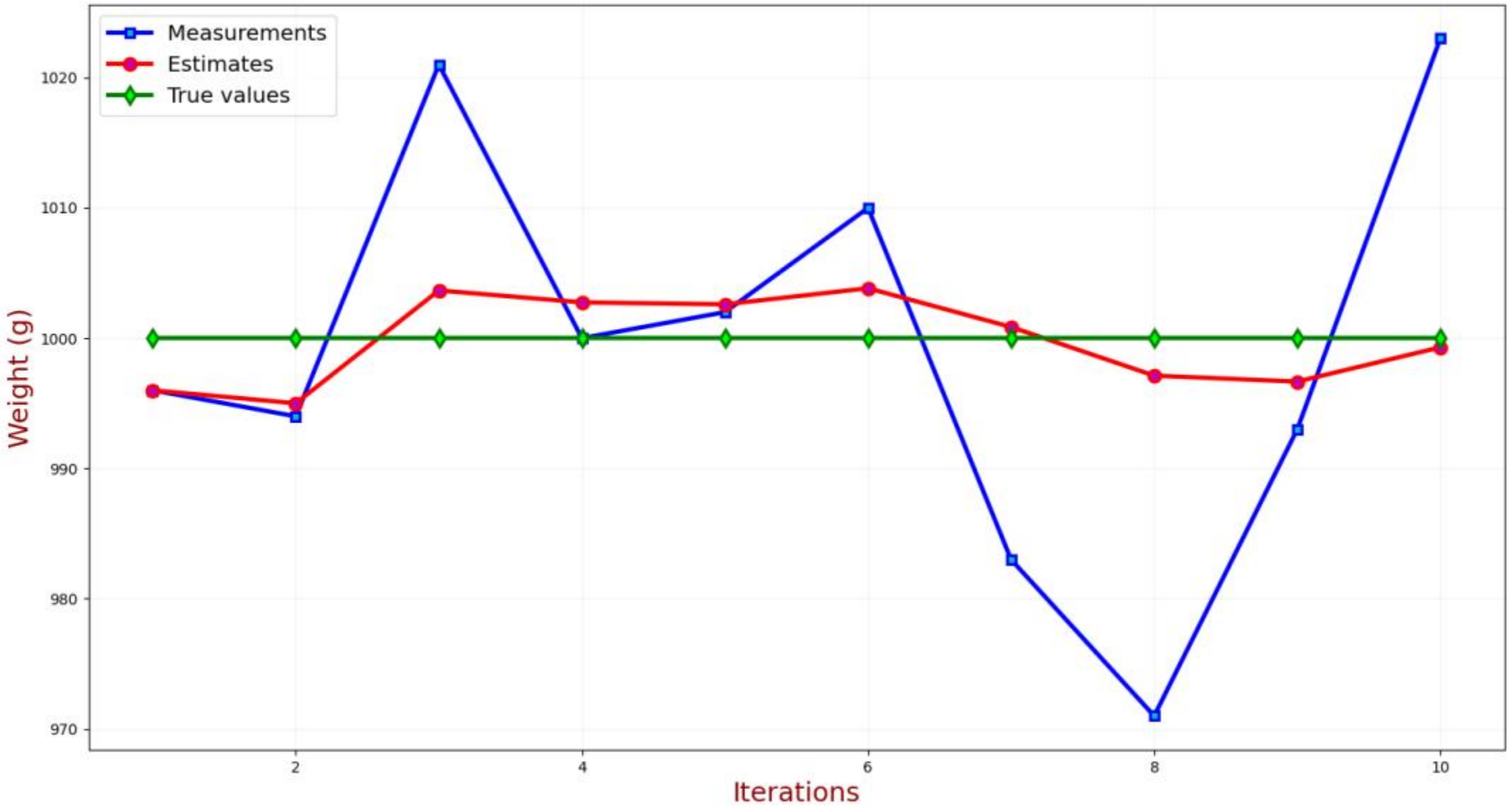
$$\hat{x}_{10,10} = 996.67 + \frac{1}{10}(1023 - 996.67) = 999.3g$$

$$\hat{x}_{11,10} = 999.3g$$

- The gain decreases with each measurement. Therefore, the contribution of each successive measurement is lower than the contribution of the previous measurement. We get pretty close to the true weight, which is 1000g. If we were making more measurements, we would get closer to the true value.
- The following table summarizes measurements and estimates:

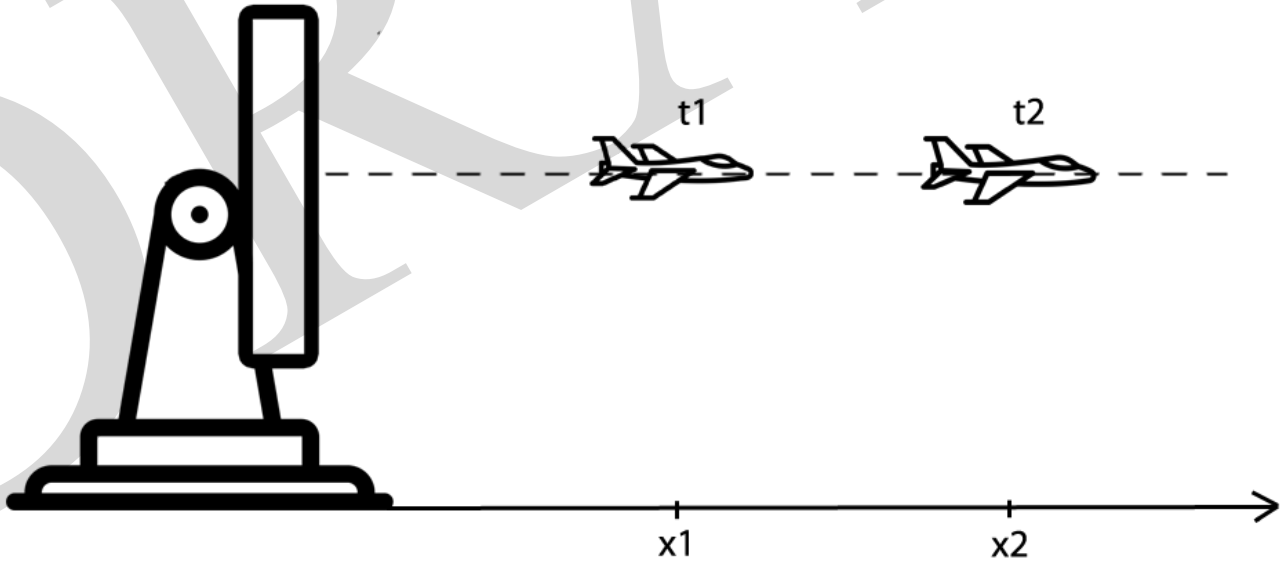
n	1	2	3	4	5	6	7	8	9	10
α_n	1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}
z_n	996	994	1021	1000	1002	1010	983	971	993	1023
$\hat{x}_{n,n}$	996	995	1003.67	1002.75	1002.6	1003.83	1000.86	997.125	996.67	999.3
$\hat{x}_{n+1,n}$	996	995	1003.67	1002.75	1002.6	1003.83	1000.86	997.125	996.67	999.3

- The following chart compares the true, measured, and estimated values.



4.2.2 $\alpha - \beta$ filter

- In this example, we track a constant-velocity aircraft in one dimension using the α-β filter.
- Let us assume a one-dimensional world. We assume an aircraft that is moving radially away from the radar (or towards the radar). In the one-dimensional world, the angle to the radar is constant, and the aircraft's altitude is constant, as shown in the following figure.



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- x_n represents the range to the aircraft at time n.
- The velocity is a derivative of the range: $\dot{x} = v = \frac{dx}{dt}$
- The radar sends a track beam in the direction of the target at a constant rate. The track-to-track interval is Δt. Two equations of motion can describe the system's dynamic model for constant velocity motion:

$$x_{n+1} = x_n + \Delta t \dot{x}_n$$
$$\dot{x}_{n+1} = \dot{x}_n$$

- The above system of equations is called a State Extrapolation Equation (also called a Transition Equation or a Prediction Equation) and is also one of the five Kalman filter equations.
- This system of equations extrapolates the current state to the next state (prediction).

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Example

Let the radar track-to-track (Δt) period be 5 seconds. Assume that at time $n-1$, the estimated range of the Unmanned Air Vehicle (UAV) is 30,000m, and the estimated UAV velocity is 40m/s.

Using the State Extrapolation Equations, we can predict the target position at time n:

$$\hat{x}_{n,n-1} = \hat{x}_{n-1,n-1} + \Delta t \hat{\dot{x}}_{n-1,n-1} = 30000 + 5 \times 40 = 30200m$$

The target velocity prediction for time n:

$$\hat{\dot{x}}_{n,n-1} = \hat{\dot{x}}_{n-1,n-1} = 40m/s$$

However, at time n, the radar measures range (z_n) of 30,110m and not 30,200m as expected. There is a 90m gap between the expected (predicted) range and the measured range. There are two possible reasons for this gap:

- The radar measurements are not precise.
- The aircraft velocity has changed. The new aircraft velocity is: $\frac{30{,}110-30{,}000}{5} = 22m/s$.

Which of the two statements is true?

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- The radar measurements are not precise.
- The aircraft velocity has changed. The new aircraft velocity is: $\frac{30{,}110-30{,}000}{5} = 22m/s$.

Which of the two statements is true?

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The State Update Equation for position and velocity

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \alpha \left(z_n - \hat{x}_{n,n-1} \right)$$

$$\hat{\dot{x}}_{n,n} = \hat{\dot{x}}_{n,n-1} + \beta \left(\frac{z_n - \hat{x}_{n,n-1}}{\Delta t} \right)$$

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Let us write down the State Update Equation for the velocity:

$$\hat{\dot{x}}_{n,n} = \hat{\dot{x}}_{n,n-1} + \beta \left(\frac{z_n - \hat{x}_{n,n-1}}{\Delta t} \right)$$

The value of the factor β depends on the precision level of the radar. Suppose that the 1σ precision of the radar is 20m. The 90 meters gap between the predicted and measured ranges most likely results from a change in the aircraft velocity. We should set the β factor to a high value in this case. If we set $\beta = 0.9$, then the estimated velocity would be:

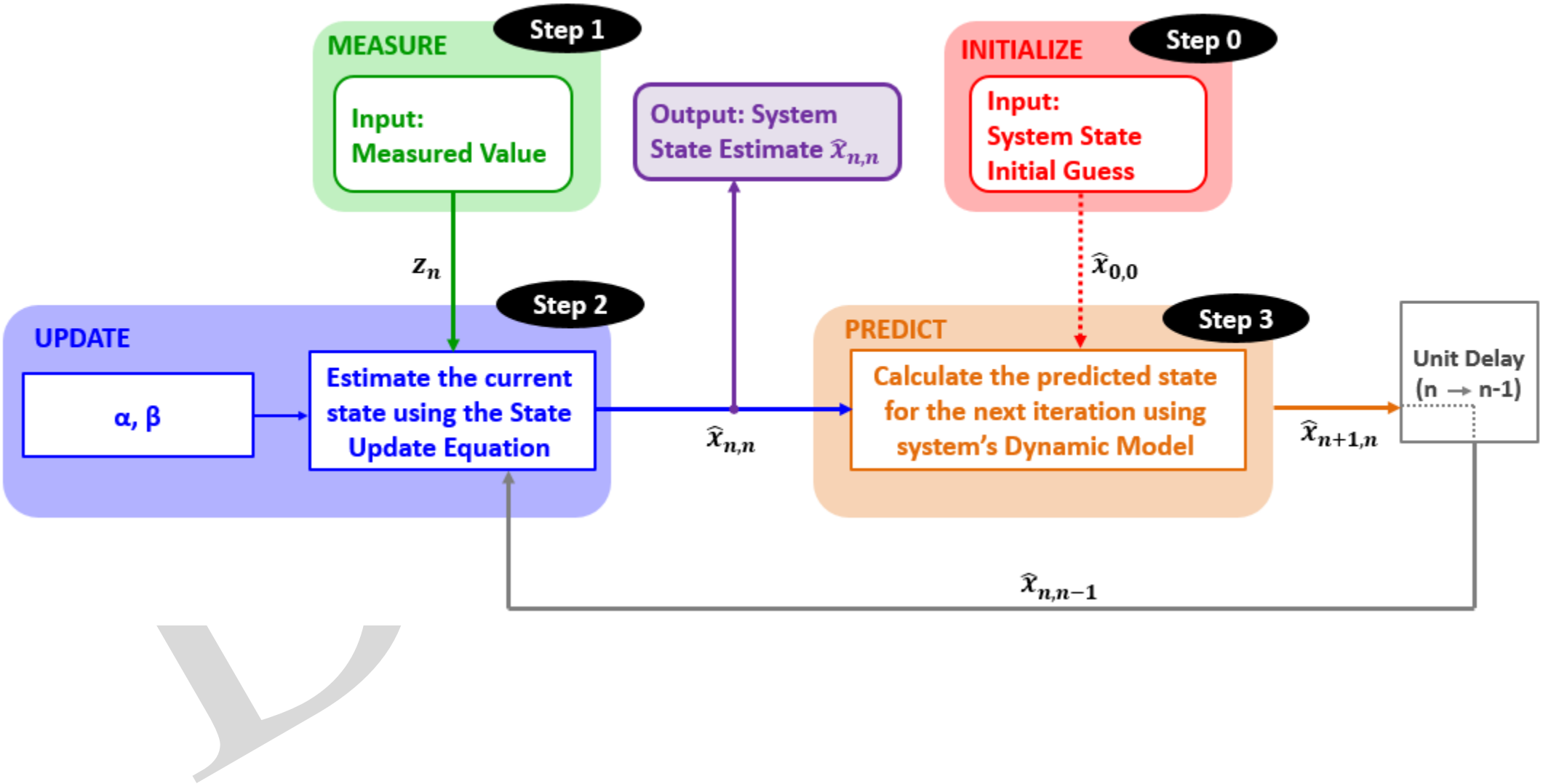
$$\hat{\dot{x}}_{n,n} = \hat{\dot{x}}_{n,n-1} + \beta \left(\frac{z_n - \hat{x}_{n,n-1}}{\Delta t} \right) = 40 + 0.9 \left(\frac{30110 - 30200}{5} \right) = 23.8 m/s$$

On the other hand, suppose that the 1σ precision of the radar is 150m. Then the 90 meters gap probably results from the radar measurement error. We should set the β factor to a low value in this case. If we set $\beta = 0.1$, then the estimated velocity would be:

$$\hat{\dot{x}}_{n,n} = \hat{\dot{x}}_{n,n-1} + \beta\left(\frac{z_n - \hat{x}_{n,n-1}}{\Delta t}\right) = 40 + 0.1\left(\frac{30110 - 30200}{5}\right) = 38.2m/s$$

If the aircraft velocity has changed from 40m/s to 22m/s, we see this after 10 track cycles (running the above equation 10 times with $\beta = 0.1$). If the gap has been caused by measurement error, then the successive measurements would be in front or behind the predicted positions. Thus on average, the target velocity would not change.

- The following chart depicts the estimation algorithm that is used in this example.



Exercise

Consider an aircraft moving radially toward (or away from) a radar in a onedimensional world.

The $\alpha - \beta$ flter parameters are:

- $\alpha = 0.2$

- $\beta = 0.1$

The track-to-track interval is 5 seconds.

The measurement results are given as follows:

n	1	2	3	4	5	6	7	8	9	10
z_n	30171	30353	30756	30799	31018	31278	31276	31379	31748	32175

Find $\hat{x}_{n,n}$.

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Iteration Zero

Initialization

The initial conditions for the time $n = 0$ are given:

$\hat{x}_{0,0} = 30000m$

$\hat{\dot{x}}_{0,0} = 40m/s$

Prediction

The initial guess should be extrapolated to the first cycle using the State Extrapolation Equations:

$\hat{x}_{n+1,n} = \hat{x}_{n,n} + \Delta t \hat{\dot{x}}_{n,n} \rightarrow \hat{x}_{1,0} = \hat{x}_{0,0} + \Delta t \hat{\dot{x}}_{0,0} = 30000 + 5 \times 40 = 30200m$

$\hat{\dot{x}}_{n+1,n} = \hat{\dot{x}}_{n,n} \rightarrow \hat{\dot{x}}_{1,0} = \hat{\dot{x}}_{0,0} = 40m/s$

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First Iteration

In the first cycle ($n = 1$), the initial guess is the prior estimate:

$\hat{x}_{n,n-1} = \hat{x}_{1,0} = 30200m$

$\hat{\dot{x}}_{n,n-1} = \hat{\dot{x}}_{1,0} = 40m/s$

Step 1

The radar measures the aircraft range:

$z_1 = 30171m$

Step 2

Calculating the current estimate using the State Update Equation:

$\hat{x}_{1,1} = \hat{x}_{1,0} + \alpha\left(z_1 - \hat{x}_{1,0}\right) = 30200 + 0.2\left(30171 - 30200\right) = 30194.2m$

$\hat{\dot{x}}_{1,1} = \hat{\dot{x}}_{1,0} + \beta\left(\frac{z_1 - \hat{x}_{1,0}}{\Delta t}\right) = 40 + 0.1\left(\frac{30171 - 30200}{5}\right) = 39.42m/s$

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Step 3

Calculating the next state estimate using the State Extrapolation Equations:

$$\hat{x}_{2,1} = \hat{x}_{1,1} + \Delta t \hat{\dot{x}}_{1,1} = 30194.2 + 5 \times 39.42 = 30391.3m$$

$$\hat{\dot{x}}_{2,1} = \hat{\dot{x}}_{1,1} = 39.42m/s$$

Second Iteration

After a unit time delay, the predicted estimate from the previous iteration becomes the prior estimate in the current iteration:

$$\hat{x}_{2,1} = 30391.3m$$

$$\hat{\dot{x}}_{2,1} = 39.42m/s$$

Step 1

The radar measures the aircraft range:

$$z_2 = 30353m$$

Step 2

Calculating the current estimate using the State Update Equation:

$$\hat{x}_{2,2} = \hat{x}_{2,1} + \alpha\left(z_2 - \hat{x}_{2,1}\right) = 30391.3 + 0.2\left(30353 - 30391.3\right) = 30383.64m$$

$$\hat{\dot{x}}_{2,2} = \hat{\dot{x}}_{2,1} + \beta\left(\frac{z_2 - \hat{x}_{2,1}}{\Delta t}\right) = 39.42 + 0.1\left(\frac{30353 - 30391.3}{5}\right) = 38.65m/s$$

Step 3

Calculating the next state estimate using the State Extrapolation Equations:

$$\hat{x}_{3,2} = \hat{x}_{2,2} + \Delta t\hat{\dot{x}}_{2,2} = 30383.64 + 5 \times 38.65 = 30576.9m$$

$$\hat{\dot{x}}_{3,2} = \hat{\dot{x}}_{2,2} = 38.65m/s$$

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Third Iteration

$$z_3 = 30756m$$

$$\hat{x}_{3,3} = 30576.9 + 0.2\,(30756 - 30576.9) = 30612.73m$$

$$\hat{\dot{x}}_{3,3} = 38.65 + 0.1\left(\frac{30756 - 30576.9}{5}\right) = 42.2m/s$$

$$\hat{x}_{4,3} = 30612.73 + 5 \times 42.2 = 30823.9m$$

$$\hat{\dot{x}}_{4,3} = 42.2m/s$$

Fourth Iteration

$$z_4 = 30799m$$

$$\hat{x}_{4,4} = 30823.9 + 0.2\,(30799 - 30823.9) = 30818.93m$$

$$\hat{\dot{x}}_{4,4} = 42.2 + 0.1\left(\frac{30799 - 30823.9}{5}\right) = 41.7m/s$$

$$\hat{x}_{5,4} = 30818.93 + 5 \times 41.7 = 31027.6m$$

$$\hat{\dot{x}}_{5,4} = 41.7m/s$$

Fifth Iteration

$$z_5 = 31018m$$

$$\hat{x}_{5,5} = 31027.6 + 0.2\,(31018 - 31027.6) = 31025.7m$$

$$\hat{\dot{x}}_{5,5} = 41.7 + 0.1\left(\frac{31018 - 31027.6}{5}\right) = 41.55m/s$$

$$\hat{x}_{6,5} = 31025.7 + 5 \times 41.55 = 31233.4m$$

$$\hat{\dot{x}}_{6,5} = 41.55m/s$$

Sixth Iteration

$$z_6 = 31278m$$

$$\hat{x}_{6,6} = 31233.4 + 0.2\,(31278 - 31233.4) = 31242.3m$$

$$\hat{\dot{x}}_{6,6} = 41.55 + 0.1\left(\frac{31278 - 31233.4}{5}\right) = 42.44m/s$$

$$\hat{x}_{7,6} = 31242.3 + 5 \times 42.44 = 31454.5m$$

$$\hat{\dot{x}}_{7,6} = 42.44m/s$$

Seventh Iteration

$$z_7 = 31276m$$

$$\hat{x}_{7,7} = 31454.5 + 0.2\,(31276 - 31454.5) = 31418.8m$$

$$\hat{\dot{x}}_{7,7} = 42.44 + 0.1\left(\frac{31276 - 31454.5}{5}\right) = 38.9m/s$$

$$\hat{x}_{8,7} = 31418.8 + 5 \times 38.9 = 31613.15m$$

$$\hat{\dot{x}}_{8,7} = 38.9m/s$$

Eighth Iteration

$$z_8 = 31379m$$

$$\hat{x}_{8,8} = 31613.15 + 0.2\,(31379 - 31613.15) = 31566.3m$$

$$\hat{\dot{x}}_{8,8} = 38.9 + 0.1\left(\frac{31379 - 31613.15}{5}\right) = 34.2m/s$$

$$\hat{x}_{9,8} = 31566.3 + 5 \times 34.2 = 31737.24m$$

$$\hat{\dot{x}}_{9,8} = 34.2m/s$$

Ninth Iteration

$$z_9 = 31748m$$

$$\hat{x}_{9,9} = 31737.24 + 0.2\,(31748 - 31737.24) = 31739.4m$$

$$\hat{\dot{x}}_{9,9} = 34.2 + 0.1\left(\frac{31748 - 31737.24}{5}\right) = 34.4m/s$$

$$\hat{x}_{10,9} = 31739.4 + 5 \times 34.4 = 31911.4m$$

$$\hat{\dot{x}}_{10,9} = 34.4m/s$$

Tenth Iteration

$$z_{10} = 32175m$$

$$\hat{x}_{10,10} = 31911.4 + 0.2\,(32175 - 31911.4) = 31964.1m$$

$$\hat{\dot{x}}_{10,10} = 34.4 + 0.1\left(\frac{32175 - 31911.4}{5}\right) = 39.67m/s$$

$$\hat{x}_{11,10} = 31964.1 + 5 \times 39.67 = 32162.45m$$

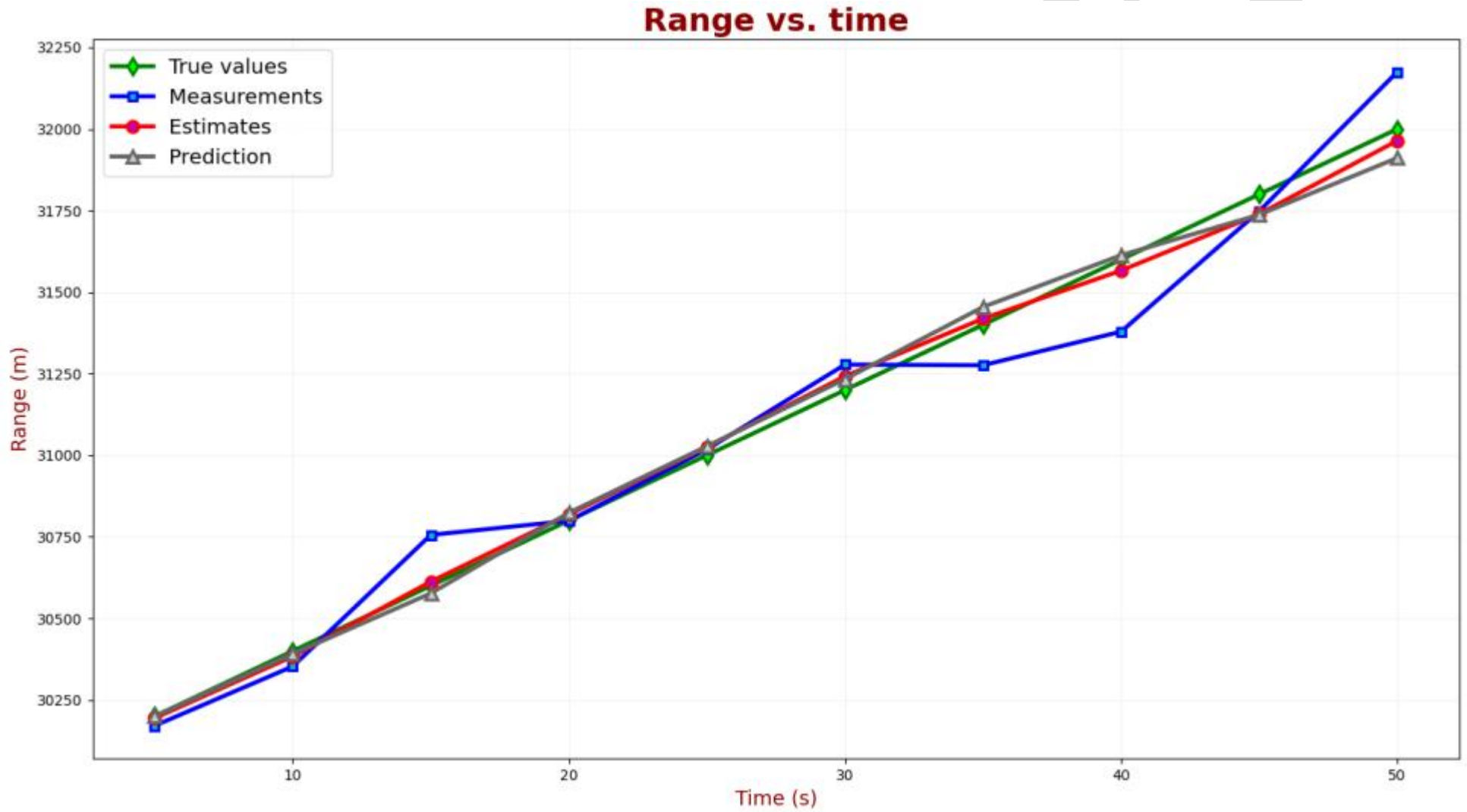
$$\hat{\dot{x}}_{11,10} = 39.67m/s$$

- The following table summarizes measurements and estimates.

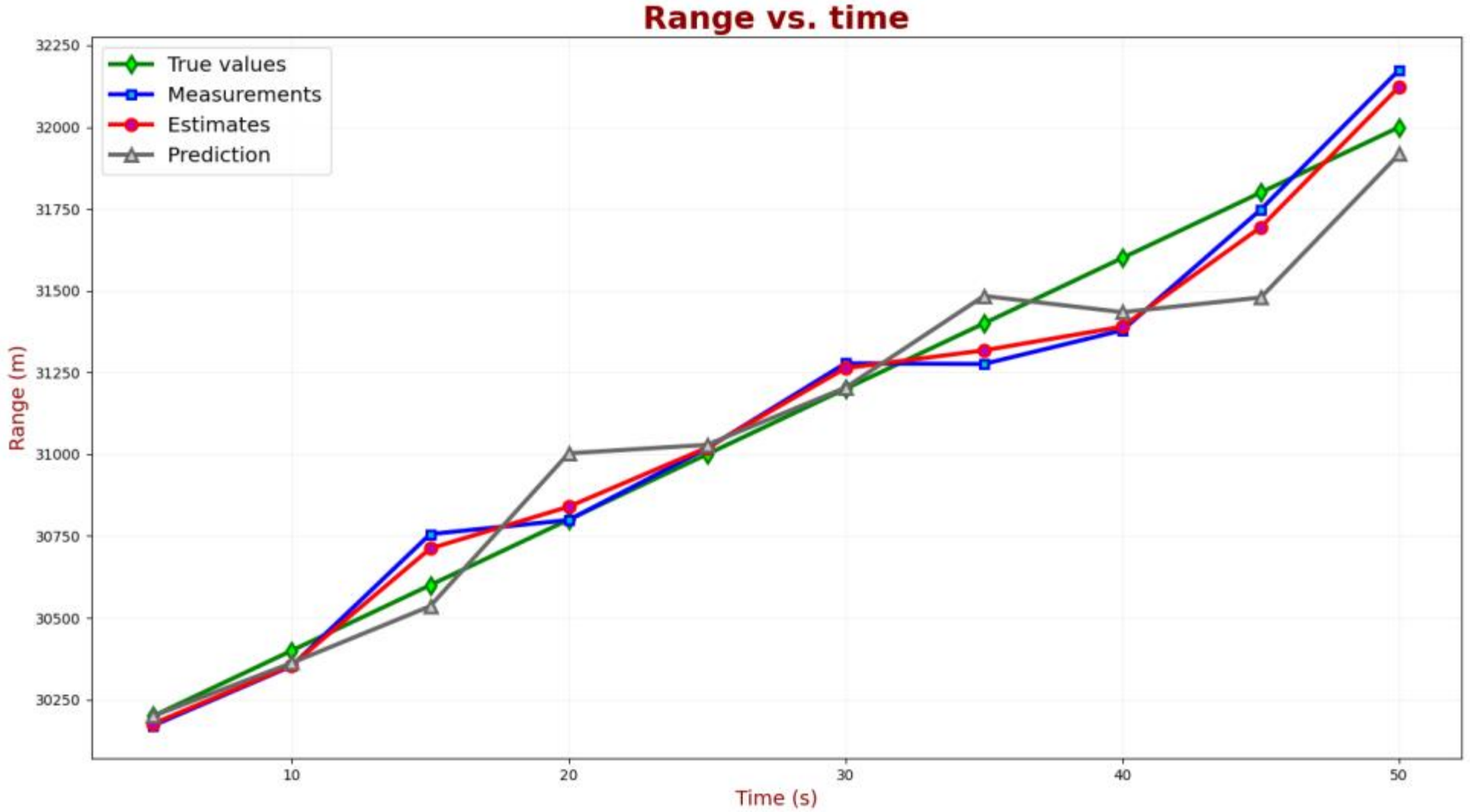
n	1	2	3	4	5	6	7	8	9	10
z_n	30171	30353	30756	30799	31018	31278	31276	31379	31748	32175
$\hat{x}_{n,n}$	30194.2	30383.64	30612.73	30818.93	31025.7	31242.3	31418.8	31566.3	31739.4	31964.1
$\dot{\hat{x}}_{n,n}$	39.42	38.65	42.2	41.7	41.55	42.44	38.9	34.2	34.4	39.67
$\hat{x}_{n+1,n}$	30391.3	30576.9	30823.9	31027.6	31233.4	31454.5	31613.15	31737.24	31911.4	32162.45
$\dot{\hat{x}}_{n+1,n}$	39.42	38.65	42.2	41.7	41.55	42.44	38.9	34.2	34.4	39.67

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- The following chart compares the true values, measured values, and estimates.



- The following chart depicts the true, measured, and estimated values for $\alpha = 0.8$ and $\beta = 0.5$.



- So, shall we always choose low values for α and β?
- The answer is NO.
- The value of α and β should depend on the measurement precision.
- If we use high-precision equipment, like laser radar, we would prefer a high α and β that follow measurements. In this case, the filter would quickly respond to a velocity change of the target.
- On the other hand, if measurement precision is low, we prefer low α and β. In this case, the filter smoothes the uncertainty (errors) in the measurements. However, the filter reaction to target velocity changes would be much slower.

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4.2.3 $\alpha - \beta - \gamma$ filter

- In this example, we track an aircraft using the $\alpha-\beta-\gamma$ flter. The aircraft is moving with a constant acceleration.
- The $\alpha - \beta - \gamma$ flter considers a target acceleration. Thus, the State Extrapolation Equations become:

The State Extrapolation Equations for position, velocity, and acceleration

$$\hat{x}_{n+1,n} = \hat{x}_{n,n} + \hat{\dot{x}}_{n,n}\Delta t + \hat{\ddot{x}}_{n,n}\frac{\Delta t^2}{2}$$
$$\hat{\dot{x}}_{n+1,n} = \hat{\dot{x}}_{n,n} + \hat{\ddot{x}}_{n,n}\Delta t$$
$$\hat{\ddot{x}}_{n+1,n} = \hat{\ddot{x}}_{n,n}$$

Where $\ddot{x}_n$ is acceleration (the second derivative of x).

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- The State Update Equations become:

The State Update Equations for position, velocity, and acceleration

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \alpha\left(z_n - \hat{x}_{n,n-1}\right)$$

$$\hat{\dot{x}}_{n,n} = \hat{\dot{x}}_{n,n-1} + \beta\left(\frac{z_n - \hat{x}_{n,n-1}}{\Delta t}\right)$$

$$\hat{\ddot{x}}_{n,n} = \hat{\ddot{x}}_{n,n-1} + \gamma\left(\frac{z_n - \hat{x}_{n,n-1}}{0.5\Delta t^2}\right)$$

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Example

Let's take the scenario from the previous example: an aircraft that moves with a constant velocity of 50m/s for 20 seconds and then accelerates with a constant acceleration of $8m/s^2$ for another 35 seconds.

The α - β -γ flter parameters are:

- $\alpha = 0.5, \beta = 0.4, \gamma = 0.1$.

The track-to-track interval is 5 seconds.

The initial conditions for the time n = 0 are given: $\hat{x}_{0,0} = 30000m$, $\hat{\dot{x}}_{0,0} = 50m/s, \hat{\ddot{x}}_{0,0} = 0m/s^2$.

The measurement results are give as follows:

n	1	2	3	4	5	6	7	8	9	10
z_n	30221	30453	30906	30999	31368	31978	32526	33379	34698	36275

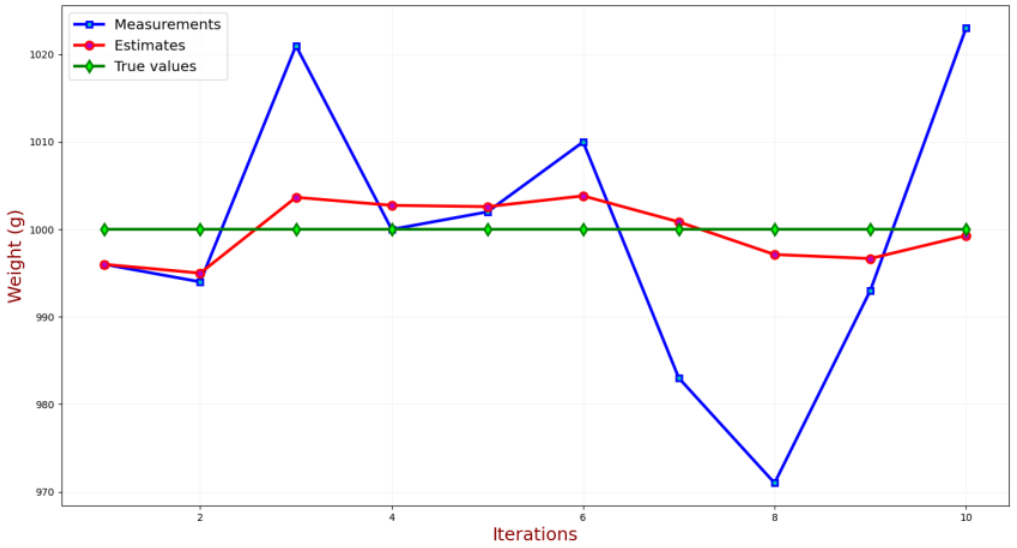
Find $\hat{x}_{n,n}$.

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4.2.4 One-dimensional Kalman Filter without process noise

Estimate as a random variable

- The difference between the estimates (the red line) and the true values (the green line) is the estimate error.
- As you can see, the estimate error becomes lower as we make additional measurements, and it converges towards zero, while the estimated value converges towards the true value.
- We don't know the estimate error, but we can estimate the state uncertainty.
- We denote the state estimate variance by p.



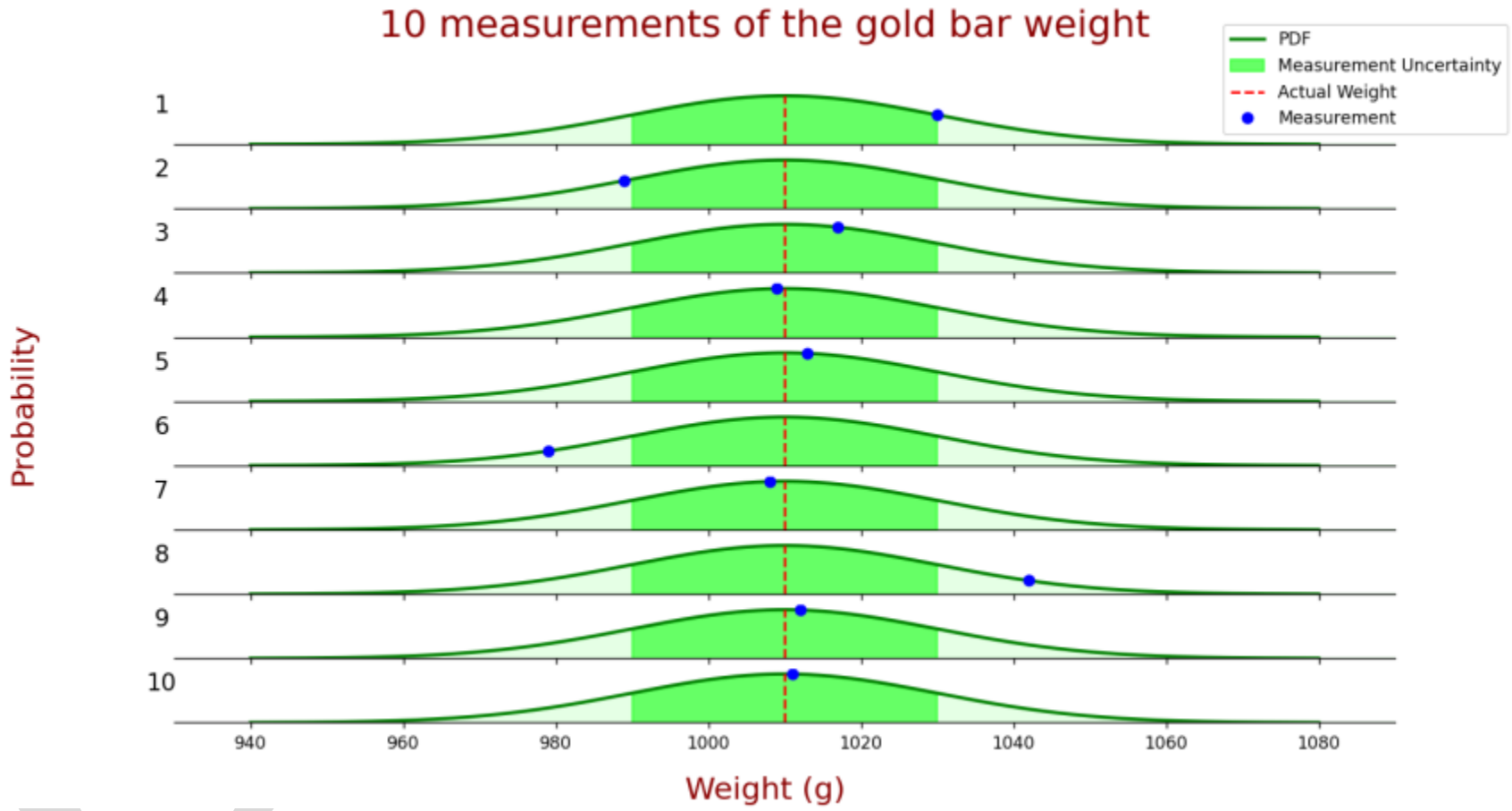
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Measurement as a random variable

- The measurement errors are the differences between the measurements (blue samples) and the true values (green line).
- Since the measurement errors are random, we can describe them by variance (σ^2).
- The standard deviation (σ) of the measurement errors is the measurement uncertainty.
- We denote the measurement variance by r.

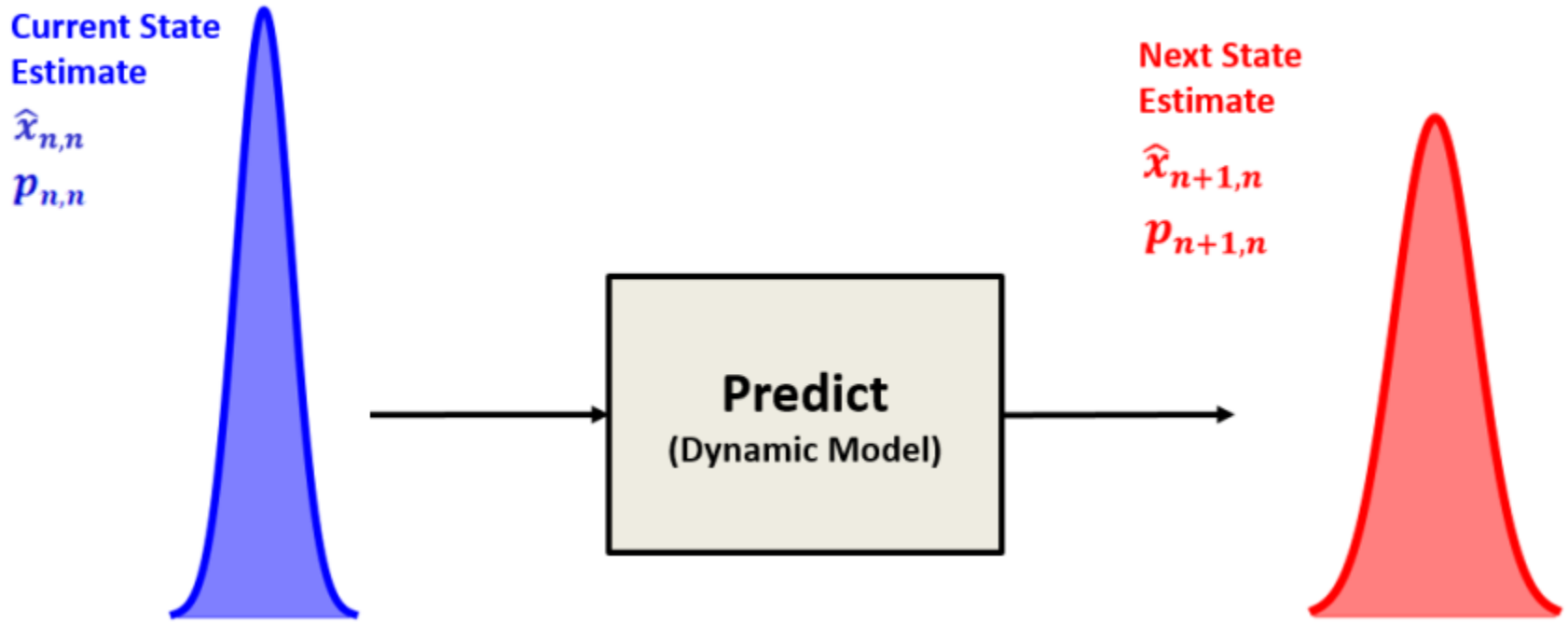
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Let's look at the weight measurements Probability Density Function (PDF). The following plot shows ten measurements of the gold bar weight.



State prediction

- Since Kalman Filter treats the estimate as a random variable, we must also extrapolate the estimation variance ($p_{n,n}$) to the next state.



- In the frst example - “The measurement of the gold bar weight”, the dynamic model of the system is constant. Thus, the estimate uncertainty extrapolation would be:

$$\hat{p}_{n+1,n} = \hat{p}_{n,n}$$

Where p is the estimate variance of the gold bar weight.

- In the second example - “Tracking the constant velocity aircraft”, the estimate uncertainty extrapolation would be:

$$p^x_{n+1,n} = p^x_{n,n} + \Delta t^2 \cdot p^v_{n,n}$$
$$p^v_{n+1,n} = p^v_{n,n}$$

Where:

p^x is the position estimate variance.

p^v is the velocity estimate variance.

Note that for any normally distributed random variable x with variance σ^2, kx is distributed normally with variance $k^2\sigma^2$, therefore the time term in the uncertainty extrapolation equation is squared.

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Expectation of the variance

$$V(X) = \sigma_x^2 = E(X^2) - \mu_X^2 \tag{A.2}$$

Where $V(X)$ is the variance of X

The proof

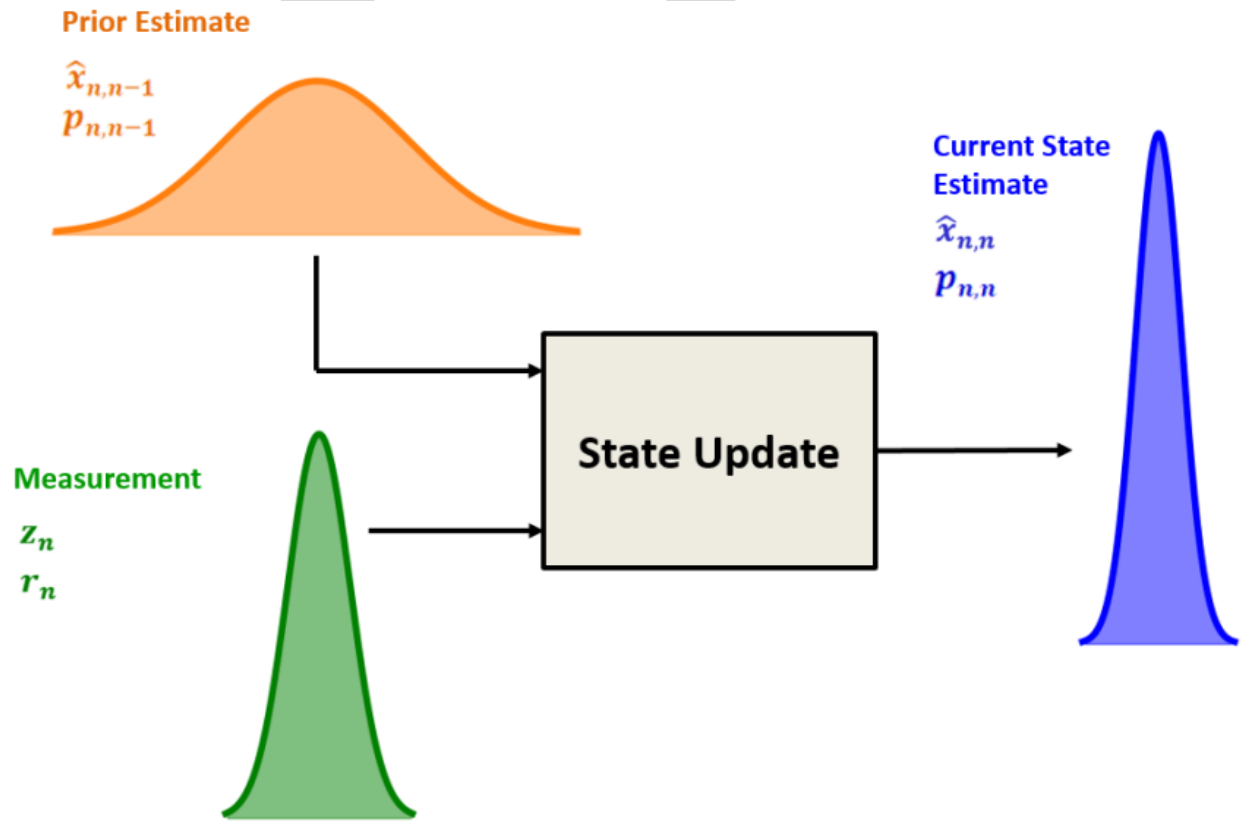
	Notes
$V(X) = \sigma_X^2 = E((X - \mu_X)^2)$	
$= E(X^2 - 2X\mu_X + \mu_X^2)$	
$= E(X^2) - E(2X\mu_X) + E(\mu_X^2)$	Applied rule number 5: $E(a \pm bX) = a \pm bE(X)$
$= E(X^2) - 2\mu_X E(X) + E(\mu_X^2)$	Applied rule number 3: $E(aX) = aE(X)$
$= E(X^2) - 2\mu_X E(X) + \mu_X^2$	Applied rule number 2: $E(a) = a$
$= E(X^2) - 2\mu_X\mu_X + \mu_X^2$	Applied rule number 1: $E(X) = \mu_X$
$= E(X^2) - \mu_X^2$	

$V(K) = \sigma_K^2 = E(K^2) - \mu_K^2$	
$= E((v\Delta t)^2) - (\mu_v \Delta t)^2$	Express the body position variance in terms of time and velocity: $x = \Delta t v$
$= E(v^2 \Delta t^2) - \mu_v^2 \Delta t^2$	
$= \Delta t^2 E(v^2) - \mu_v^2 \Delta t^2$	Applied rule number 3: $E(aX) = aE(X)$
$= \Delta t^2 (E(v^2) - \mu_v^2)$	
$= \Delta t^2 V(v)$	Applied expectation of variance rule: $V(X) = E(X^2) - \mu_X^2$

State update

- To estimate the current state of the system, we combine two random variables:

- The prior state estimate (the current state estimate predicted at the previous state).
- The measurement.



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- The Kalman Filter is an optimal flter. It combines the prior state estimate with the measurement in a way that minimizes the uncertainty of the current state estimate.
- The current state estimate is a weighted mean of the measurement and the prior state estimate:

$$\hat{x}_{n,n} = w_1 z_n + w_2 \hat{x}_{n,n-1}$$

$$w_1 + w_2 = 1$$

where w_1 and w_2 are the weights of the measurement and the prior state estimate.

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- We can write $\hat{x}_{n,n}$ as follows:

$$\hat{x}_{n,n} = w_1 z_n + (1 - w_1)\hat{x}_{n,n-1}$$

- The relation between variances is given by:

$$p_{n,n} = w_1^2 r_n + (1 - w_1)^2 p_{n,n-1}$$

Where:

$p_{n,n}$	is the variance of the optimum combined estimate
$p_{n,n-1}$	is the variance of the prior estimate $\hat{x}_{n,n-1}$
r_n	is the variance of the measurement z_n

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- Since we are looking for an optimum estimate, we want to minimize $p_{n,n}$.
- To fnd w_1 that minimizes $p_{n,n}$, we differentiate $p_{n,n}$ with respect to w_1 and set the result to zero.

$$\frac{dp_{n,n}}{dw_1} = 2w_1 r_n - 2(1 - w_1)p_{n,n-1} = 0$$

- Hence,

$$w_1 r_n = p_{n,n-1} - w_1 p_{n,n-1}$$

$$w_1 p_{n,n-1} + w_1 r_n = p_{n,n-1}$$

$$w_1 = \frac{p_{n,n-1}}{p_{n,n-1} + r_n}$$

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- Let us substitute the result into the current state estimation $\hat{x}_{n,n}$.

$$\hat{x}_{n,n} = w_1 z_n + (1 - w_1)\hat{x}_{n,n-1} = w_1 z_n + \hat{x}_{n,n-1} - w_1 \hat{x}_{n,n-1} = \hat{x}_{n,n-1} + w_1 (z_n - \hat{x}_{n,n-1})$$

State Update Equation

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + \frac{p_{n,n-1}}{p_{n,n-1} + r_n} (z_n - \hat{x}_{n,n-1})$$

- The weight of the innovation is called the Kalman Gain (denoted by K_n).

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- The Kalman Gain Equation is the fourth Kalman Filter equation. In one dimension, the Kalman Gain Equation is the following:

State Update Equation

$$K_n = \frac{Variance \quad in \quad Estimate}{Variance \quad in \quad Estimate \quad + \quad Variance \quad in \quad Measurement}$$

$$K_n = \frac{p_{n,n-1}}{p_{n,n-1} + r_n}$$

Where:

$p_{n,n-1}$ is the extrapolated estimate variance

r_n is the measurement variance

- The Kalman Gain is a number between zero and one:

$$0 \leq K_n \leq 1$$

- Finally, we need to fnd the variance of the current state estimate. We've seen that the relation between variances is given by:

$$p_{n,n} = w_1^2 r_n + (1 - w_1)^2 p_{n,n-1}$$

$$p_{n,n} = K_n^2 r_n + (1 - K_n)^2 p_{n,n-1}$$

$$(1 - K_n) = \left(1 - \frac{p_{n,n-1}}{p_{n,n-1} + r_n}\right) = \left(\frac{p_{n,n-1} + r_n - p_{n,n-1}}{p_{n,n-1} + r_n}\right) = \left(\frac{r_n}{p_{n,n-1} + r_n}\right)$$

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Equation	Notes
$p_{n,n} = \left(\frac{p_{n,n-1}}{p_{n,n-1}+r_n}\right)^2 r_n + \left(\frac{r_n}{p_{n,n-1}+r_n}\right)^2 p_{n,n-1}$	Substitute K_n and $(1-K_n)$
$= \frac{p_{n,n-1}^2 r_n}{(p_{n,n-1}+r_n)^2} + \frac{r_n^2 p_{n,n-1}}{(p_{n,n-1}+r_n)^2}$	Expand
$= \frac{p_{n,n-1} r_n}{p_{n,n-1}+r_n}\left(\frac{p_{n,n-1}}{p_{n,n-1}+r_n} + \frac{r_n}{p_{n,n-1}+r_n}\right)$	
$= (1-K_n)\, p_{n,n-1}\Big(K_n + (1-K_n)\Big)$	Substitute K_n and $(1-K_n)$
$= (1-K_n)\, p_{n,n-1}$	

- This equation updates the estimate variance of the current state. It is called the Covariance Update Equation.

Covariance Update Equation

$$p_{n,n} = (1 - K_n)\, p_{n,n-1}$$

- It is clear from the equation that the estimate uncertainty is constantly decreasing with each flter iteration, since (1 − Kn) ≤ 1.
- When the measurement uncertainty is high, the Kalman gain is low. Therefore, the convergence of the estimate uncertainty would be slow.
- On the other hand, the Kalman gain is high when the measurement uncertainty is low. Therefore, the estimate uncertainty would quickly converge toward zero.

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- We combine all of these pieces into a single algorithm.
- The filter inputs are:

- Initialization

The initialization is performed only once, and it provides two parameters:

+ Initial System State ($\hat{x}_{0,0}$)

+ Initial State Variance ($p_{0,0}$)

Even if the initialization parameters are not precise, the Kalman flter can converge close to the true value.

- Measurement

The measurement is performed for every flter cycle, and it provides two parameters:

+ Measured System State (z_n)

+ Measurement Variance (r_n)

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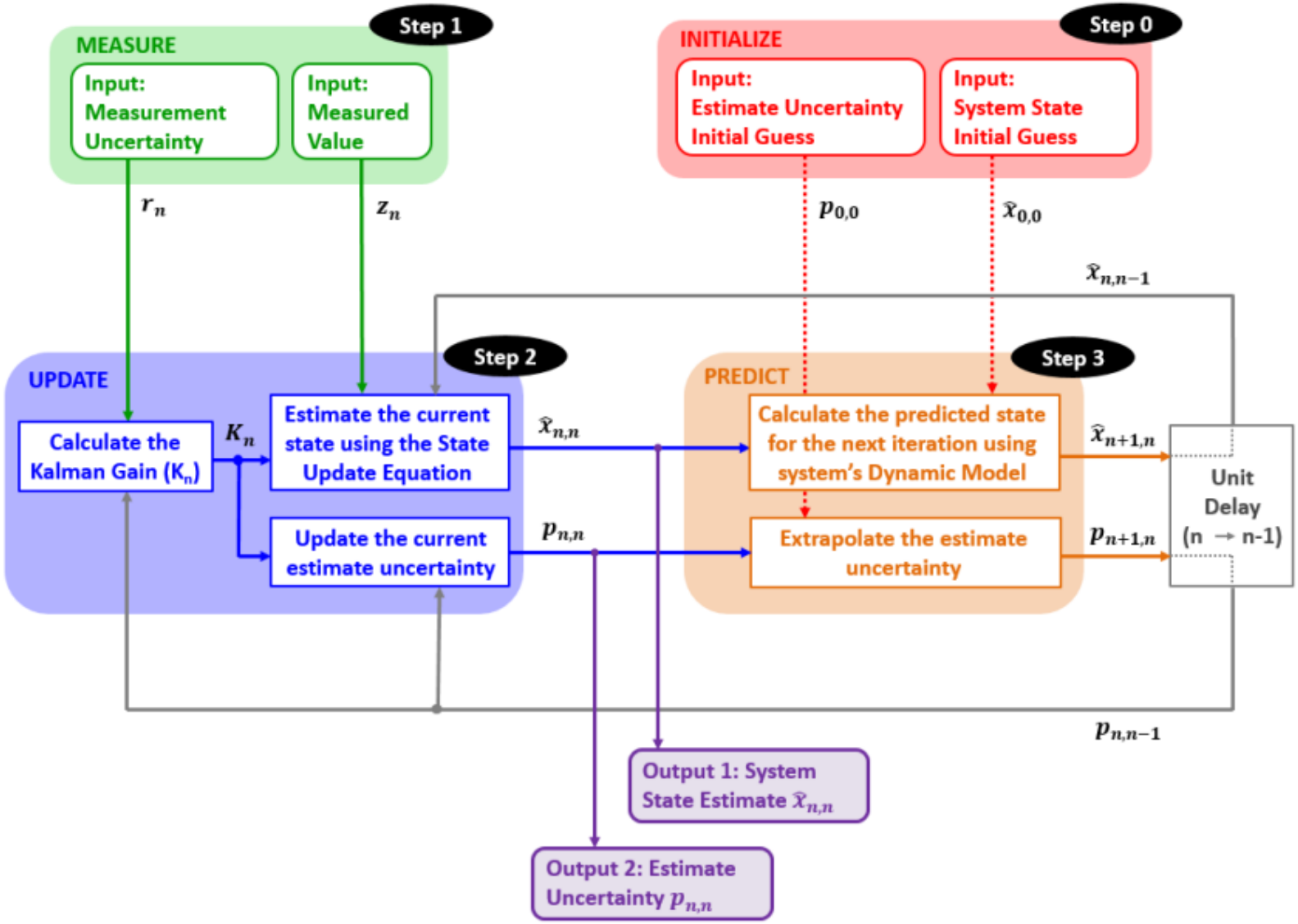
- The filter outputs are:
  - System State Estimate ($\hat{x}_{n,n}$)
  - Estimate Variance ($p_{n,n}$)
- The following table summarizes the five Kalman Filter equations.

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	Equation	Equation Name	Alternative names used in the literature
State Update	$\hat{x}_{n,n} = \hat{x}_{n,n-1} + K_n(z_n - \hat{x}_{n,n})$	State Update	Filtering Equation
	$p_{n,n} = (1 - K_n)p_{n,n-1}$	Covariance Update	Corrector Equation
	$K_n = \frac{p_{n,n-1}}{p_{n,n-1} + r_n}$	Kalman Gain	Weight Equation
State Predict	$\hat{x}_{n+1,n} = \hat{x}_{n,n}$ (for constant dynamics)	State Extrapolation	Predictor Equation Transition Equation Prediction Equation Dynamic Model State Space Model
	$\hat{x}_{n+1,n} = \hat{x}_{n,n} + \Delta t\hat{\dot{x}}_{n,n}$ $\hat{\dot{x}}_{n+1,n} = \hat{\dot{x}}_{n,n}$ (for constant velocity dynamics)		
	$p^x_{n+1,n} = p^x_{n,n}$ (for constant dynamics)	Covariance Extrapolation	Predictor Covariance Equation
	$p^x_{n+1,n} = p^x_{n,n} + \Delta t^2 p^v_{n,n}$ $p^v_{n+1,n} = p^v_{n,n}$ (for constant velocity dynamics)		

- The equations above don't include the process noise.
- The State Extrapolation Equation and the Covariance Extrapolation Equation depend on the system dynamics.

- The following fgure provides a detailed description of the Kalman Filter's block diagram.



Kalman Gain intuition

- Let's rewrite the state update equation:

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + K_n\left(z_n - \hat{x}_{n,n-1}\right) = \left(1 - K_n\right)\hat{x}_{n,n-1} + K_n z_n$$

- As you can see, the Kalman Gain (K_n) is the measurement weight, and the ($1 - K_n$) term is the weight of the current state estimate.
- The Kalman Gain is close to zero when the measurement uncertainty is high and the estimate uncertainty is low.
- On the other hand, when the measurement uncertainty is low, and the estimate uncertainty is high, the Kalman Gain is close to one.

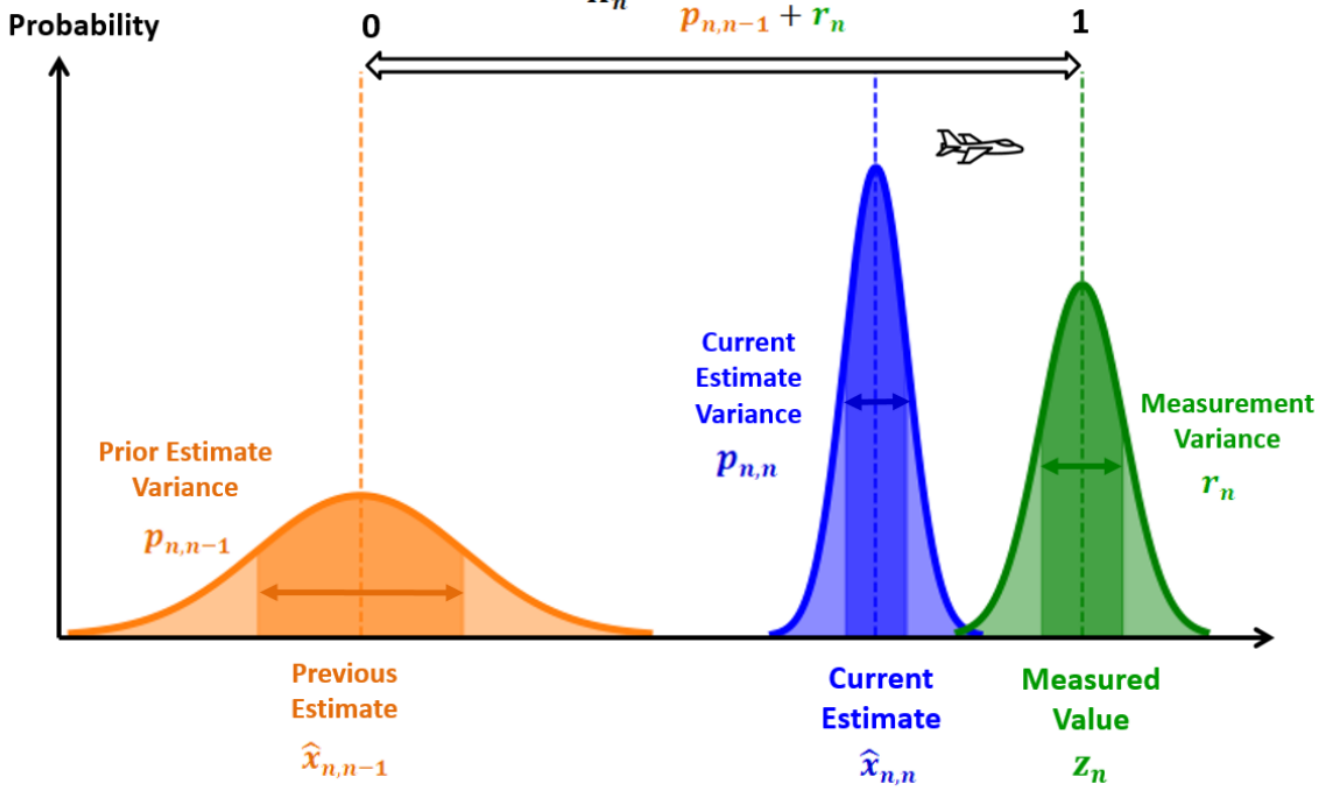
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High Kalman Gain

- A low measurement uncertainty relative to the estimate uncertainty would result in a high Kalman Gain (close to 1). Therefore the new estimate would be close to the measurement.

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + K_n(z_n - \hat{x}_{n,n-1})$$
$$p_{n,n} = (1 - K_n)p_{n,n-1}$$

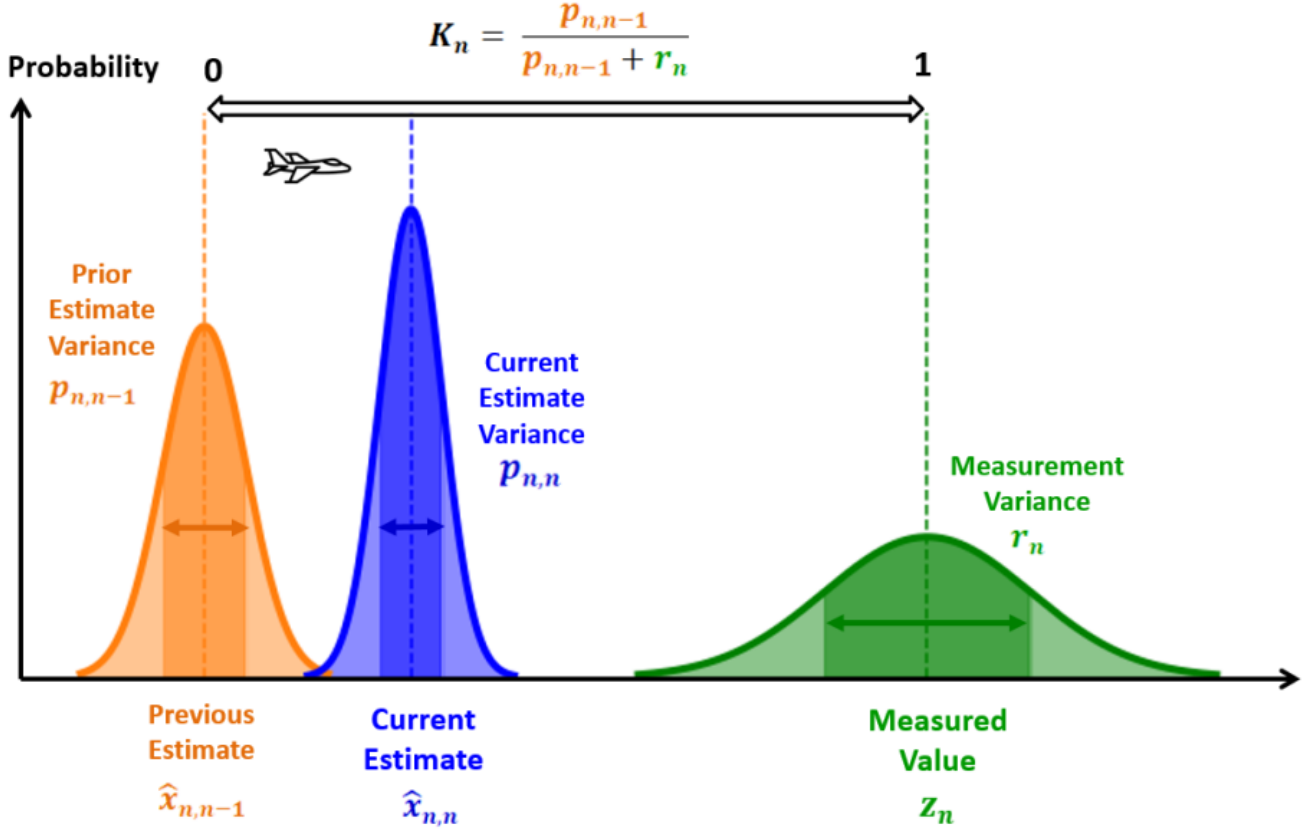
$$K_n = \frac{p_{n,n-1}}{p_{n,n-1} + r_n}$$



Low Kalman Gain

A high measurement uncertainty relative to the estimate uncertainty would result in a low Kalman Gain (close to 0). Therefore the new estimate would be close to the prior estimate.

$$\hat{x}_{n,n} = \hat{x}_{n,n-1} + K_n(z_n - \hat{x}_{n,n-1})$$
$$p_{n,n} = (1 - K_n)p_{n,n-1}$$



Example

- Assume that we would like to estimate the height of a building using an imprecise altimeter.
- We know that building height doesn't change over time, at least during the short measurement process.
- The true building height is 50 meters.
- The altimeter measurement error (standard deviation) is 5 meters.
- The ten measurements are: 49.03m, 48.44m, 55.21m, 49.98m, 50.6m, 52.61m, 45.87m, 42.64m, 48.26m, 55.84m.

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Iteration Zero

Initialization

One can estimate the height of the building simply by looking at it.

The estimated height of the building for the initialization purpose is:

$\hat{x}_{0,0} = 60m$

Now we shall initialize the estimate variance. A human estimation error (standard deviation) is about 15 meters: $\sigma = 15$. Consequently the variance is 225: $\sigma^2 = 225$.

$p_{0,0} = 225$

Prediction

Now, we shall predict the next state based on the initialization values.

Since our system's Dynamic Model is constant, i.e., the building doesn't change its height:

$\hat{x}_{1,0} = \hat{x}_{0,0} = 60m$

The extrapolated estimate variance also doesn't change:

$p_{1,0} = p_{0,0} = 225$

First Iteration

Step 1 - Measure

The first measurement is:

$z_1 = 49.03m$

Since the standard deviation (σ) of the altimeter measurement error is 5, the variance (σ^2) would be 25, thus, the measurement uncertainty is: $r_2 = 25$

Step 2 - Update

Kalman Gain calculation:

$$K_1 = \frac{p_{1,0}}{p_{1,0} + r_1} = \frac{225}{225 + 25} = 0.9$$

Estimating the current state:

$$\hat{x}_{1,1} = \hat{x}_{1,0} + K_1 (z_1 - \hat{x}_{1,0}) = 60 + 0.9 (49.03 - 60) = 50.13m$$

Update the current estimate variance:

$$p_{1,1} = (1 - K_1) p_{1,0} = (1 - 0.9) 225 = 22.5$$

Step 3 - Predict

Since the dynamic model of our system is constant, i.e., the building doesn't change its height:

$\hat{x}_{2,1} = \hat{x}_{1,1} = 50.13m$

The extrapolated estimate variance also doesn't change:

$p_{2,1} = p_{1,1} = 22.5$

Second Iteration

After a unit time delay, the predicted estimate from the previous iteration becomes the prior estimate in the current iteration:

$\hat{x}_{2,1} = 50.13m$

The extrapolated estimate variance becomes the prior estimate variance:

$p_{2,1} = 22.5$

Step 1 - Measure

The second measurement is:

$z_2 = 48.44m$

The measurement variance is:

$r_2 = 25$

Step 2 - Update

Kalman Gain calculation:

$$K_2 = \frac{p_{2,1}}{p_{2,1} + r_2} = \frac{22.5}{22.5 + 25} = 0.47$$

Estimating the current state:

$$\hat{x}_{2,2} = \hat{x}_{2,1} + K_2 (z_2 - x_{2,1}) = 50.13 + 0.47 (48.44 - 50.13) = 49.33m$$

Update the current estimate variance:

$$p_{2,2} = (1 - K_2) p_{2,1} = (1 - 0.47) 22.5 = 11.84$$

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Step 3 - Predict

Since the dynamic model of our system is constant, i.e., the building doesn't change its height:

$\hat{x}_{3,2} = \hat{x}_{2,2} = 49.33m$

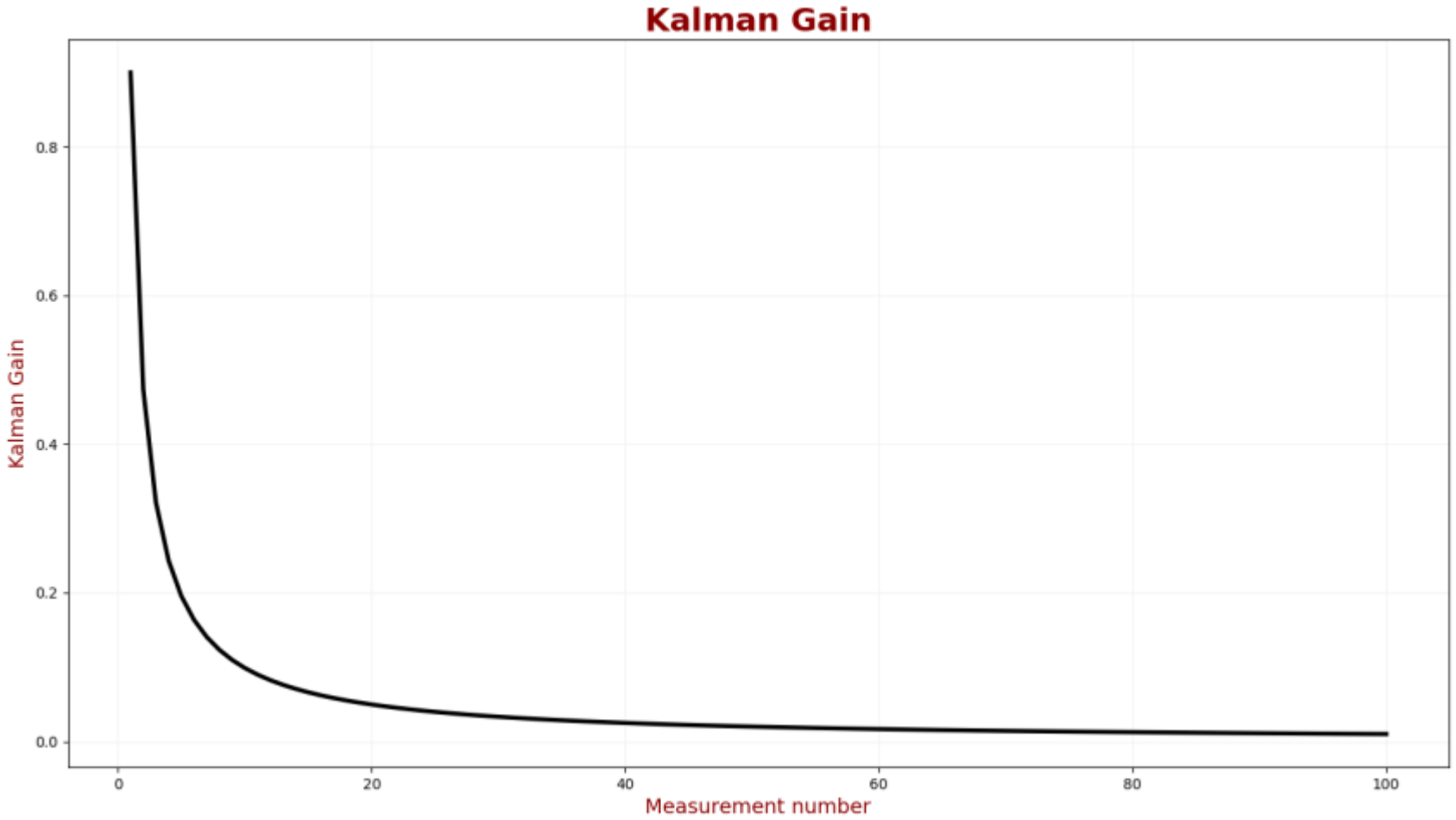
The extrapolated estimate variance also doesn't change:

$p_{3,2} = p_{2,2} = 11.84$

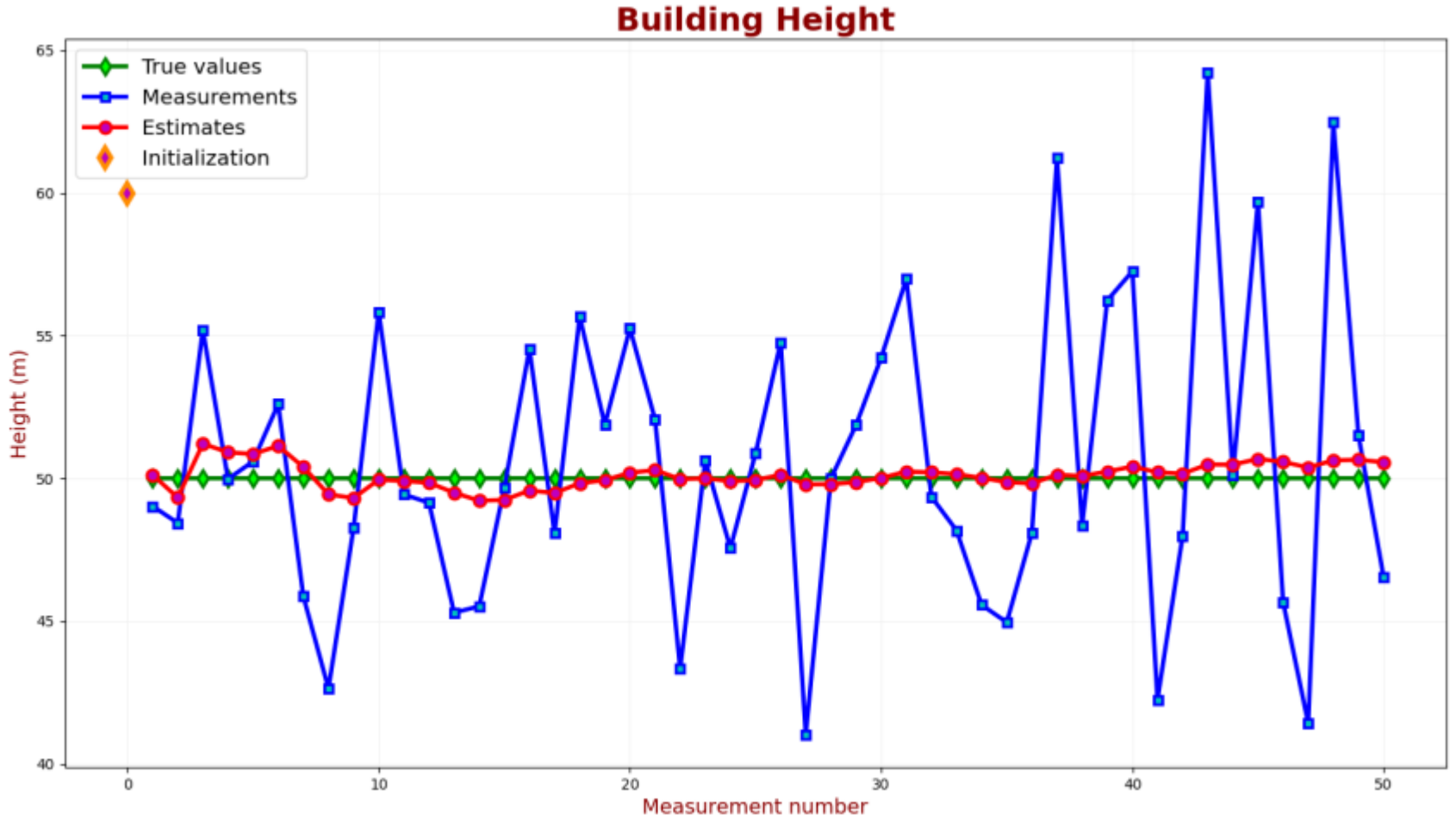
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n	z_n	Current state estimates $(K_n, \hat{x}_{n,n}, p_{n,n})$	Prediction $(\hat{x}_{n+1,n}, p_{n+1,n})$
3	55.21m	$K_3 = \frac{11.84}{11.84 + 25} = 0.32$ $\hat{x}_{3,3} = 49.33 + 0.32\,(55.21 - 49.33) = 51.22m$ $p_{3,3} = (1 - 0.32)\,11.84 = 8.04$	$\hat{x}_{4,3} = \hat{x}_{3,3} = 51.22m$ $p_{4,3} = p_{3,3} = 8.04$
4	49.98m	$K_4 = \frac{8.04}{8.04 + 25} = 0.24$ $\hat{x}_{4,4} = 51.22 + 0.24\,(49.98 - 51.22) = 50.92m$ $p_{4,4} = (1 - 0.24)\,8.04 = 6.08$	$\hat{x}_{5,4} = \hat{x}_{4,4} = 50.92m$ $p_{5,4} = p_{4,4} = 6.08$

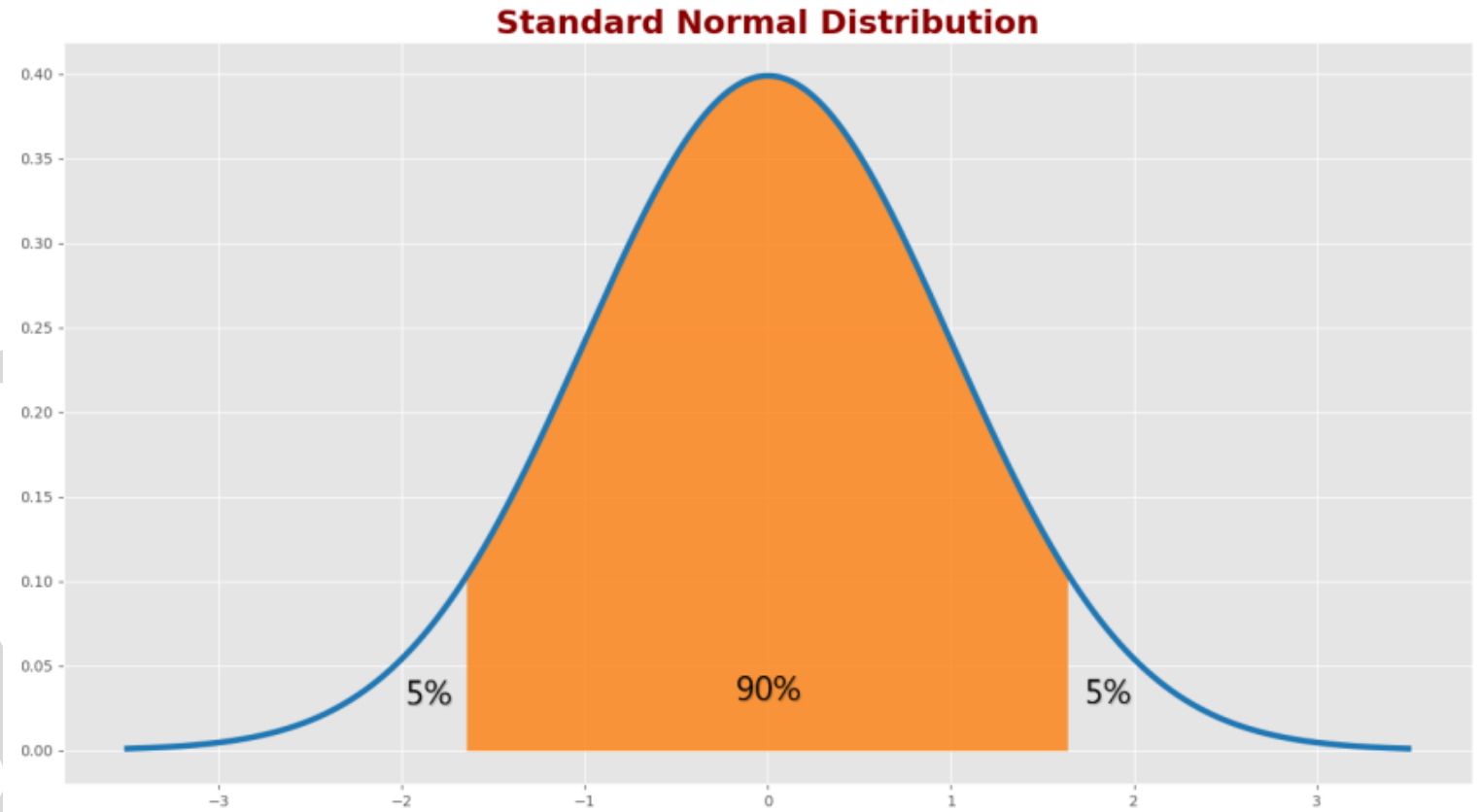
- The Kalman Gain should gradually decrease until it reaches a steady state.



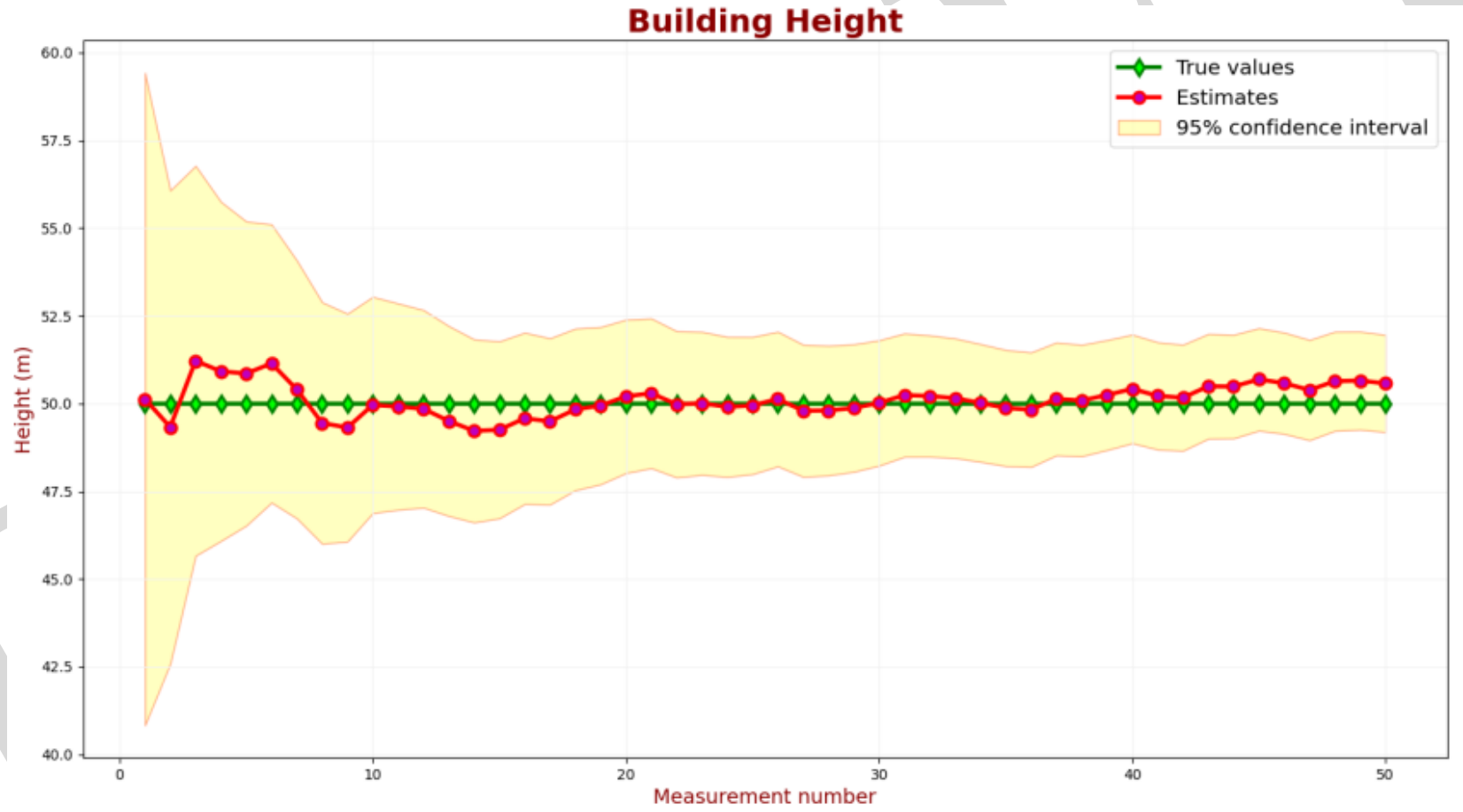
- True value, measured values and estimates



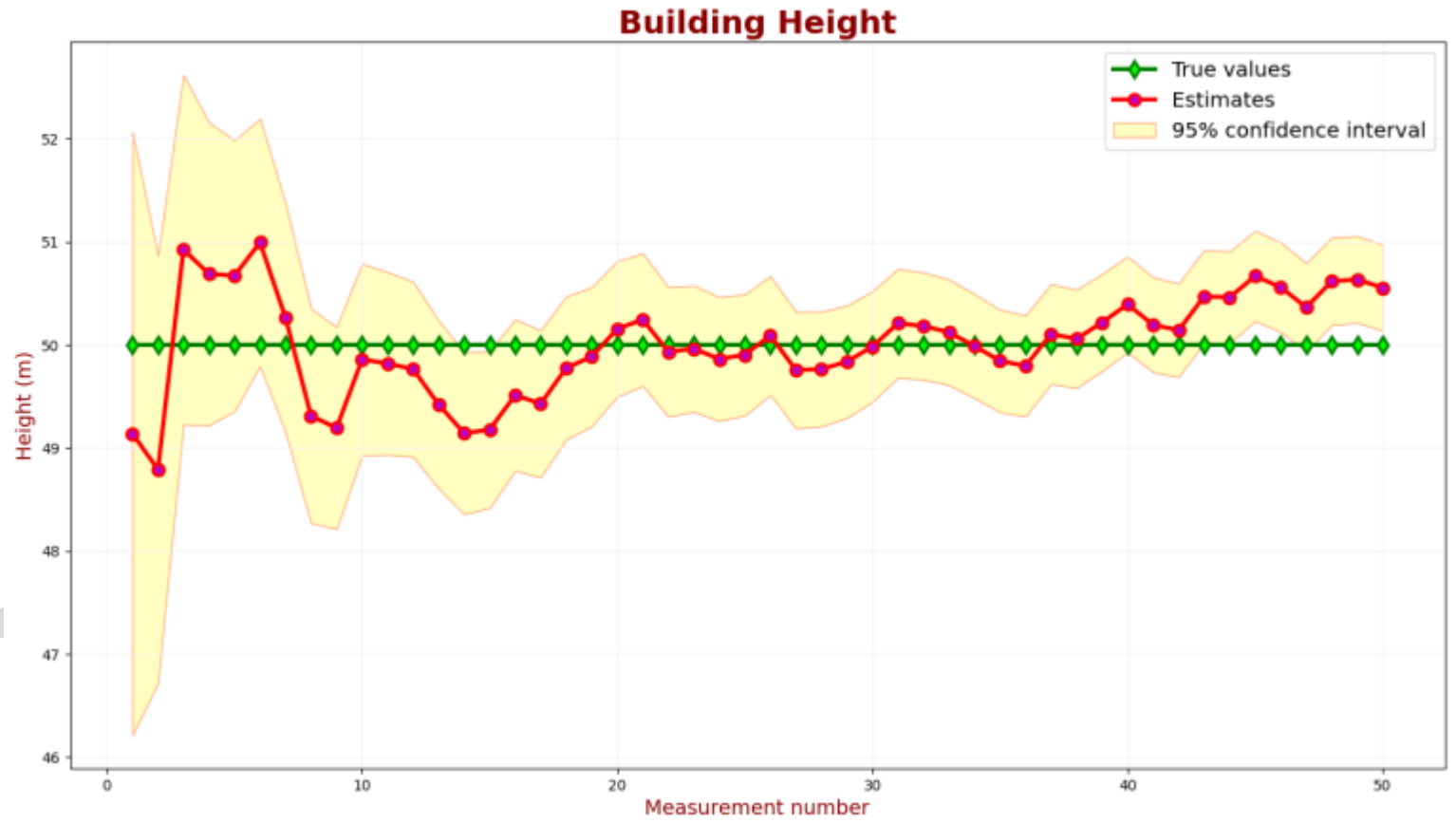
- Another important parameter is estimation uncertainty.
- We want the Kalman Filter (KF) estimates to be precise; therefore, we are interested in low estimation uncertainty.
- Assume that for a building height measurement application, there is a requirement for 95% confidence.



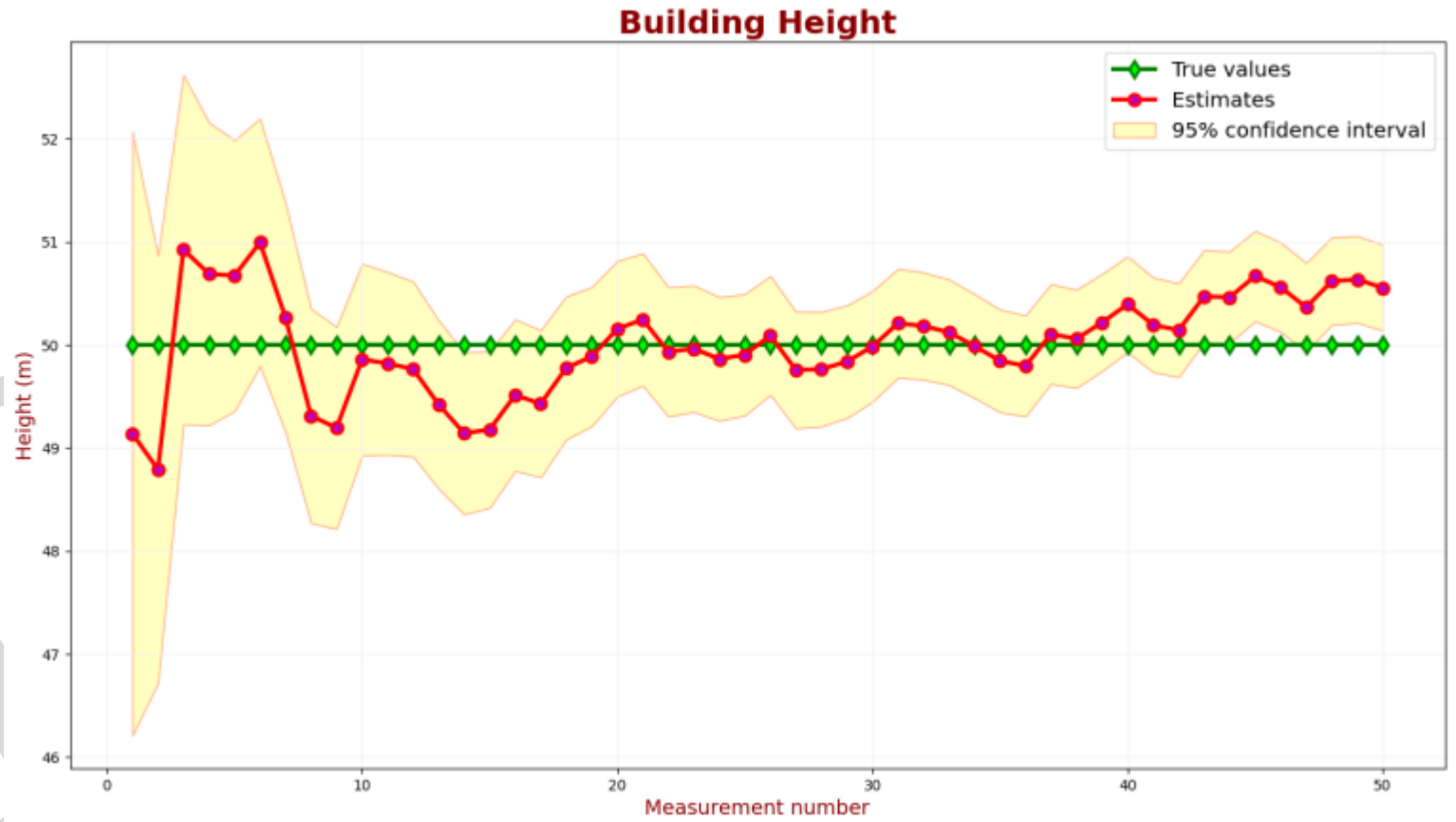
- The following chart shows the KF estimates and the true values with 95% confidence intervals.
- We can see that the uncertainty is too high.



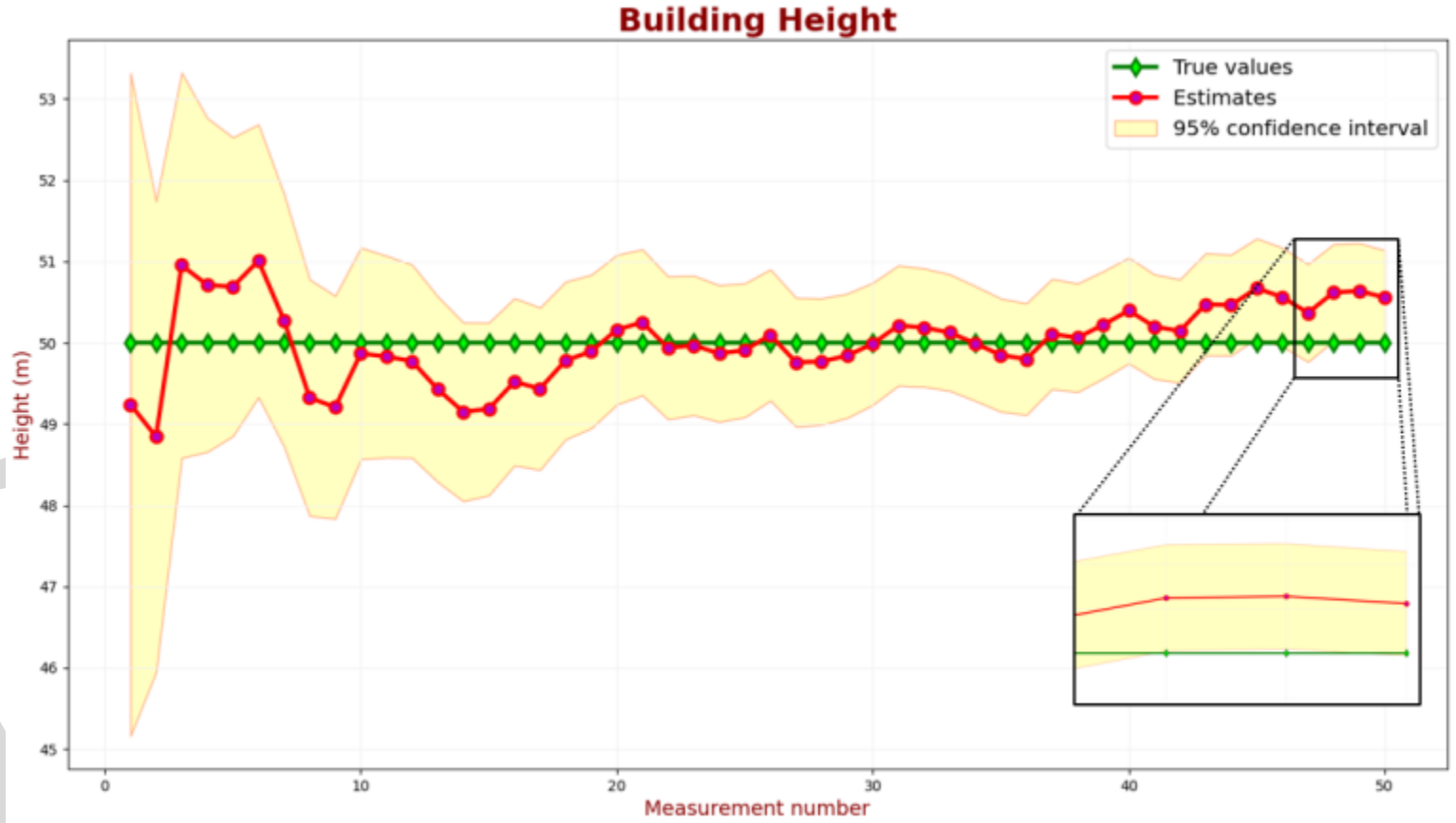
- The following chart describes the KF output for a low measurement uncertainty parameter.



- The following chart describes the KF output for a low measurement uncertainty parameter.
- Although we've decreased the uncertainty of the estimates, many green samples are outside the 95% confidence region. The Kalman Filter is overconfident and too optimistic about its accuracy.



- Let us fnd the measurement uncertainty that yields the desired estimate uncertainty.
- The above chart shows that 2 out of 50 samples slightly exceed the 95% confidence region. This performance satisfes our requirements.



4.2.5 One-dimensional Kalman Filter with process noise

- In the real world, there are uncertainties in the system dynamic model.
- The uncertainty of the dynamic model is called the Process Noise.
- The Process Noise Variance is denoted by the letter q.
- The Covariance Extrapolation Equation shall include the Process Noise Variance.
- The Covariance Extrapolation Equation for constant dynamics is:

$$p_{n+1,n} = p_{n,n} + q_n$$

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	Equation	Equation Name	Alternative names used in the literature
State Update	$\hat{x}_{n,n} = \hat{x}_{n,n-1} + K_n(z_n - \hat{x}_{n,n})$	State Update	Filtering Equation
	$p_{n,n} = (1 - K_n)p_{n,n-1}$	Covariance Update	Corrector Equation
	$K_n = \frac{p_{n,n-1}}{p_{n,n-1} + r_n}$	Kalman Gain	Weight Equation
State Predict	$\hat{x}_{n+1,n} = \hat{x}_{n,n}$ (for constant dynamics) $\hat{x}_{n+1,n} = \hat{x}_{n,n} + \Delta t\hat{\dot{x}}_{n,n}$ $\hat{\dot{x}}_{n+1,n} = \hat{\dot{x}}_{n,n}$ (for constant velocity dynamics)	State Extrapolation	Predictor Equation Transition Equation Prediction Equation Dynamic Model State Space Model
	$p^x_{n+1,n} = p^x_{n,n} + q_n$ (for constant dynamics) $p^x_{n+1,n} = p^x_{n,n} + \Delta t^2 p^v_{n,n}$ $p^v_{n+1,n} = p^v_{n,n} + q_n$ (for constant velocity dynamics)	Covariance Extrapolation	Predictor Covariance Equation

- The State Extrapolation Equation and the Covariance Extrapolation Equation depend on the system dynamics.

Example

- We want to estimate the temperature of the liquid in a tank.
- We assume that at a steady state, the liquid temperature is constant. However, some fluctuations in the true liquid temperature are possible. We can describe the system dynamics by the following equation:

$$x_n = T + w_n$$

where T is the constant temperature; w_n is a random process noise with variance q.

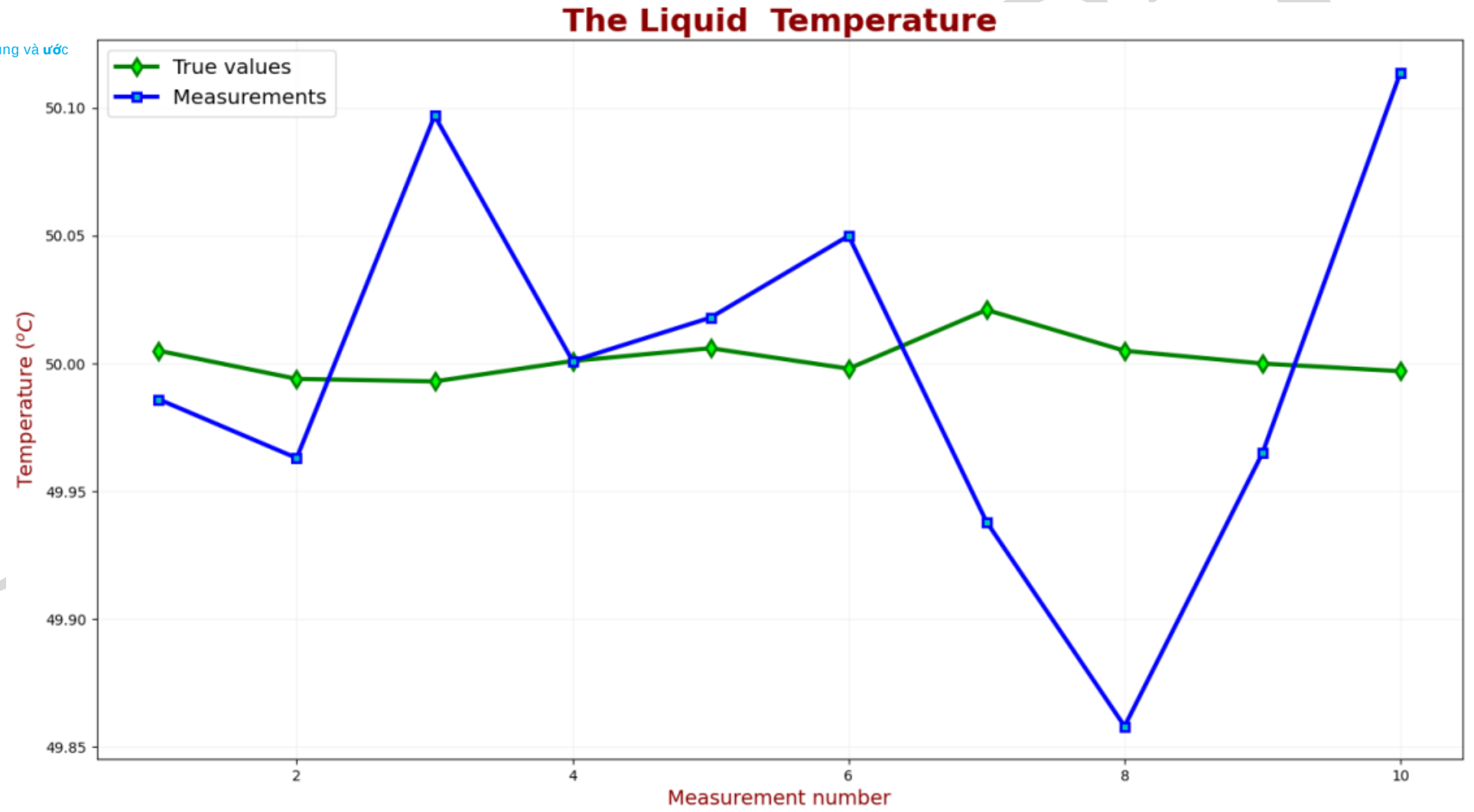
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- Let us assume a true temperature of 50 degrees Celsius.
- We assume that the model is accurate. Thus we set the process noise variance (q) to 0.0001.
- The measurement error (standard deviation) is 0.1 degrees Celsius.
- The measurements are taken every 5 seconds.
- The true liquid temperature values at the measurement points are: 50.005°C, 49.994°C, 49.993°C, 50.001°C, 50.006°C, 49.998°C, 50.021°C, 50.005°C, 50°C, and 49.997°C.
- The measurements are: 49.986°C, 49.963°C, 50.09°C, 50.001°C, 50.018°C, 50.05°C, 49.938°C, 49.858°C, 49.965°C, and 50.114°C.

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- The following chart compares the true liquid temperature and the measurements.

chênh lệch giữa giá trị đúng và ước lượng ko quá lớn.



Iteration Zero

Before the first iteration, we must initialize the Kalman Filter and predict the following state (which is the first state).

Initialization

We don't know the true temperature of the liquid in a tank, and our guess is 60^oC.

$\hat{x}_{0,0} = 60^oC$

Our guess is imprecise, so we set our initialization estimate error σ to 100. The **Estimate Variance** of the initialization is the error variance (σ^2):

$p_{0,0} = 100^2 = 10,000$

This variance is very high. We get faster Kalman Filter convergence if we initialize with a more meaningful value.

Prediction

Since our model has constant dynamics, the predicted estimate is equal to the current estimate:

$\hat{x}_{1,0} = 60^oC$

The extrapolated estimate variance:

$p_{1,0} = p_{0,0} + q = 10000 + 0.0001 = 10000.0001$

First Iteration

Step 1 - Measure

The measurement value:

$z_1 = 49.986^oC$

Since the measurement error is 0.1 (σ), the variance (σ^2) would be 0.01; thus, the measurement variance is:

$r_1 = 0.01$

Step 2 - Update

Kalman Gain calculation:

$$K_1 = \frac{p_{1,0}}{p_{1,0} + r_1} = \frac{10000.0001}{10000.0001 + 0.01} = 0.999999$$

The Kalman Gain is almost 1; thus, our estimate error is much bigger than the measurement error. Thus the weight of the estimate is negligible, while the measurement weight is almost 1.

Estimating the current state:

$$\hat{x}_{1,1} = \hat{x}_{1,0} + K_1 (z_1 - \hat{x}_{1,0}) = 60 + 0.999999 (49.986 - 60) = 49.986^oC$$

Update the current estimate variance:

$$p_{1,1} = (1 - K_1) p_{1,0} = (1 - 0.999999) 10000.0001 = 0.01$$

Step 3 - Predict

Since our system's Dynamic Model is constant, i.e., the liquid temperature doesn't change:

$\hat{x}_{2,1} = \hat{x}_{1,1} = 49.986^oC$

The extrapolated estimate variance is:

$p_{2,1} = p_{1,1} + q = 0.01 + 0.0001 = 0.0101$

Second Iteration

Step 1 - Measure

The second measurement is:

$z_2 = \ 49.963^oC$

Since the measurement error is 0.1 (σ), the variance (σ^2) would be 0.01; thus, the measurement variance is:

$r_2 = 0.01$

Step 2 - Update

Kalman Gain calculation:

$$K_2 = \frac{p_{2,1}}{p_{2,1} + r_2} = \frac{0.0101}{0.0101 + 0.01} = 0.5$$

The Kalman Gain is 0.5, i.e., the weight of the estimate and the measurement weight are equal.

Estimating the current state:

$$\hat{x}_{2,2} = \hat{x}_{2,1} + K_2 \left(z_2 - x_{2,1}\right) = 50.13 + 0.47 \left(48.44 - 50.13\right) = 49.33m$$

Update the current estimate variance:

$$p_{2,2} = \left(1 - K_2\right) p_{2,1} = \left(1 - 0.5\right) 0.0101 = 0.005$$

Step 3 - Predict

Since the dynamic model of the system is constant, i.e., the liquid temperature doesn't change:

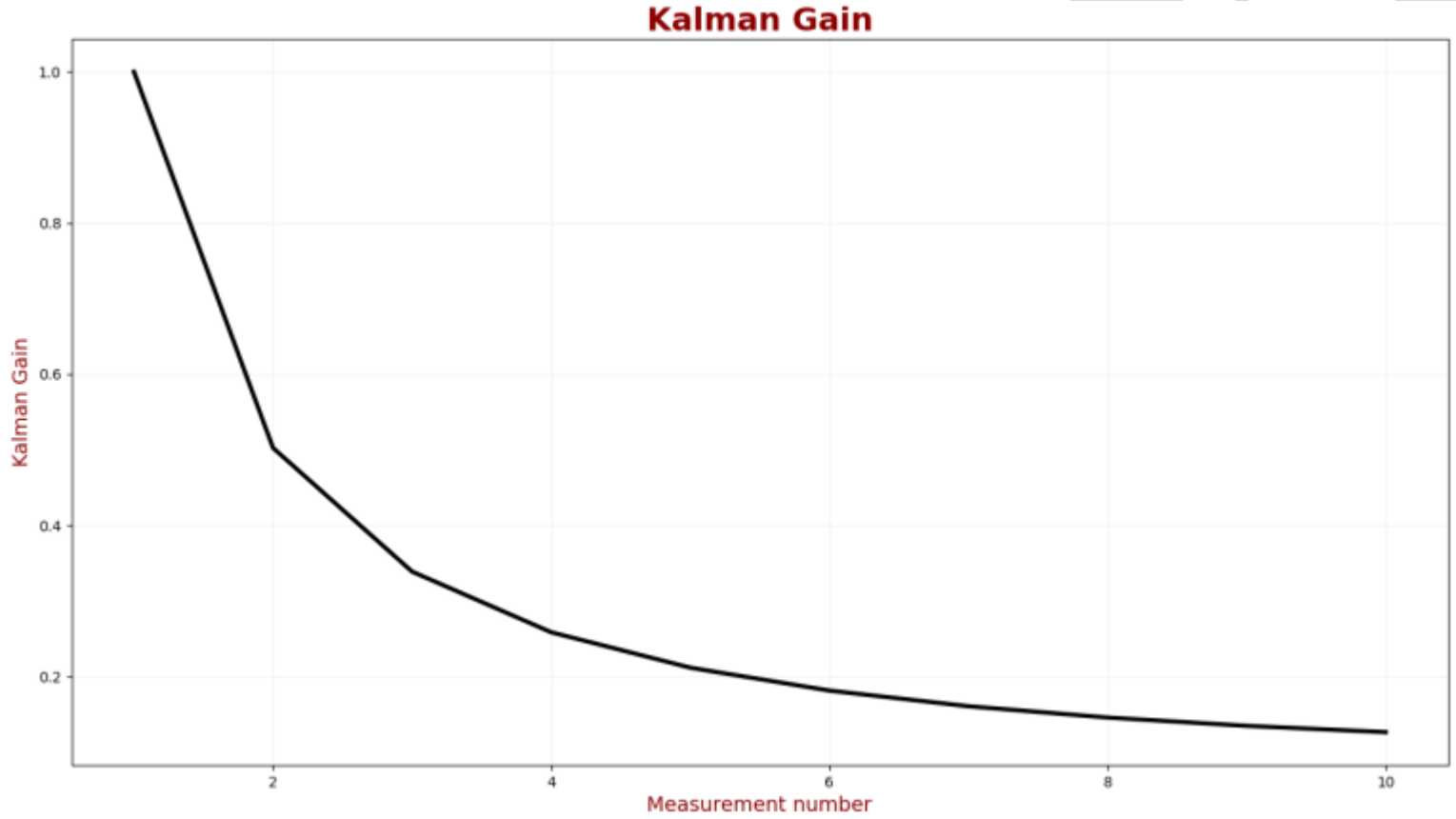
$$\hat{x}_{3,2} = \hat{x}_{2,2} = 49.974^oC$$

The extrapolated estimate variance is:

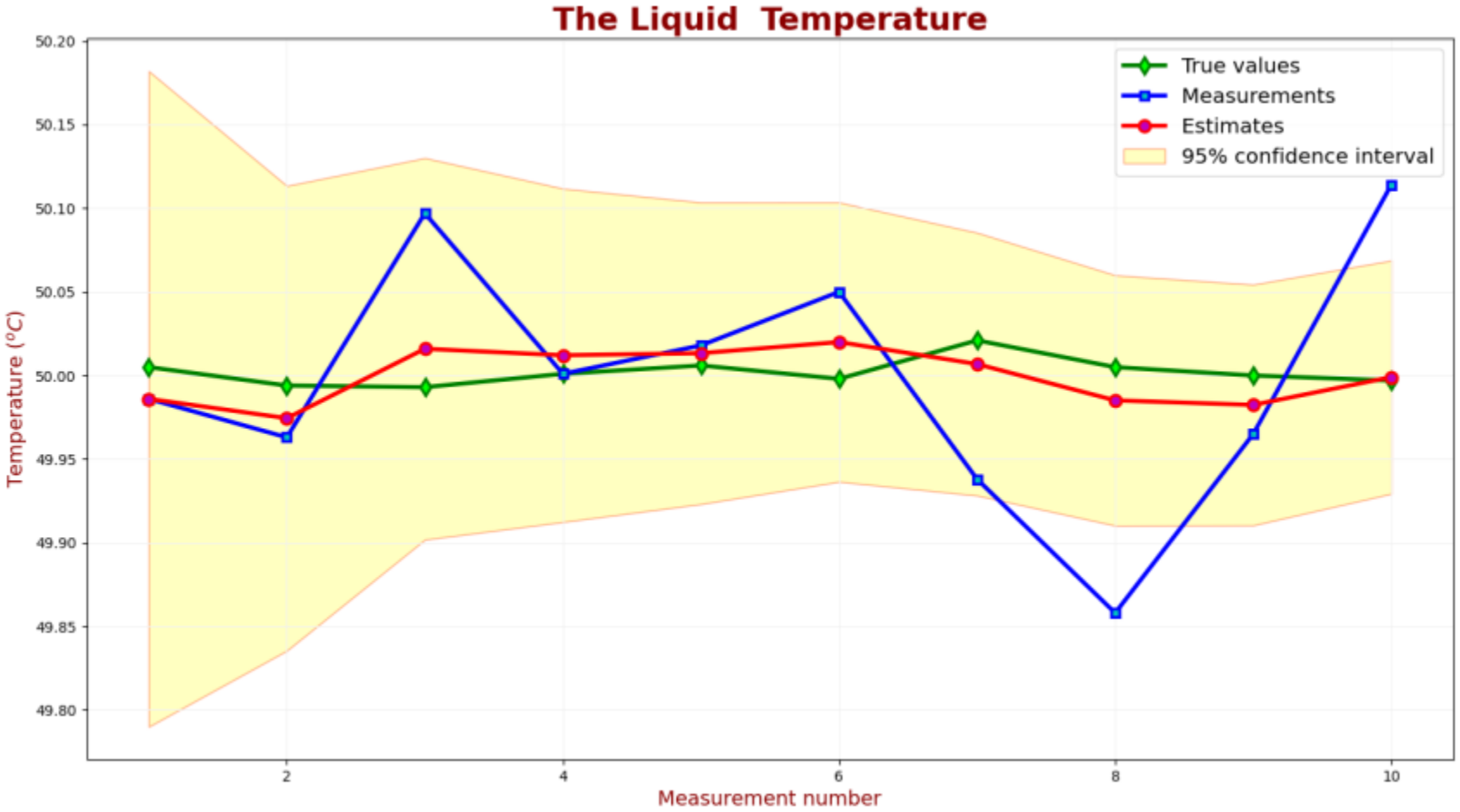
$$p_{3,2} = p_{2,2} + q = 0.005 + 0.0001 = 0.0051$$

3	50.09^oC	$K_3 = \frac{0.0051}{0.0051 + 0.01} = 0.3388$ $\hat{x}_{3,3} = 49.974 + 0.3388\,(50.09 - 49.974) = 50.016^oC$ $p_{3,3} = (1 - 0.3388)\,0.0051 = 0.0034$	$\hat{x}_{4,3} = \hat{x}_{3,3} = 50.016^oC$ $p_{4,3} = 0.0034 + 0.0001 = 0.0035$
4	50.001^oC	$K_4 = \frac{0.0035}{0.0035 + 0.01} = 0.2586$ $\hat{x}_{4,4} = 50.016 + 0.2586\,(50.001 - 50.016) = 50.012^oC$ $p_{4,4} = (1 - 0.2586)\,0.0035 = 0.0026$	$\hat{x}_{5,4} = \hat{x}_{4,4} = 50.012^oC$ $p_{5,4} = 0.0026 + 0.0001 = 0.0027$

- The following chart describes the Kalman Gain.



- The following chart compares the true value, measured values, and estimates. The confidence interval is 95%.



4.3 Multivariate Kalman Filter

- This part deals with a Linear Kalman Filter (LKF). The LKF assumes that the system dynamics are linear. The next part describes non-linear Kalman Filters.
- Until now, we've dealt with one-dimensional processes. But many dynamic processes have two, three, or even more dimensions.
- It is common practice to describe a multidimensional process with a single equation in matrix form.

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4.3.1 State Extrapolation Equation

State Extrapolation Equation

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_n + \boldsymbol{w}_n$$

Where:

$\hat{\boldsymbol{x}}_{n+1,n}$	is a predicted system state vector at time step $n+1$
$\hat{\boldsymbol{x}}_{n,n}$	is an estimated system state vector at time step n
$\boldsymbol{u}_n$	is a **control variable** or **input variable** - a measurable (deterministic) input to the system
$\boldsymbol{w}_n$	is a **process noise** or disturbance - an unmeasurable input that affects the state
$\boldsymbol{F}$	is a **state transition matrix**
$\boldsymbol{G}$	is a **control matrix** or **input transition matrix** (mapping control to state variables)

Variable	Description	Dimension
$\boldsymbol{x}$	state vector	$n_x \times 1$
$\boldsymbol{F}$	state transition matrix	$n_x \times n_x$
$\boldsymbol{u}$	input variable	$n_u \times 1$
$\boldsymbol{G}$	control matrix	$n_x \times n_u$
$\boldsymbol{w}$	process noise vector	$n_x \times 1$

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Example - airplane - no control input

$$\hat{\boldsymbol{x}}_n = \begin{bmatrix} \hat{x}_n \\ \hat{y}_n \\ \hat{z}_n \\ \hat{\dot{x}}_n \\ \hat{\dot{y}}_n \\ \hat{\dot{z}}_n \\ \hat{\ddot{x}}_n \\ \hat{\ddot{y}}_n \\ \hat{\ddot{z}}_n \end{bmatrix}$$

$$\begin{cases} \hat{x}_{n+1,n} = \hat{x}_{n,n} + \hat{\dot{x}}_{n,n}\Delta t + \frac{1}{2}\hat{\ddot{x}}_{n,n}\Delta t^2 \\ \hat{y}_{n+1,n} = \hat{y}_{n,n} + \hat{\dot{y}}_{n,n}\Delta t + \frac{1}{2}\hat{\ddot{y}}_{n,n}\Delta t^2 \\ \hat{z}_{n+1,n} = \hat{z}_{n,n} + \hat{\dot{z}}_{n,n}\Delta t + \frac{1}{2}\hat{\ddot{z}}_{n,n}\Delta t^2 \\ \hat{\dot{x}}_{n+1,n} = \hat{\dot{x}}_{n,n} + \hat{\ddot{x}}_{n,n}\Delta t \\ \hat{\dot{y}}_{n+1,n} = \hat{\dot{y}}_{n,n} + \hat{\ddot{y}}_{n,n}\Delta t \\ \hat{\dot{z}}_{n+1,n} = \hat{\dot{z}}_{n,n} + \hat{\ddot{z}}_{n,n}\Delta t \\ \hat{\ddot{x}}_{n+1,n} = \hat{\ddot{x}}_{n,n} \\ \hat{\ddot{y}}_{n+1,n} = \hat{\ddot{y}}_{n,n} \\ \hat{\ddot{z}}_{n+1,n} = \hat{\ddot{z}}_{n,n} \end{cases}$$

$$\boldsymbol{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 \\ 0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$
\begin{bmatrix}
\hat{x}_{n+1,n} \\
\hat{y}_{n+1,n} \\
\hat{z}_{n+1,n} \\
\hat{\dot{x}}_{n+1,n} \\
\hat{\dot{y}}_{n+1,n} \\
\hat{\dot{z}}_{n+1,n} \\
\hat{\ddot{x}}_{n+1,n} \\
\hat{\ddot{y}}_{n+1,n} \\
\hat{\ddot{z}}_{n+1,n}
\end{bmatrix}
=
\begin{bmatrix}
1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 & 0 & 0 \\
0 & 1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 & 0 \\
0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 & 0.5\Delta t^2 \\
0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\begin{bmatrix}
\hat{x}_{n,n} \\
\hat{y}_{n,n} \\
\hat{z}_{n,n} \\
\hat{\dot{x}}_{n,n} \\
\hat{\dot{y}}_{n,n} \\
\hat{\dot{z}}_{n,n} \\
\hat{\ddot{x}}_{n,n} \\
\hat{\ddot{y}}_{n,n} \\
\hat{\ddot{z}}_{n,n}
\end{bmatrix}
$$

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Example - airplane - with control input

$$\hat{\boldsymbol{x}}_n = \begin{bmatrix} \hat{x}_n \\ \hat{y}_n \\ \hat{z}_n \\ \hat{\dot{x}}_n \\ \hat{\dot{y}}_n \\ \hat{\dot{z}}_n \end{bmatrix} \quad \boldsymbol{u}_n = \begin{bmatrix} \ddot{x}_n \\ \ddot{y}_n \\ \ddot{z}_n \end{bmatrix} \quad \boldsymbol{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{G} = \begin{bmatrix} 0.5\Delta t^2 & 0 & 0 \\ 0 & 0.5\Delta t^2 & 0 \\ 0 & 0 & 0.5\Delta t^2 \\ \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix}$$

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_{n,n}$$

$$\begin{bmatrix} \hat{x}_{n+1,n} \\ \hat{y}_{n+1,n} \\ \hat{z}_{n+1,n} \\ \hat{\dot{x}}_{n+1,n} \\ \hat{\dot{y}}_{n+1,n} \\ \hat{\dot{z}}_{n+1,n} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_{n,n} \\ \hat{y}_{n,n} \\ \hat{z}_{n,n} \\ \hat{\dot{x}}_{n,n} \\ \hat{\dot{y}}_{n,n} \\ \hat{\dot{z}}_{n,n} \end{bmatrix} + \begin{bmatrix} 0.5\Delta t^2 & 0 & 0 \\ 0 & 0.5\Delta t^2 & 0 \\ 0 & 0 & 0.5\Delta t^2 \\ \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix} \begin{bmatrix} \ddot{x}_n \\ \ddot{y}_n \\ \ddot{z}_n \end{bmatrix}$$

Example – falling object

vật rơi

$$\hat{\boldsymbol{x}}_n = \begin{bmatrix} \hat{h}_n \\ \hat{\dot{h}}_n \end{bmatrix} \qquad \boldsymbol{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \qquad \boldsymbol{G} = \begin{bmatrix} 0.5\Delta t^2 \\ \Delta t \end{bmatrix} \qquad \boldsymbol{u}_n = \begin{bmatrix} g \end{bmatrix}$$

$$\begin{cases} \hat{h}_{n+1,n} = \hat{h}_{n,n} + \hat{\dot{h}}_{n,n}\Delta t + 0.5\Delta t^2 g \\ \hat{\dot{h}}_{n+1,n} = \hat{\dot{h}}_{n,n} + \Delta t g \end{cases}$$

mô hình hóa

$$\begin{bmatrix} \hat{h}_{n+1,n} \\ \hat{\dot{h}}_{n+1,n} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{h}_{n,n} \\ \hat{\dot{h}}_{n,n} \end{bmatrix} + \begin{bmatrix} 0.5\Delta t^2 \\ \Delta t \end{bmatrix} \begin{bmatrix} g \end{bmatrix}$$

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4.3.2 Covariance Extrapolation Equation

Covariance Extrapolation Equation

$$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$$

Where:

$\boldsymbol{P}_{n,n}$	is the squared uncertainty of an estimate (covariance matrix) of the current state
$\boldsymbol{P}_{n+1,n}$	is the squared uncertainty of a prediction (covariance matrix) for the next state
$\boldsymbol{F}$	is the state transition matrix that we derived in Appendix C ("Modeling linear dynamic systems")
$\boldsymbol{Q}$	is the process noise matrix

Process noise matrix Q

Process Noise Uncertainty

$$\boldsymbol{Q_n} = E\left(\boldsymbol{w_n}\boldsymbol{w_n^T}\right)$$

Where:

$\boldsymbol{Q_n}$ is the covariance matrix of the process noise

$\boldsymbol{w_n}$ is the process noise

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- As you already know, the system dynamics is described by:

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\hat{\boldsymbol{u}}_{n,n} + \boldsymbol{w}_n$$

dự đoán phương sai thì để wn vào còn tính toán thì bỏ ra.
wn là nhiễu tiến trình, chúng ta ko đo được.

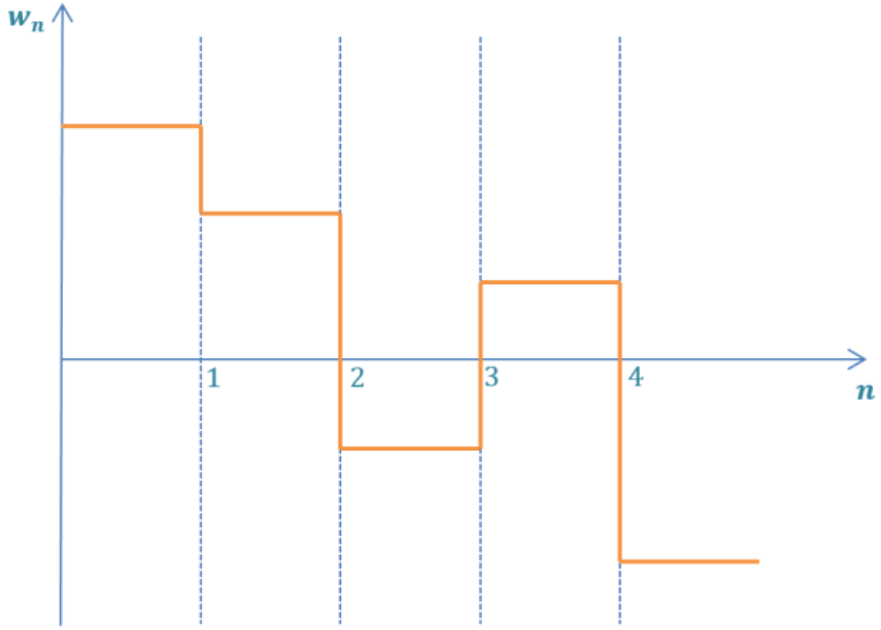
- In the multidimensional case, the process noise is a covariance matrix denoted by **Q**.
- We've seen that the process noise variance has a critical influence on the Kalman Filter performance.
- The process noise can be independent between different state variables. In this case, the process noise covariance matrix **Q** is a diagonal matrix:

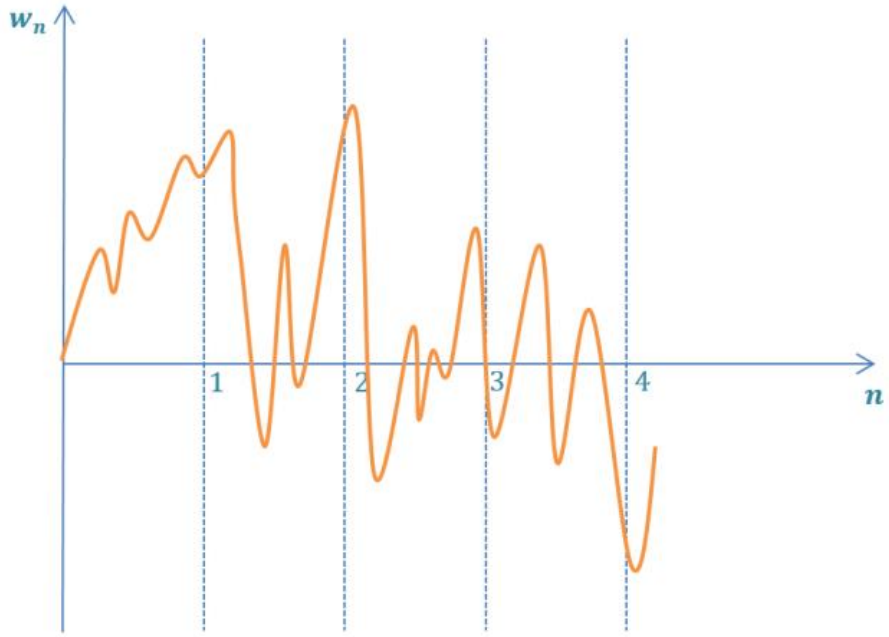
$$\boldsymbol{Q} = \begin{bmatrix} q_{11} & 0 & \cdots & 0 \\ 0 & q_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{kk} \end{bmatrix}$$

- The process noise can also be dependent.

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- There are two models for the environmental process noise.
- Discrete noise model
- Continuous noise model





- If the dynamic model doesn't include a control input, we can project the random variance in acceleration σ^2 a on our dynamic model using the state transition matrix.

$$\boldsymbol{Q}_a = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sigma_a^2 \qquad \boldsymbol{Q} = \boldsymbol{F}\boldsymbol{Q}_a\boldsymbol{F}^T \qquad \boldsymbol{F} = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{Q} = \boldsymbol{F}\boldsymbol{Q}_a\boldsymbol{F}^T \qquad \boldsymbol{Q} = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \Delta t & 1 & 0 \\ \frac{\Delta t^2}{2} & \Delta t & 1 \end{bmatrix} \sigma_a^2 \qquad \boldsymbol{Q} = \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & \frac{\Delta t^2}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 & \Delta t \\ \frac{\Delta t^2}{2} & \Delta t & 1 \end{bmatrix} \sigma_a^2$$

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- If the dynamic model includes a control input, we can compute the **Q** matrix even faster. We can project the random variance in acceleration σ_a^2 on our dynamic model using the control matrix.

$$\boldsymbol{Q} = \boldsymbol{G}\sigma_a^2\boldsymbol{G}^T$$

$$\boldsymbol{G} = \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix}$$

$$\boldsymbol{Q} = \boldsymbol{G}\sigma_a^2\boldsymbol{G}^T = \sigma_a^2\boldsymbol{G}\boldsymbol{G}^T = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix} \begin{bmatrix} \frac{\Delta t^2}{2} & \Delta t \end{bmatrix} = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix}$$

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4.3.3 Measurement equation

Measurement Equation

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n$$

Where:

- $\boldsymbol{z}_n$ is a measurement vector
- $\boldsymbol{x}_n$ is a true system state (hidden state)
- $\boldsymbol{v}_n$ is a random noise vector
- $\boldsymbol{H}$ is an observation matrix

Variable	Description	Dimension
$\boldsymbol{x}$	state vector	$n_x \times 1$
$\boldsymbol{z}$	measurements vector	$n_z \times 1$
$\boldsymbol{H}$	observation matrix	$n_z \times n_x$
$\boldsymbol{v}$	measurement noise vector	$n_z \times 1$

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Scaling

- A range meter sends a signal toward a destination and receives a reflected echo. The measurement is the time delay between the transmission and reception of the signal. The system state is the range.
- In this case, we need to perform a scaling:

$$\boldsymbol{z}_n = \left[\frac{2}{c}\right] \boldsymbol{x}_n + \boldsymbol{v}_n$$

$$\boldsymbol{H} = \left[\frac{2}{c}\right]$$

Where:

c	is the speed of light
$\boldsymbol{x}_n$	is the range
$\boldsymbol{z}_n$	is the measured time delay

xn là giá trị dự đoán, zn là giá trị đo. Còn H là ma trận chuyển đổi đơn vị của 2 giá trị đó. Ở đây, xn là quãng đường, còn zn là thời gian

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State selection

- Sometimes certain states are measured while others are not. For example, the first, third, and fifth states of a five-dimensional state vector are measurable, while the second and fourth states are not measurable:

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \boldsymbol{v}_n = \begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} + \boldsymbol{v}_n$$

Combination of states

Sometimes some combination of states can be measured instead of each separate state. For example, maybe the lengths of the sides of a triangle are the states, and only the total perimeter can be measured:

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \boldsymbol{v}_n = (x_1 + x_2 + x_3) + \boldsymbol{v}_n$$

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State selection

- Sometimes certain states are measured while others are not. For example, the first, third, and fifth states of a five-dimensional state vector are measurable, while the second and fourth states are not measurable:

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \boldsymbol{v}_n = \begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} + \boldsymbol{v}_n$$

Combination of states

Sometimes some combination of states can be measured instead of each separate state. For example, maybe the lengths of the sides of a triangle are the states, and only the total perimeter can be measured:

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \boldsymbol{v}_n = (x_1 + x_2 + x_3) + \boldsymbol{v}_n$$

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4.3.4 State Update Equation

Estimation Uncertainty

$$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n(\boldsymbol{z}_n - \boldsymbol{H}\hat{\boldsymbol{x}}_{n,n-1})$$

Where:

$\hat{\boldsymbol{x}}_{n,n}$	is an estimated system state vector at time step n
$\hat{\boldsymbol{x}}_{n,n-1}$	is a predicted system state vector at time step $n-1$
$\boldsymbol{K}_n$	is a Kalman Gain
$\boldsymbol{z}_n$	is a measurement
$\boldsymbol{H}$	is an observation matrix

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Variable	Description	Dimension
$\boldsymbol{x}$	state vector	$n_x \times 1$
$\boldsymbol{z}$	measurements vector	$n_z \times 1$
$\boldsymbol{H}$	observation matrix	$n_z \times n_x$
$\boldsymbol{K}$	Kalman gain	$n_x \times n_z$

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4.3.5 Covariance Update Equation

Covariance Update Equation

$$\boldsymbol{P}_{n,n} = (\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})\,\boldsymbol{P}_{n,n-1}\,(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$$

Where:

$\boldsymbol{P}_{n,n}$	is the covariance matrix of the current state estimation
$\boldsymbol{P}_{n,n-1}$	is the prior estimate covariance matrix of the current state (predicted at the previous state)
$\boldsymbol{K}_n$	is a Kalman Gain
$\boldsymbol{H}$	is the observation matrix
$\boldsymbol{R}_n$	is the measurement noise covariance matrix
$\boldsymbol{I}$	is an Identity Matrix (the $n \times n$ square matrix with ones on the main diagonal and zeros elsewhere)

4.3.6 The Kalman Gain

Kalman Gain

$$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\boldsymbol{H}^T\left(\boldsymbol{H}\boldsymbol{P}_{n,n-1}\boldsymbol{H}^T + \boldsymbol{R}_n\right)^{-1} \tag{8.66}$$

Where:

$\boldsymbol{K}_n$	is the Kalman Gain
$\boldsymbol{P}_{n,n-1}$	is the prior estimate covariance matrix of the current state (predicted at the previous step)
$\boldsymbol{H}$	is the observation matrix
$\boldsymbol{R}_n$	is the measurement noise covariance matrix

Simplifed Covariance Update Equation

- In many textbooks, you can fnd a simplifed form of the Covariance Update Equation:

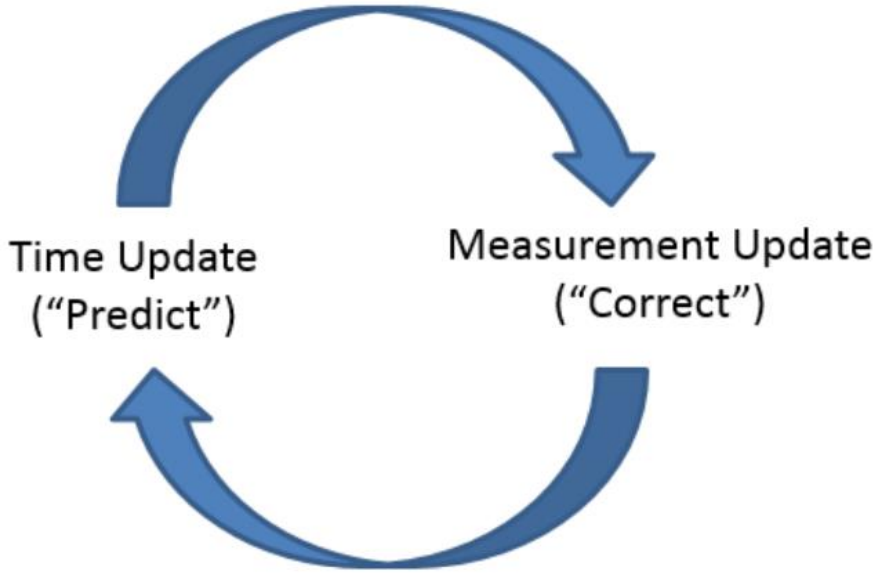
$$\boldsymbol{P}_{n,n} = (\boldsymbol{I} - \boldsymbol{K}_n \boldsymbol{H})\, \boldsymbol{P}_{n,n-1}$$

R **Warning!** This equation is much more elegant and easier to remember and it performs well in many cases. However, **a minor error in computing the Kalman Gain (due to round-off) can lead to huge computation errors. The subtraction $(\boldsymbol{I} - \boldsymbol{K}_n \boldsymbol{H})$ can lead to nonsymmetric matrices due to floating-point errors. This equation is numerically unstable!**

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4.3.7 Summary

- We have derived all five Kalman Filter equations in matrix notation. The Kalman Filter operates in a "predict-correct" loop, as shown in the diagram below.
- Once initialized, the Kalman Filter predicts the system state at the next step. It also provides the uncertainty of the prediction.
- Once the measurement is received, the Kalman Filter updates (or corrects) the prediction and the uncertainty of the current state.



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Time Update ("Predict")

1. Extrapolate the state

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_n$$

2. Extrapolate uncertainty

$$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$$

Measurement Update ("Correct")

1. Compute the Kalman Gain

$$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\boldsymbol{H}^T\left(\boldsymbol{H}\boldsymbol{P}_{n,n-1}\boldsymbol{H}^T + \boldsymbol{R}_n\right)^{-1}$$

2. Update estimate with measurement

$$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n\left(\boldsymbol{z}_n - \boldsymbol{H}\hat{\boldsymbol{x}}_{n,n-1}\right)$$

3. Update the estimate uncertainty

$$\boldsymbol{P}_{n,n} = (\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})\boldsymbol{P}_{n,n-1}(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$$



Initial Estimate: $\hat{\boldsymbol{x}}_{0,0}$, $\boldsymbol{P}_{0,0}$

	Equation	Name	Alternative names in the literature
Predict	$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_n$	State Extrapolation	Predictor Equation Transition Equation Prediction Equation Dynamic Model State Space Model
	$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$	Covariance Extrapolation	Predictor Covariance Equation
Update	$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n$ $\times(\boldsymbol{z}_n - \boldsymbol{H}\hat{\boldsymbol{x}}_{n,n-1})$	State Update	Filtering Equation
	$\boldsymbol{P}_{n,n} = (\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})\,\boldsymbol{P}_{n,n-1}$ $\times\,(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$	Covariance Update	Corrector Equation
	$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\boldsymbol{H}^T$ $\times\left(\boldsymbol{H}\boldsymbol{P}_{n,n-1}\boldsymbol{H}^T + \boldsymbol{R}_n\right)^{-1}$	Kalman Gain	Weight Equation
Auxiliary	$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n$	Measurement Equation	
	$\boldsymbol{R}_n = E\left(\boldsymbol{v}_n\boldsymbol{v}_n^T\right)$	Measurement Covariance	Measurement Error
	$\boldsymbol{Q}_n = E\left(\boldsymbol{w}_n\boldsymbol{w}_n^T\right)$	Process Noise Covariance	Process Noise Error
	$\boldsymbol{P}_{n,n} = E\left(\boldsymbol{e}_n\boldsymbol{e}_n^T\right)$ $= E\left(\left(\boldsymbol{x}_n - \hat{\boldsymbol{x}}_{n,n}\right)\left(\boldsymbol{x}_n - \hat{\boldsymbol{x}}_{n,n}\right)^T\right)$	Estimation Covariance	Estimation Error

Term	Name	Alternative term	Dimensions
$\boldsymbol{x}$	State Vector		$n_x \times 1$
$\boldsymbol{z}$	Measurements Vector	$\boldsymbol{y}$	$n_z \times 1$
$\boldsymbol{F}$	State Transition Matrix	$\boldsymbol{\Phi}, \boldsymbol{A}$	$n_x \times n_x$
$\boldsymbol{u}$	Input Variable		$n_u \times 1$
$\boldsymbol{G}$	Control Matrix	$\boldsymbol{B}$	$n_x \times n_u$
$\boldsymbol{P}$	Estimate Covariance	$\boldsymbol{\Sigma}$	$n_x \times n_x$
$\boldsymbol{Q}$	Process Noise Covariance		$n_x \times n_x$
$\boldsymbol{R}$	Measurement Covariance		$n_z \times n_z$
$\boldsymbol{w}$	Process Noise Vector	$\boldsymbol{y}$	$n_x \times 1$
$\boldsymbol{v}$	Measurement Noise Vector		$n_z \times 1$
$\boldsymbol{H}$	Observation Matrix	$\boldsymbol{C}$	$n_z \times n_x$
$\boldsymbol{K}$	Kalman Gain		$n_x \times n_z$
$\boldsymbol{n}$	Discrete-Time Index	k	

Dimensions notation:

- n_x is a number of states in a state vector
- n_z is a number of measured states
- n_u is a number of elements of the input variable

4.3.8 Example

Rocket altitude estimation

- In this example, we estimate the altitude of a rocket.
- The rocket is equipped with an onboard altimeter that provides altitude measurements. In addition to an altimeter, the rocket is equipped with an accelerometer that measures the rocket's acceleration.
- The accelerometer serves as a control input to the Kalman Filter.
- We assume constant acceleration dynamics.

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- Accelerometers don't sense gravity. An accelerometer at rest on a table measures 1g upwards, while an accelerometer in free fall measures zero acceleration. Thus, we need to subtract the gravitational acceleration constant g from each accelerometer measurement.
- The accelerometer measurement at time step n is:

$$a_n = \ddot{x} - g + \epsilon$$

where

$\ddot{x}$ is the actual acceleration of the object (the second derivative of the object position)

g is the gravitational acceleration constant; $g = -9.8\frac{m}{s^2}$

ϵ is the accelerometer measurement error

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State extrapolation equation

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_n + \boldsymbol{w}_n$$

$$\boldsymbol{x}_n = \begin{bmatrix} x_n \\ \dot{x}_n \end{bmatrix}$$

$$\begin{bmatrix} \hat{x}_{n+1,n} \\ \hat{\dot{x}}_{n+1,n} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_{n,n} \\ \hat{\dot{x}}_{n,n} \end{bmatrix} + \begin{bmatrix} 0.5\Delta t^2 \\ \Delta t \end{bmatrix} (a_n + g)$$

$$\boldsymbol{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \qquad \boldsymbol{G} = \begin{bmatrix} 0.5\Delta t^2 \\ \Delta t \end{bmatrix}$$

$$\boldsymbol{u}_n = (a_n + g)$$

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Covariance extrapolation equation

$$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$$

$$\boldsymbol{P} = \begin{bmatrix} p_x & p_{x\dot{x}} \\ p_{\dot{x}x} & p_{\dot{x}} \end{bmatrix}$$

- p_x is the variance of the altitude estimation
- $p_{\dot{x}}$ is the variance of the velocity estimation
- The off-diagonal entries are covariances

$$\boldsymbol{Q} = \begin{bmatrix} \sigma_x^2 & \sigma_{x\dot{x}}^2 \\ \sigma_{\dot{x}x}^2 & \sigma_{\dot{x}}^2 \end{bmatrix} \qquad \boldsymbol{Q} = \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix} \epsilon^2$$

Where:

Δt is the time between successive measurements

ϵ^2 is the random variance in accelerometer measurement

$$\begin{bmatrix} p_{x_{n+1,n}} & p_{x\dot{x}_{n+1,n}} \\ p_{\dot{x}x_{n+1,n}} & p_{\dot{x}_{n+1,n}} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} p_{x_{n,n}} & p_{x\dot{x}_{n,n}} \\ p_{\dot{x}x_{n,n}} & p_{\dot{x}_{n,n}} \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ \Delta t & 1 \end{bmatrix} + \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix} \epsilon^2$$

$$V(v) = \sigma_v^2 = E\left(v^2\right) - \mu_v^2 = E\left((a\Delta t)^2\right) - (\mu_a \Delta t)^2$$
$$= \Delta t^2 \left(E\left(a^2\right) - \mu_a^2\right) = \Delta t^2 \sigma_a^2$$

$$V(x) = \sigma_x^2 = E\left(x^2\right) - \mu_x^2 = E\left(\left(\frac{1}{2}\, a\Delta t^2\right)^2\right) - \left(\frac{1}{2}\, \mu_a \Delta t^2\right)^2$$

$$= \frac{\Delta t^4}{4}\left(E\left(a^2\right) - \mu_a^2\right) = \frac{\Delta t^4}{4}\, \sigma_a^2$$

$$COV(x, v) = COV(v, x) = E\left(xv\right) - \mu_x \mu_v$$

$$= E\left(\frac{1}{2}\, a\Delta t^2 a\Delta t\right) - \left(\frac{1}{2}\, \mu_a \Delta t^2 \mu_a \Delta t\right) = \frac{\Delta t^3}{2}\left(E\left(a^2\right) - \mu_a^2\right) = \frac{\Delta t^3}{2}\, \sigma_a^2$$

Measurement equation

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n + \boldsymbol{v}_n$$

$$\boldsymbol{z}_n = \boldsymbol{H}\boldsymbol{x}_n$$

$$\boldsymbol{z}_n = \boldsymbol{H}\begin{bmatrix} x_n \\ \dot{x}_n \end{bmatrix} \quad \boldsymbol{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\boldsymbol{R}_n = \begin{bmatrix} \sigma_{x_m}^2 \end{bmatrix}$$

$$\boldsymbol{R}_1 = \boldsymbol{R}_2 ... \boldsymbol{R}_{n-1} = \boldsymbol{R}_n = \boldsymbol{R}$$

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Kalman Gain

$$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\boldsymbol{H}^T\left(\boldsymbol{H}\boldsymbol{P}_{n,n-1}\boldsymbol{H}^T + \boldsymbol{R}_n\right)^{-1}$$

$$\begin{bmatrix} K_{1,1_n} \\ K_{2,1_n} \end{bmatrix} = \begin{bmatrix} p_{x_{n,n-1}} & p_{x\dot{x}_{n,n-1}} \\ p_{\dot{x}x_{n,n-1}} & p_{\dot{x}_{n,n-1}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \times \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_{x_{n,n-1}} & p_{x\dot{x}_{n,n-1}} \\ p_{\dot{x}x_{n,n-1}} & p_{\dot{x}_{n,n-1}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} \sigma^2_{x_m} \end{bmatrix} \right)^{-1}$$

State update equation

$$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n(\boldsymbol{z}_n - \boldsymbol{H}\hat{\boldsymbol{x}}_{n,n-1})$$

Covariance update equation

$$\boldsymbol{P}_{n,n} = (\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})\,\boldsymbol{P}_{n,n-1}\,(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H})^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$$

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Numerical example

Let us assume a vertically boosting rocket with constant acceleration. The rocket is equipped with an altimeter that provides altitude measurements and an accelerometer that serves as a control input.

- The measurements period: $\Delta t = 0.25s$
- The rocket acceleration: $\ddot{x} = 30\frac{m}{s^2}$
- The altimeter measurement error standard deviation: $\sigma_{x_m} = 20m$
- The accelerometer measurement error standard deviation: $\epsilon = 0.1\frac{m}{s^2}$
- The state transition matrix $\boldsymbol{F}$ would be:

$$\boldsymbol{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix}$$

- The control matrix $\boldsymbol{G}$ would be:

$$\boldsymbol{G} = \begin{bmatrix} 0.5\Delta t^2 \\ \Delta t \end{bmatrix} = \begin{bmatrix} 0.0313 \\ 0.25 \end{bmatrix}$$

- The process noise matrix $\boldsymbol{Q}$ would be:

$$\boldsymbol{Q} = \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix} \sigma_a^2 = \begin{bmatrix} \frac{0.25^4}{4} & \frac{0.25^3}{2} \\ \frac{0.25^3}{2} & 0.25^2 \end{bmatrix} 0.1^2$$

- The measurement uncertainty $\boldsymbol{R}$ would be:

$$\boldsymbol{R}_n = \boldsymbol{R} = \begin{bmatrix} \sigma_{x_m}^2 \end{bmatrix} = 400$$

- The following table contains the set of 30 noisy measurements of the altitude h_n and acceleration a_n:

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
$h_n(m)$	6.43	1.3	39.43	45.89	41.44	48.7	78.06	80.08
$a_n(m/s^2)$	39.81	39.67	39.81	39.84	40.05	39.85	39.78	39.65

9	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
61.77	75.15	110.39	127.83	158.75	156.55	213.32	229.82	262.8
39.67	39.78	39.59	39.87	39.85	39.59	39.84	39.9	39.63

18	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**
297.57	335.69	367.92	377.19	411.18	460.7	468.39	553.9	583.97
39.59	39.76	39.79	39.73	39.93	39.83	39.85	39.94	39.86

27	**28**	**29**	**30**
655.15	723.09	736.85	787.22
39.76	39.86	39.74	39.94

Iteration Zero

Initialization

We don't know the rocket's location; we set the initial position and velocity to 0.

$$\hat{\boldsymbol{x}}_{0,0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We also don't know the rocket's acceleration, but we can assume it's greater than zero. Let's assume:

$$\boldsymbol{u}_0 = 0$$

Since our initial state vector is a guess, we set a very high estimate uncertainty. The high estimate uncertainty results in high Kalman Gain, giving a high weight to the measurement.

$$\boldsymbol{P}_{0,0} = \begin{bmatrix} 500 & 0 \\ 0 & 500 \end{bmatrix}$$

Prediction

Now we can predict the next state based on the initialization values.

$$\hat{\boldsymbol{x}}_{1,0} = \boldsymbol{F}\hat{\boldsymbol{x}}_{0,0} + \boldsymbol{G}\boldsymbol{u}_0 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.0313 \\ 0.25 \end{bmatrix} 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{P}_{1,0} = \boldsymbol{F}\boldsymbol{P}_{0,0}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 500 & 0 \\ 0 & 500 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.25 & 1 \end{bmatrix} + \begin{bmatrix} \frac{0.25^4}{4} & \frac{0.25^3}{2} \\ \frac{0.25^3}{2} & 0.25^2 \end{bmatrix} 0.1^2$$

$$= \begin{bmatrix} 531.25 & 125 \\ 125 & 500 \end{bmatrix}$$

First Iteration

Step 1 - Measure

The measurement values:

$\boldsymbol{z}_1 = 6.43$

$\boldsymbol{u}_1 = 39.81$

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Step 2 - Update

The Kalman Gain calculation:

$$\boldsymbol{K}_1 = \boldsymbol{P}_{1,0}\boldsymbol{H}^T\left(\boldsymbol{H}\boldsymbol{P}_{1,0}\boldsymbol{H}^T + \boldsymbol{R}\right)^{-1} = \begin{bmatrix} 0.57 \\ 0.13 \end{bmatrix}$$

Estimate the current state:

$$\hat{\boldsymbol{x}}_{1,1} = \hat{\boldsymbol{x}}_{1,0} + \boldsymbol{K}_1(\boldsymbol{z}_1 - \boldsymbol{H}\hat{\boldsymbol{x}}_{1,0}) = \begin{bmatrix} 3.67 \\ 0.86 \end{bmatrix}$$

Update the current estimate uncertainty:

$$\boldsymbol{P}_{1,1} = \left(\boldsymbol{I} - \boldsymbol{K}_1\boldsymbol{H}\right)\boldsymbol{P}_{1,0}\left(\boldsymbol{I} - \boldsymbol{K}_1\boldsymbol{H}\right)^T + \boldsymbol{K}_1\boldsymbol{R}\boldsymbol{K}_1^T = \begin{bmatrix} 228.2 & 53.7 \\ 53.7 & 483.2 \end{bmatrix}$$

Step 3 - Predict

$$\hat{\boldsymbol{x}}_{2,1} = \boldsymbol{F}\hat{\boldsymbol{x}}_{1,1} + \boldsymbol{G}\boldsymbol{u}_1 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3.67 \\ 0.86 \end{bmatrix} + \begin{bmatrix} 0.0313 \\ 0.25 \end{bmatrix}(39.81 + (-9.8)) = \begin{bmatrix} 4.82 \\ 8.36 \end{bmatrix}$$

$$\boldsymbol{P}_{2,1} = \boldsymbol{F}\boldsymbol{P}_{1,1}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 285.2 & 174.5 \\ 174.5 & 483.2 \end{bmatrix}$$

Our prediction uncertainty is still very high.

Second Iteration

Step 1 - Measure

The measurement values:

$z_2 = 1.3$

$u_2 = 39.67$

Step 2 - Update

The Kalman Gain calculation:

$$\boldsymbol{K}_2 = \boldsymbol{P}_{2,1}\boldsymbol{H}^T \left(\boldsymbol{H}\boldsymbol{P}_{2,1}\boldsymbol{H}^T + \boldsymbol{R}\right)^{-1} = \begin{bmatrix} 0.42 \\ 0.26 \end{bmatrix}$$

Estimate the current state:

$$\hat{\boldsymbol{x}}_{2,2} = \hat{\boldsymbol{x}}_{2,1} + \boldsymbol{K}_2(\boldsymbol{z}_2 - \boldsymbol{H}\hat{\boldsymbol{x}}_{2,1}) = \begin{bmatrix} 3.36 \\ 7.47 \end{bmatrix}$$

Update the current estimate uncertainty:

$$\boldsymbol{P}_{2,2} = \left(\boldsymbol{I} - \boldsymbol{K}_2\boldsymbol{H}\right)\boldsymbol{P}_{2,1}\left(\boldsymbol{I} - \boldsymbol{K}_2\boldsymbol{H}\right)^T + \boldsymbol{K}_2\boldsymbol{R}\boldsymbol{K}_2^T = \begin{bmatrix} 166.5 & 101.9 \\ 101.9 & 438.8 \end{bmatrix}$$

Step 3 - Predict

$$\hat{\boldsymbol{x}}_{3,2} = \boldsymbol{F}\hat{\boldsymbol{x}}_{2,2} + \boldsymbol{G}\boldsymbol{u}_2 = \begin{bmatrix} 6.16 \\ 14.93 \end{bmatrix}$$

$$\boldsymbol{P}_{3,2} = \boldsymbol{F}\boldsymbol{P}_{2,2}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 244.9 & 211.6 \\ 211.6 & 438.8 \end{bmatrix}$$

Thirtieth Iteration

Step 1 - Measure

The measurement values:

$\boldsymbol{z}_{30} = 787.22$

$\boldsymbol{u}_{30} = 39.94$

Step 2 - Update

The Kalman Gain calculation:

$$\boldsymbol{K}_{30} = \boldsymbol{P}_{30,29}\boldsymbol{H}^T \left(\boldsymbol{H}\boldsymbol{P}_{30,29}\boldsymbol{H}^T + \boldsymbol{R}\right)^{-1} = \begin{bmatrix} 0.12 \\ 0.02 \end{bmatrix}$$

The Kalman Gain for altitude converged to 0.12, which means that the estimation weight is much higher than the measurement weight.

Estimate the current state:

$$\hat{\boldsymbol{x}}_{30,30} = \hat{\boldsymbol{x}}_{30,29} + \boldsymbol{K}_{30}(\boldsymbol{z}_{30} - \boldsymbol{H}\hat{\boldsymbol{x}}_{30,29}) = \begin{bmatrix} 797.07 \\ 215.7 \end{bmatrix}$$

Update the current estimate uncertainty:

$$\boldsymbol{P}_{30,30} = (\boldsymbol{I} - \boldsymbol{K}_{30}\boldsymbol{H})\,\boldsymbol{P}_{30,29}\,(\boldsymbol{I} - \boldsymbol{K}_{35}\boldsymbol{H})^T + \boldsymbol{K}_{30}\boldsymbol{R}\boldsymbol{K}_{30}^T = \begin{bmatrix} 49.3 & 9.7 \\ 9.7 & 2.6 \end{bmatrix}$$

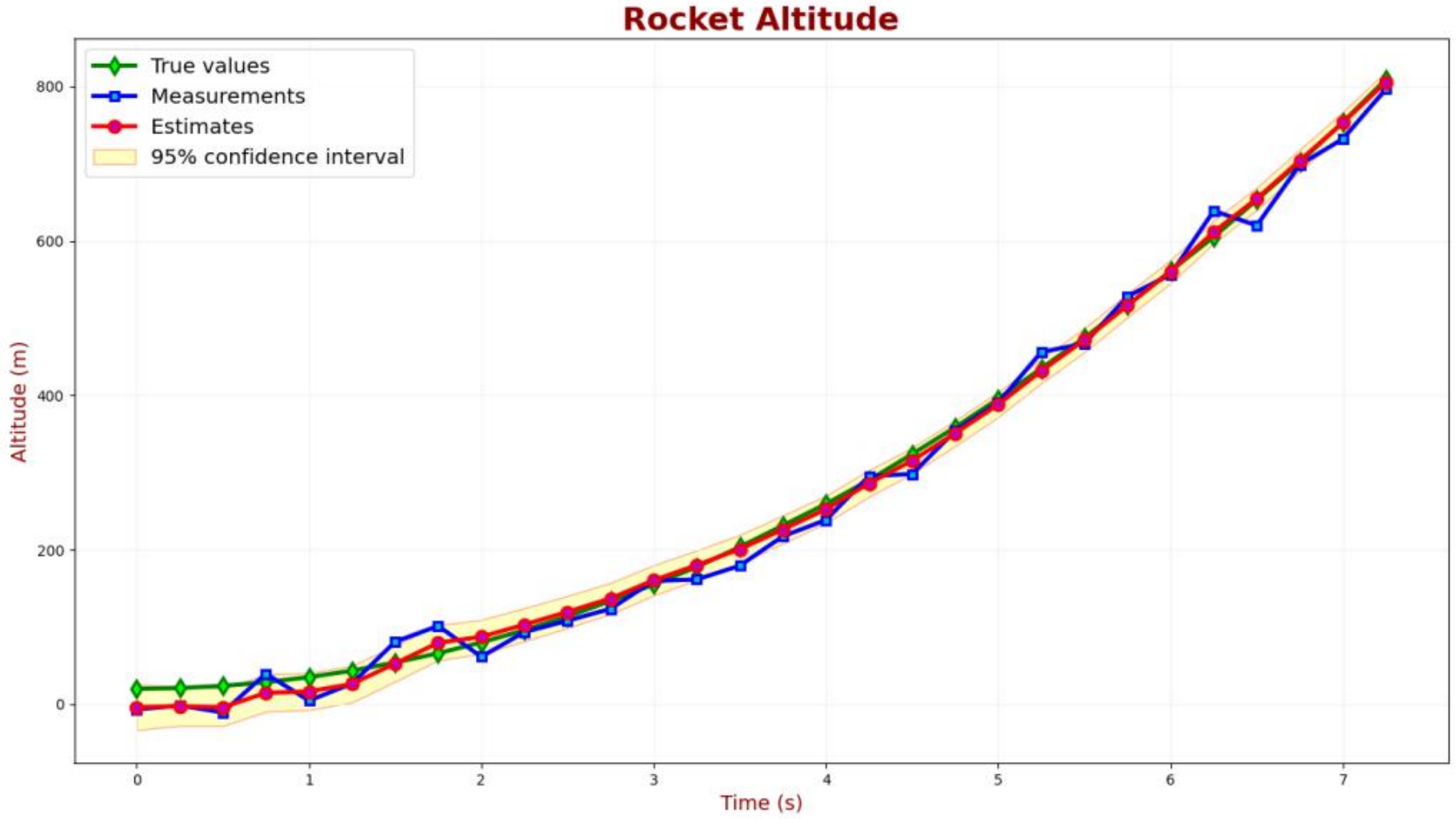
At this point, the altitude variance $p_x = 49.3$, which means that the standard deviation of the estimate is $\sqrt{49.3}m = 7.02m$ (remember that the standard deviation of the measurement is $20m$).

Step 3 - Predict

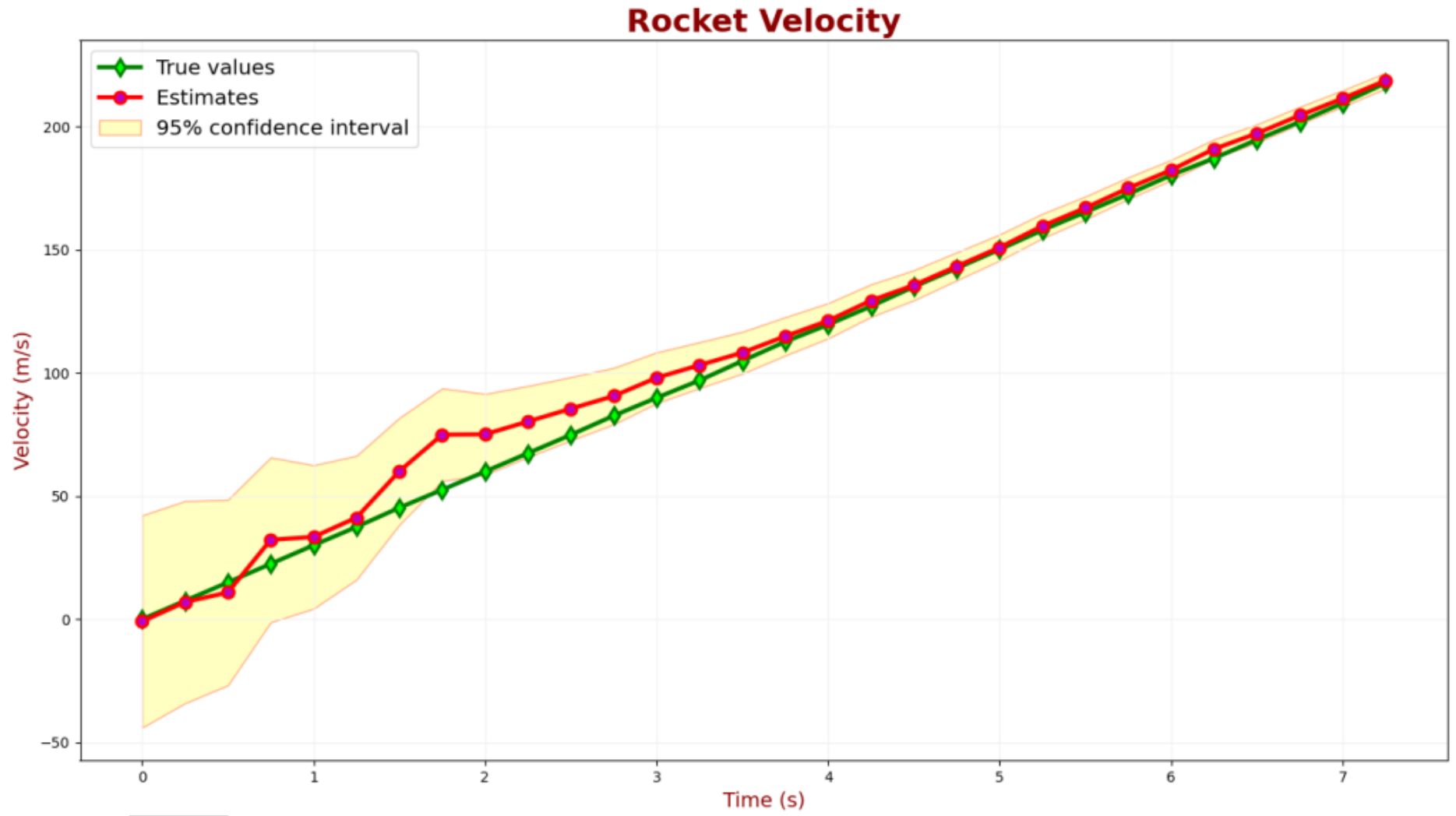
$$\hat{\boldsymbol{x}}_{31,30} = \boldsymbol{F}\hat{\boldsymbol{x}}_{30,30} + \boldsymbol{G}\boldsymbol{u}_{30} = \begin{bmatrix} 851.9 \\ 223.2 \end{bmatrix}$$

$$\boldsymbol{P}_{31,30} = \boldsymbol{F}\boldsymbol{P}_{30,30}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 54.3 & 10.4 \\ 10.4 & 2.6 \end{bmatrix}$$

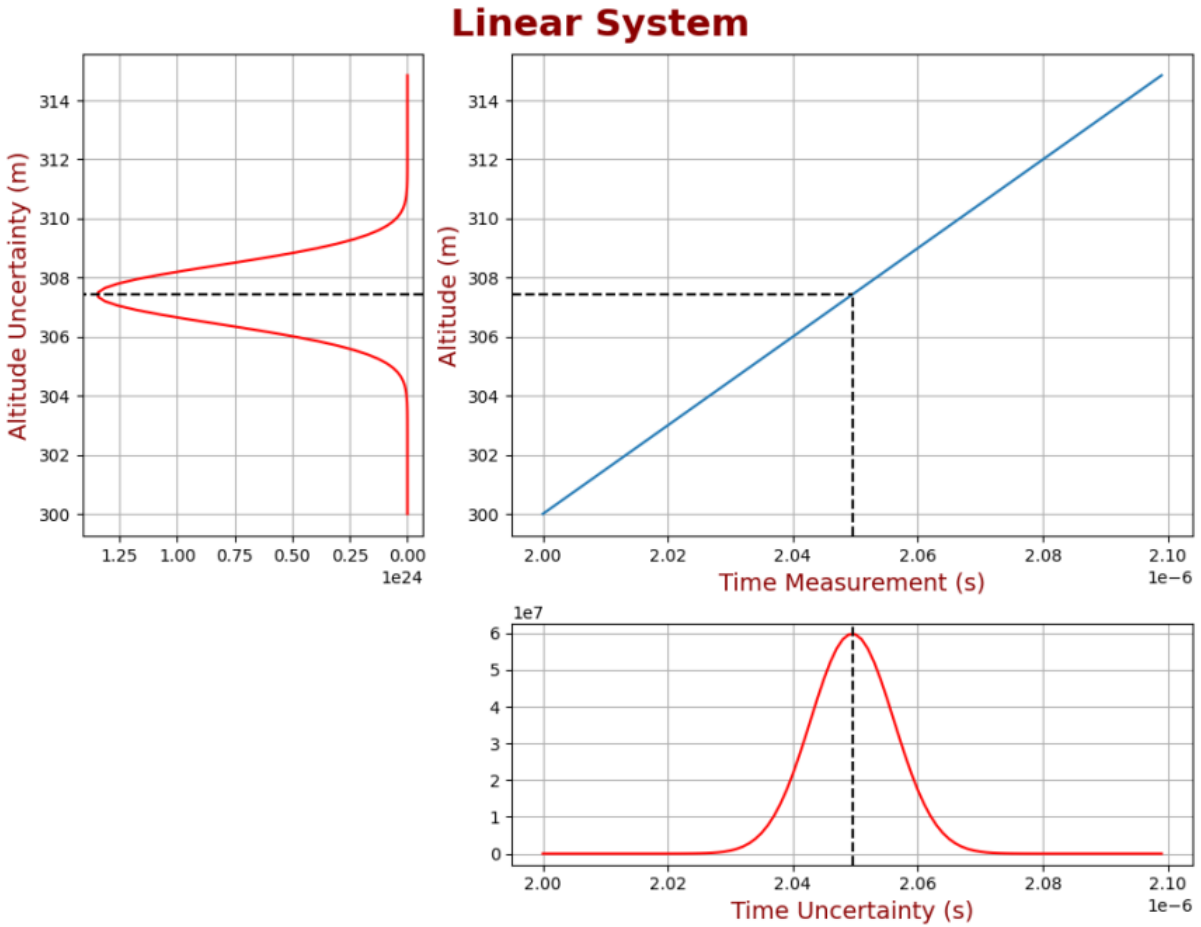
- The following chart compares the true, measured, and estimated values of the rocket altitude.

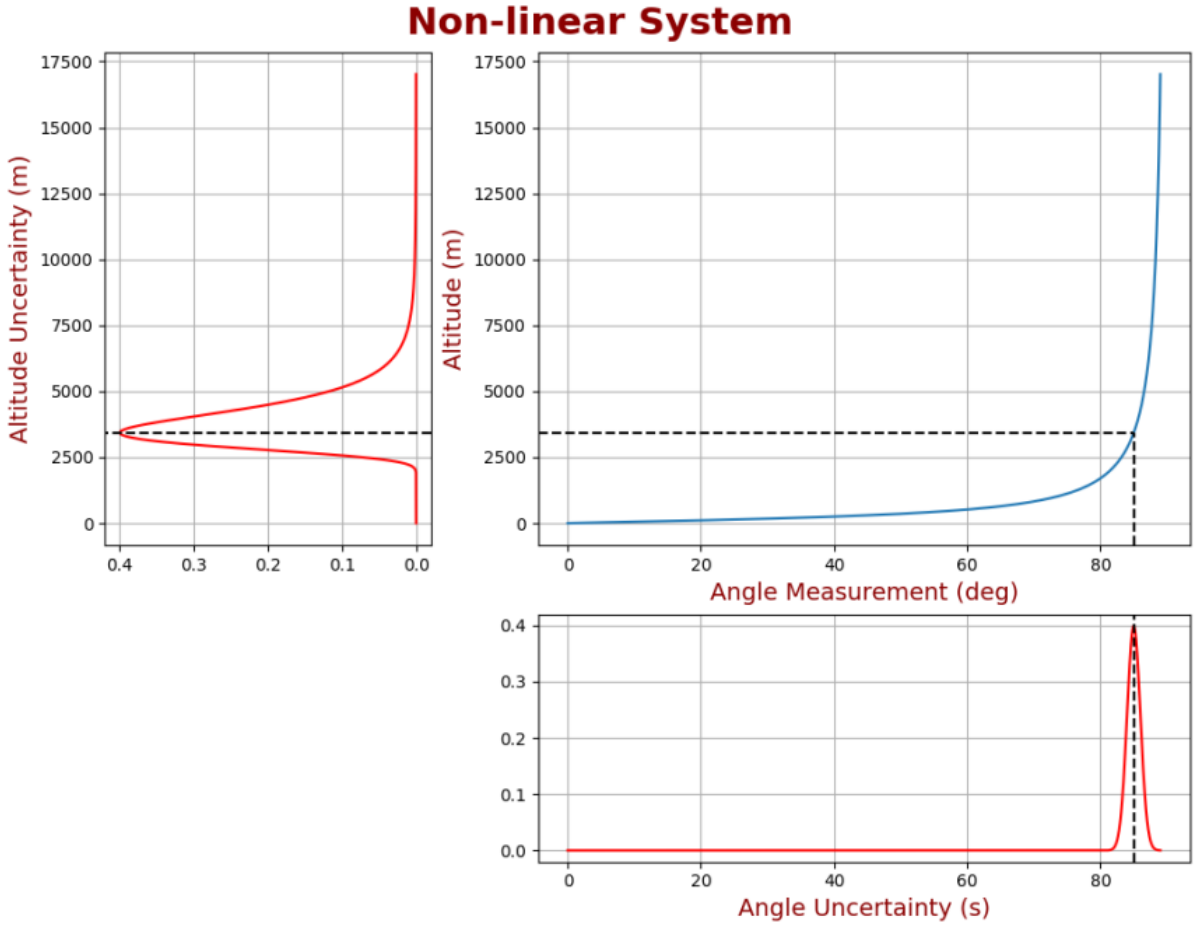


- The following chart compares the true value, measured values, and estimates of the rocket velocity.



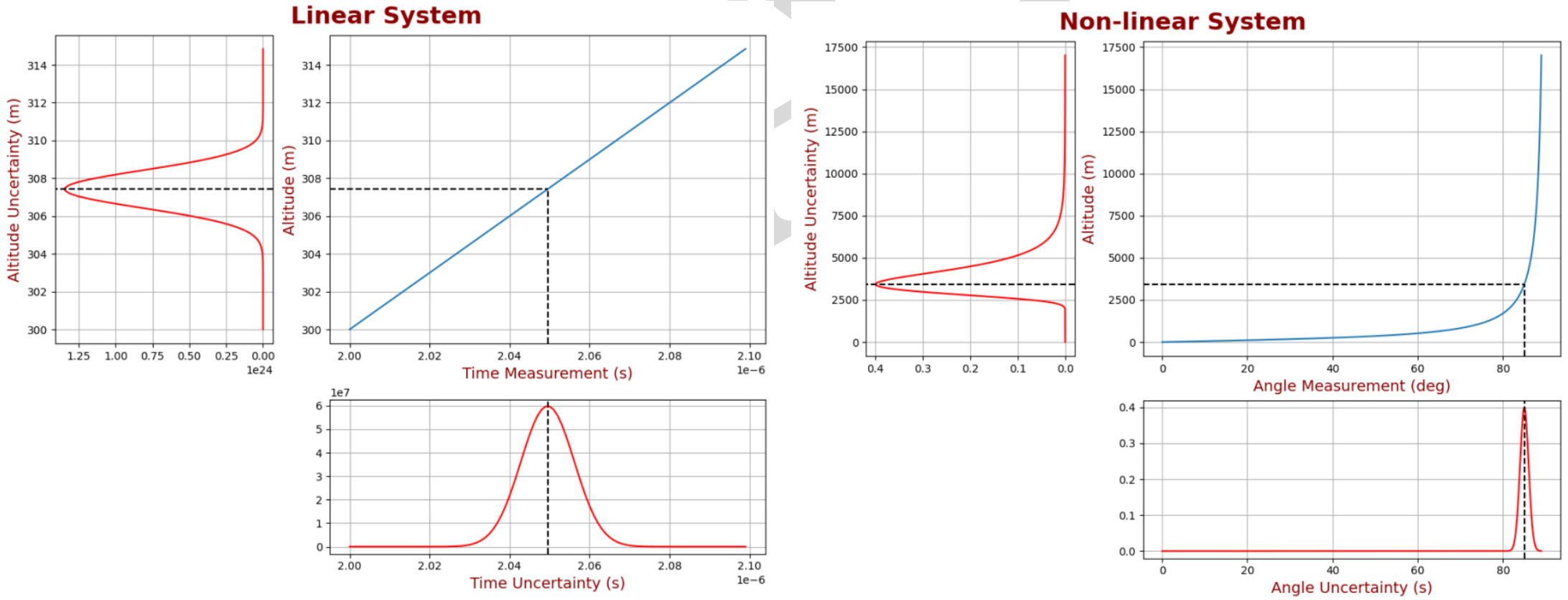
4.4 Extended Kalman Filter (EKF)





4.4.1 Non-linearity problem

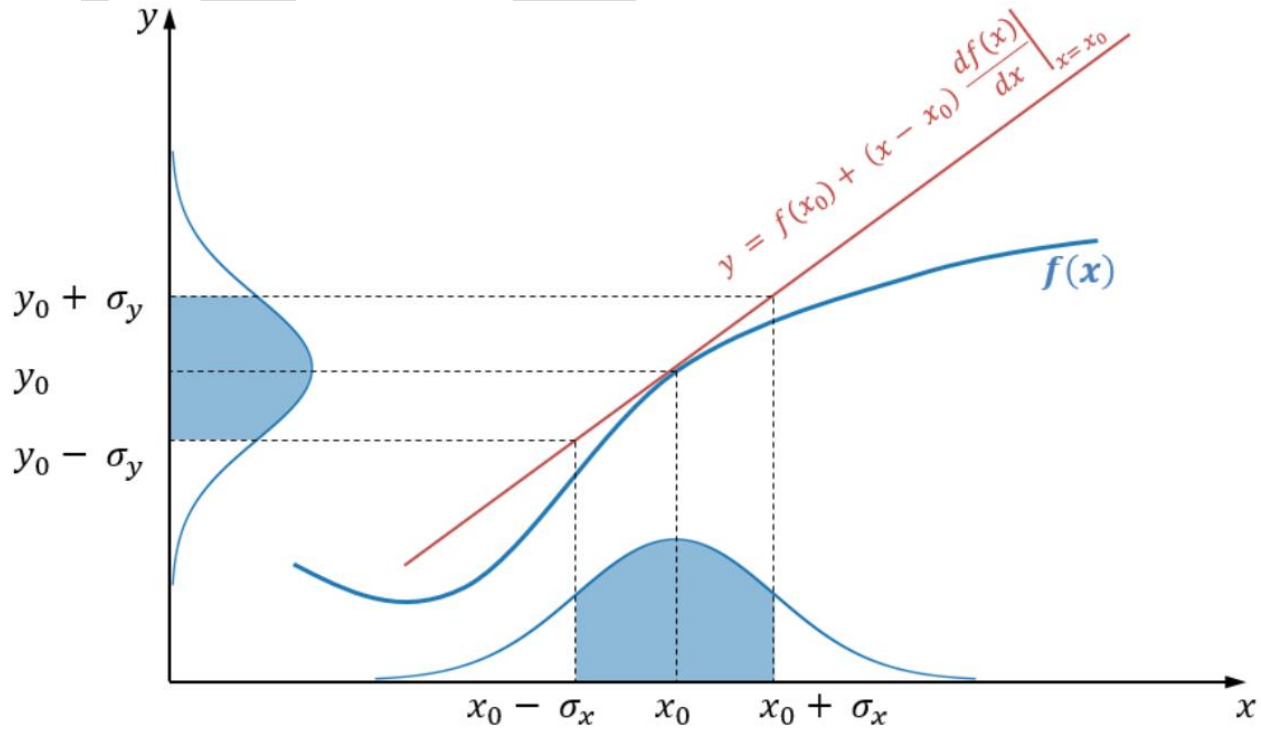
- The Kalman Filter algorithm assumes that the distribution of all random variables is Gaussian. For non-linear systems, this assumption does not hold anymore. The algorithm is not stable and yields signifcant estimation errors.



4.4.2 Analytic linearization

- The EKF performs analytic linearization of the model at each point in time. The following fgure exemplifes a one-dimensional case.
- We want to find a tangent line for a point x = x0. Using the tangent line, we can project the uncertainty to the y axis and keep its' shape Gaussian.

$$y = f(x_0) + (x - x_0)\left.\frac{df(x)}{dx}\right|_{x=x_0}$$



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4.4.3 Multivariate uncertainty projection

- For multi-dimensional problems, we propagate the multivariate Gaussian random variable (represented by covariance matrix) using a linear approximation of the multi-dimensional function.
- For the non-linear system dynamics, the multivariate analytical linearization is given by:

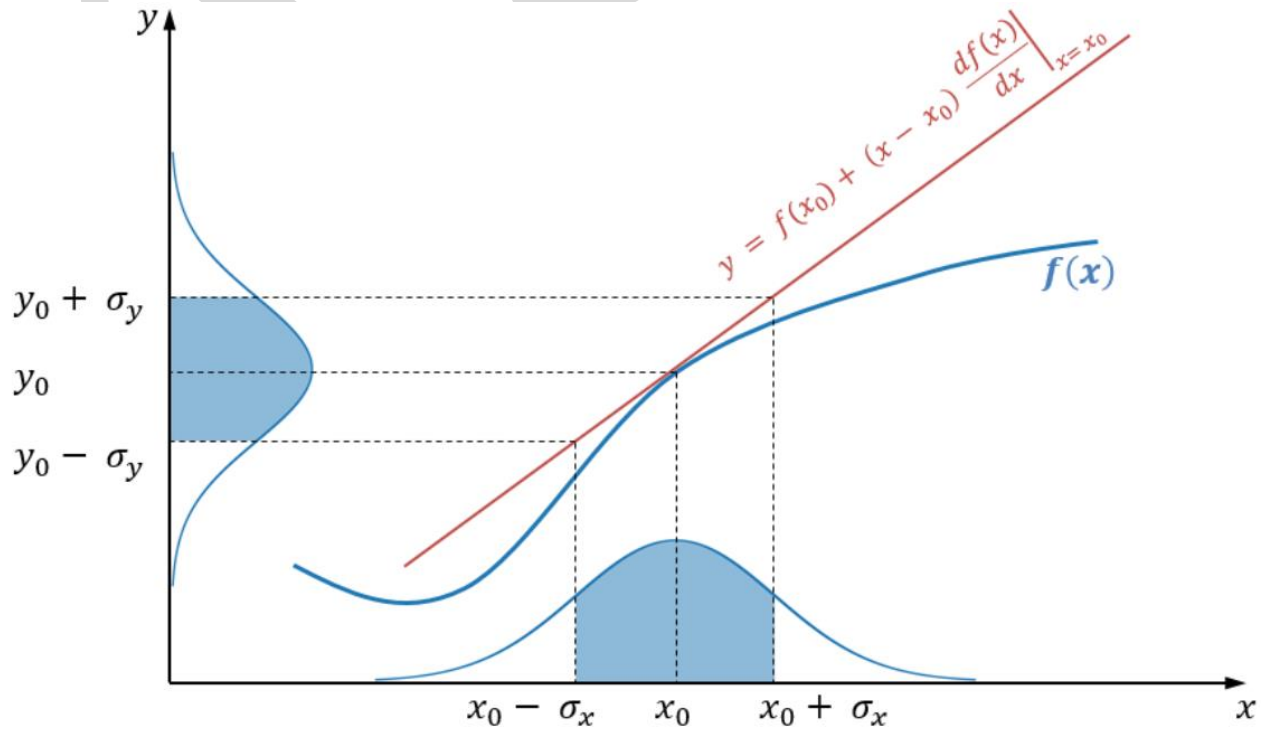
$$\boldsymbol{P}_{out} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \boldsymbol{P}_{in} \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \right)^{T}$$

Where:

$\boldsymbol{P}_{in}$ is an input covariance

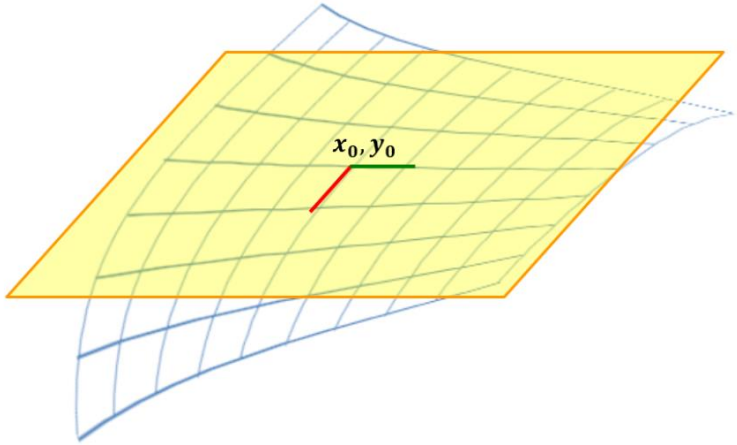
$\boldsymbol{P}_{out}$ is a projected covariance

$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}$ is a state transition matrix Jacobian



- Jacobian matrix is a matrix of partial derivatives:

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$



- Similarly, for the state-to-measurement non-linear relation, the multivariate analytical linearization is given by:

$$\boldsymbol{P}_{out} = \frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}} \boldsymbol{P}_{in} \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}} \right)^T$$

Where:

$\boldsymbol{P}_{in}$ is an input covariance

$\boldsymbol{P}_{out}$ is a projected covariance

$\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}$ is an observation matrix Jacobian

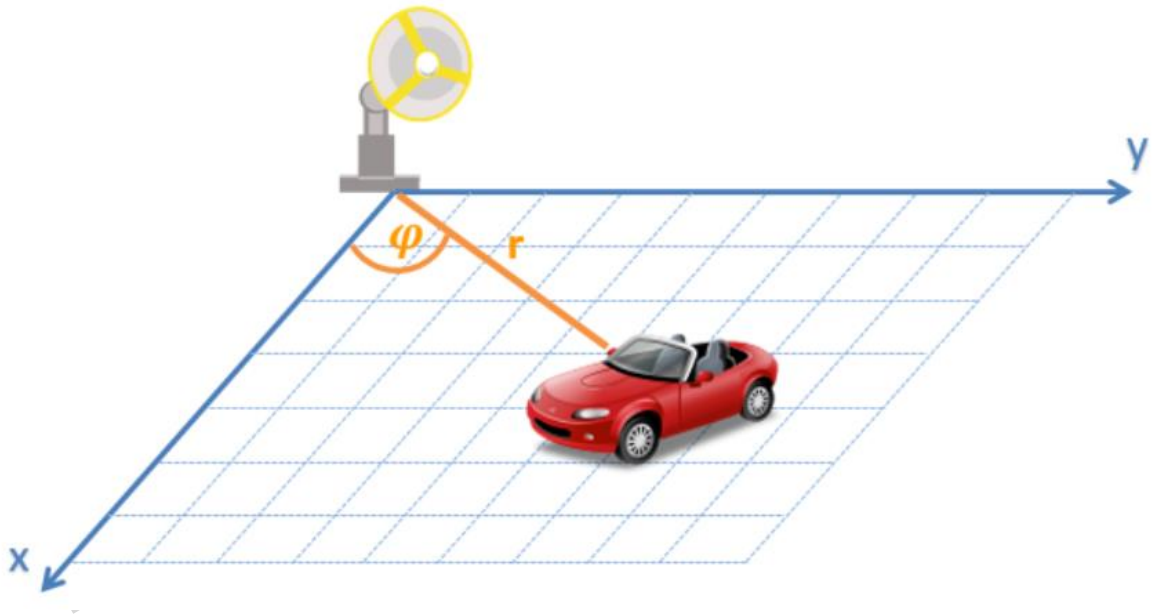
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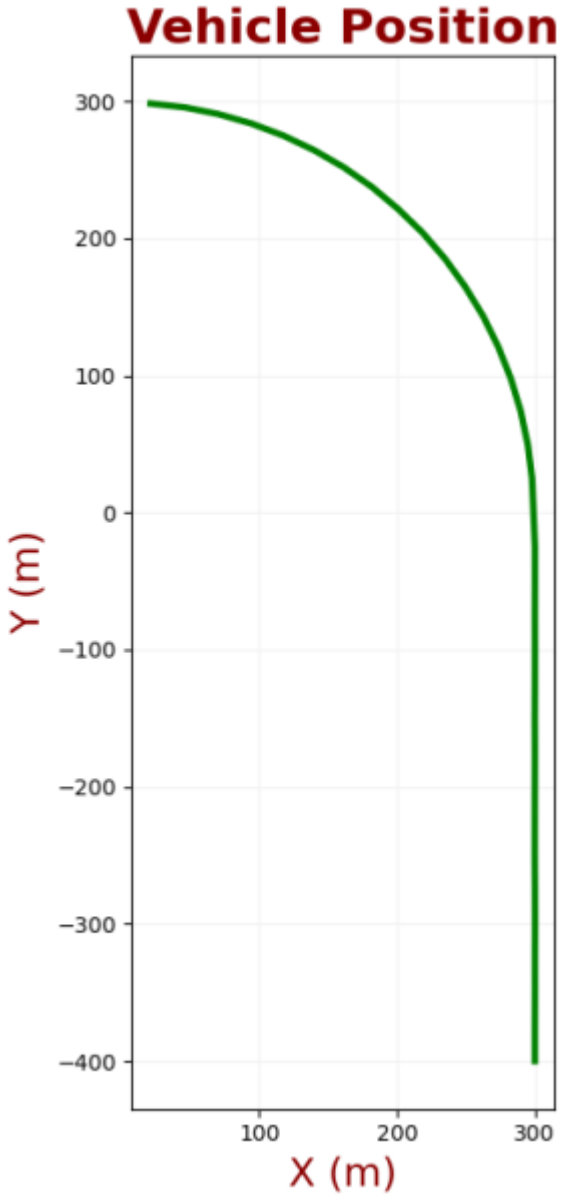
4.4.4 EKF equations summary

	Equation	LKF Equation	EKF Equation
Predict	State Extrapolation	$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n} + \boldsymbol{G}\boldsymbol{u}_n$	$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{f}(\hat{\boldsymbol{x}}_{n,n}) + \boldsymbol{G}\boldsymbol{u}_n$
	Covariance Extrapolation	$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$	$\boldsymbol{P}_{n+1,n} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\boldsymbol{P}_{n,n}\left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)^T + \boldsymbol{Q}$
Update	State Update	$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n\left(\boldsymbol{z}_n - \boldsymbol{H}\hat{\boldsymbol{x}}_{n,n-1}\right)$	$\hat{\boldsymbol{x}}_{n,n} = \hat{\boldsymbol{x}}_{n,n-1} + \boldsymbol{K}_n\left(\boldsymbol{z}_n - \boldsymbol{h}(\hat{\boldsymbol{x}}_{n,n-1})\right)$
	Covariance Update	$\boldsymbol{P}_{n,n} = \left(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H}\right)\boldsymbol{P}_{n,n-1} \times \left(\boldsymbol{I} - \boldsymbol{K}_n\boldsymbol{H}\right)^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$	$\boldsymbol{P}_{n,n} = \left(\boldsymbol{I} - \boldsymbol{K}_n\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\right)\boldsymbol{P}_{n,n-1} \times \left(\boldsymbol{I} - \boldsymbol{K}_n\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\right)^T + \boldsymbol{K}_n\boldsymbol{R}_n\boldsymbol{K}_n^T$
	Kalman Gain	$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\boldsymbol{H}^T \times \left(\boldsymbol{H}\boldsymbol{P}_{n,n-1}\boldsymbol{H}^T + \boldsymbol{R}_n\right)^{-1}$	$\boldsymbol{K}_n = \boldsymbol{P}_{n,n-1}\left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\right)^T \times \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\boldsymbol{P}_{n,n-1}\left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\right)^T + \boldsymbol{R}_n\right)^{-1}$

4.4.5 Numerical example

- The vehicle moves straight in the Y direction with a constant velocity. After traveling 400 meters, the vehicle turns left with a turning radius of 300 meters. During the turning maneuver, the vehicle experiences acceleration due to the circular motion (angular acceleration).





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- The measurements period: $\Delta t = 1s$
- The random acceleration standard deviation: $\sigma_a = 0.2\frac{m}{s^2}$
- The range measurement error standard deviation: $\sigma_{r_m} = 5m$
- The bearing angle measurement error standard deviation: $\sigma_{\varphi_m} = 0.0087rad$
- The state transition matrix $\boldsymbol{F}$ is:

$$\boldsymbol{F} = \begin{bmatrix} 1 & \Delta t & 0.5\Delta t^2 & 0 & 0 & 0 \\ 0 & 1 & \Delta t & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \Delta t & 0.5\Delta t^2 \\ 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0.5 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- The measurement variance $\boldsymbol{R}$ is:

$$\boldsymbol{R}_n = \begin{bmatrix} \sigma_{r_m}^2 & 0 \\ 0 & \sigma_{\varphi_m}^2 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 0.0087^2 \end{bmatrix}$$

- The process noise matrix $\boldsymbol{Q}$ is:

$$\boldsymbol{Q} = \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & \frac{\Delta t^2}{2} & 0 & 0 & 0 \\ \frac{\Delta t^3}{2} & \Delta t^2 & \Delta t & 0 & 0 & 0 \\ \frac{\Delta t^2}{2} & \Delta t & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & \frac{\Delta t^2}{2} \\ 0 & 0 & 0 & \frac{\Delta t^3}{2} & \Delta t^2 & \Delta t \\ 0 & 0 & 0 & \frac{\Delta t^2}{2} & \Delta t & 1 \end{bmatrix} \sigma_a^2 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & 1 & 1 \\ 0 & 0 & 0 & \frac{1}{2} & 1 & 1 \end{bmatrix} 0.2^2$$

- The following table contains the set of 35 noisy measurements:

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
r(m)	502.55	477.34	457.21	442.94	427.27	406.05	400.73	377.32
φ(rad)	-0.9316	-0.8977	-0.8512	-0.8114	-0.7853	-0.7392	-0.7052	-0.6478
9	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
360.27	345.93	333.34	328.07	315.48	301.41	302.87	304.25	294.46
-0.59	-0.5183	-0.4698	-0.3952	-0.3026	-0.2445	-0.1626	-0.0937	0.0085
18	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**
294.29	299.38	299.37	300.68	304.1	301.96	300.3	301.9	296.7
0.0856	0.1675	0.2467	0.329	0.4149	0.504	0.5934	0.667	0.7537
27	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**
297.07	295.29	296.31	300.62	292.3	298.11	298.07	298.92	298.04
0.8354	0.9195	1.0039	1.0923	1.1546	1.2564	1.3274	1.409	1.5011

Iteration Zero

Initialization

We don't know the vehicle location, so we approximate the initial position at about 100m from the true vehicle position ($\hat{x}_{0,0} = 400m, \hat{y}_{0,0} = -300m$)

$$\hat{\boldsymbol{x}}_{0,0} = \begin{bmatrix} 400 \\ 0 \\ 0 \\ -300 \\ 0 \\ 0 \end{bmatrix}$$

Since our initial state vector is a guess, we set a very high estimate uncertainty. The high estimate uncertainty results in a high Kalman Gain by giving a high weight to the measurement.

$$\boldsymbol{P}_{0,0} = \begin{bmatrix} 500 & 0 & 0 & 0 & 0 & 0 \\ 0 & 500 & 0 & 0 & 0 & 0 \\ 0 & 0 & 500 & 0 & 0 & 0 \\ 0 & 0 & 0 & 500 & 0 & 0 \\ 0 & 0 & 0 & 0 & 500 & 0 \\ 0 & 0 & 0 & 0 & 0 & 500 \end{bmatrix}$$

Prediction

$$\hat{\boldsymbol{x}}_{1,0} = \boldsymbol{F}\hat{\boldsymbol{x}}_{0,0} = \begin{bmatrix} 400 \\ 0 \\ 0 \\ -300 \\ 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{P}_{1,0} = \boldsymbol{F}\boldsymbol{P}_{0,0}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 1125 & 750 & 250 & 0 & 0 & 0 \\ 750 & 1000 & 500 & 0 & 0 & 0 \\ 250 & 500 & 500 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1125 & 750 & 250 \\ 0 & 0 & 0 & 750 & 1000 & 500 \\ 0 & 0 & 0 & 250 & 500 & 500 \end{bmatrix}$$

First Iteration

Step 1 - Measure

The measurement values:

$$\boldsymbol{z}_1 = \begin{bmatrix} 502.55 \\ -0.9316 \end{bmatrix}$$

Step 2 - Update

Observation matrix ($\boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})$) calculation.

$$\boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0}) = \begin{bmatrix} \sqrt{x_{1,0}^2 + y_{1,0}^2} \\ tan^{-1}\dfrac{y_{1,0}}{x_{1,0}} \end{bmatrix} = \begin{bmatrix} \sqrt{400^2 + (-300)^2} \\ tan^{-1}\dfrac{-300}{400} \end{bmatrix} = \begin{bmatrix} 500 \\ 0.644 \end{bmatrix}$$

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Observation matrix Jacobian $\left(\frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}}\right)$ calculation:

$$\frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{x_{1,0}}{\sqrt{x_{1,0}^2 + y_{1,0}^2}} & 0 & 0 & \frac{y_{1,0}}{\sqrt{x_{1,0}^2 + y_{1,0}^2}} & 0 & 0 \\ \frac{-y_{1,0}}{x_{1,0}^2 + y_{1,0}^2} & 0 & 0 & \frac{x_{1,0}}{x_{1,0}^2 + y_{1,0}^2} & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.8 & 0 & 0 & -0.6 & 0 & 0 \\ 0.0012 & 0 & 0 & 0.0016 & 0 & 0 \end{bmatrix}$$

The Kalman Gain calculation:

$$\boldsymbol{K}_1 = \boldsymbol{P}_{1,0} \left(\frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}}\right)^T \left(\frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}} \boldsymbol{P}_{1,0} \left(\frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}}\right)^T + \boldsymbol{R}\right)^{-1} = \begin{bmatrix} 0.783 & 295 \\ 0.522 & 196.7 \\ 0.174 & 65.6 \\ -0.587 & 393.3 \\ -0.391 & 262.2 \\ -0.13 & 87.4 \end{bmatrix}$$

Estimate the current state:

$$\hat{\boldsymbol{x}}_{1,1} = \hat{\boldsymbol{x}}_{1,0} + \boldsymbol{K}_1 \left(\boldsymbol{z}_1 - \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0}) \right) = \begin{bmatrix} 317 \\ -55.3 \\ -18.4 \\ -414.8 \\ -76.5 \\ -25.5 \end{bmatrix}$$

Update the current estimate covariance:

$$\boldsymbol{P}_{1,1} = \left(\boldsymbol{I} - \boldsymbol{K}_1 \frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}} \right) \boldsymbol{P}_{1,0} \left(\boldsymbol{I} - \boldsymbol{K}_1 \frac{\partial \boldsymbol{h}(\hat{\boldsymbol{x}}_{1,0})}{\partial \boldsymbol{x}} \right)^T + \boldsymbol{K}_1 \boldsymbol{R} \boldsymbol{K}_1^T$$

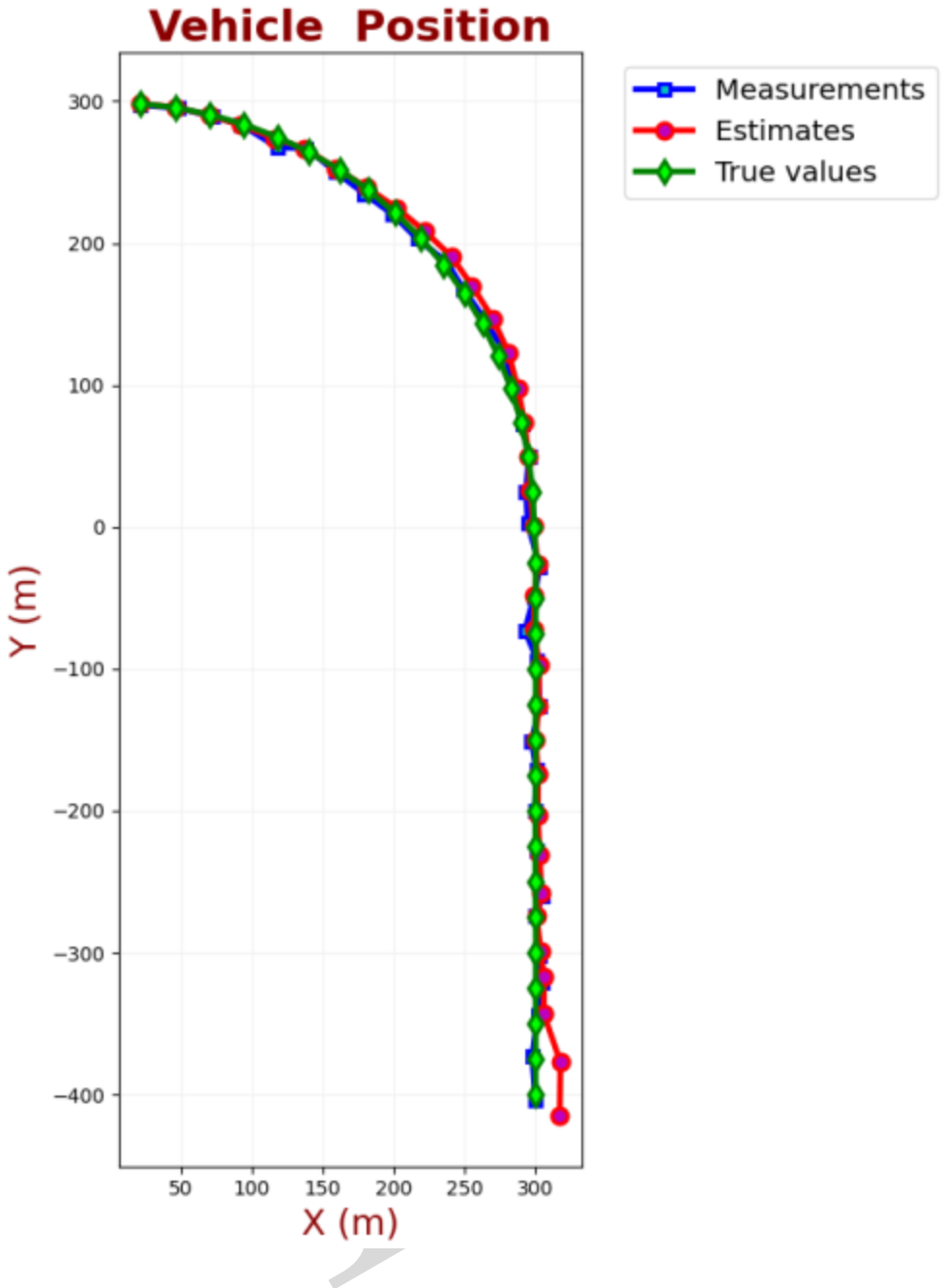
$$= \begin{bmatrix} 22.39 & 14.93 & 4.98 & -2.75 & -1.84 & -0.61 \\ 14.93 & 509.97 & 336.67 & -1.84 & -1.22 & -0.41 \\ 4.98 & 336.67 & 445.58 & -0.61 & -0.41 & -0.14 \\ -2.75 & -1.84 & -0.61 & 22.39 & 14.93 & 4.98 \\ -1.84 & -1.22 & -0.41 & 14.93 & 509.97 & 336.67 \\ -0.61 & -0.41 & -0.14 & 4.98 & 336.67 & 445.58 \end{bmatrix}$$

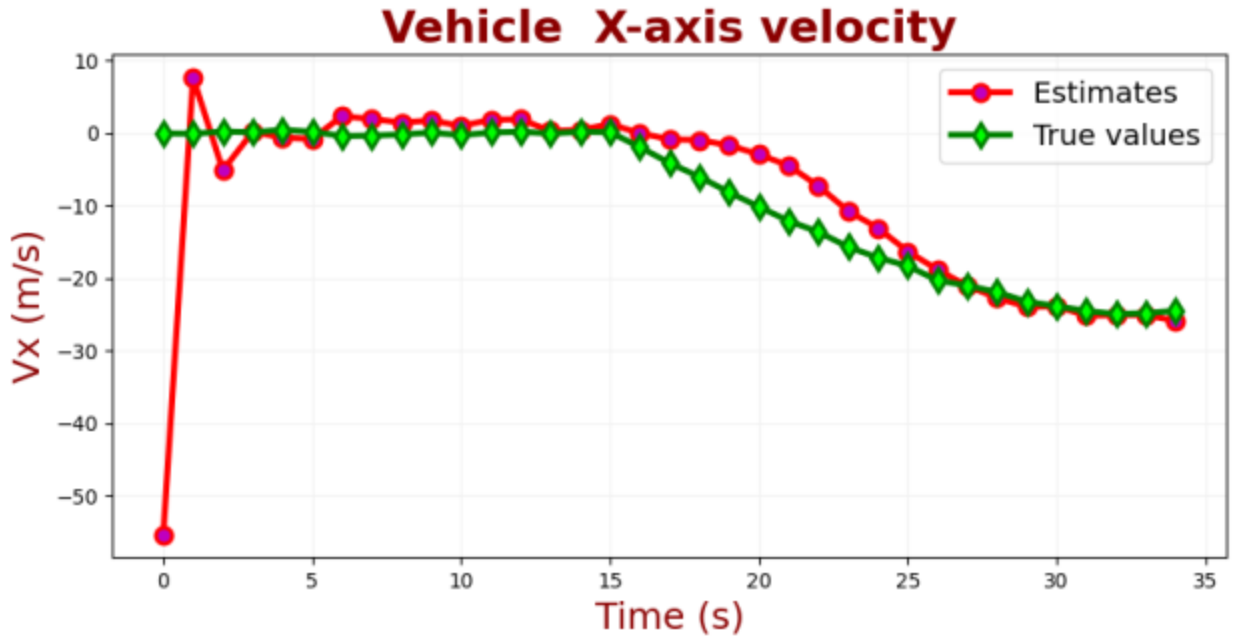
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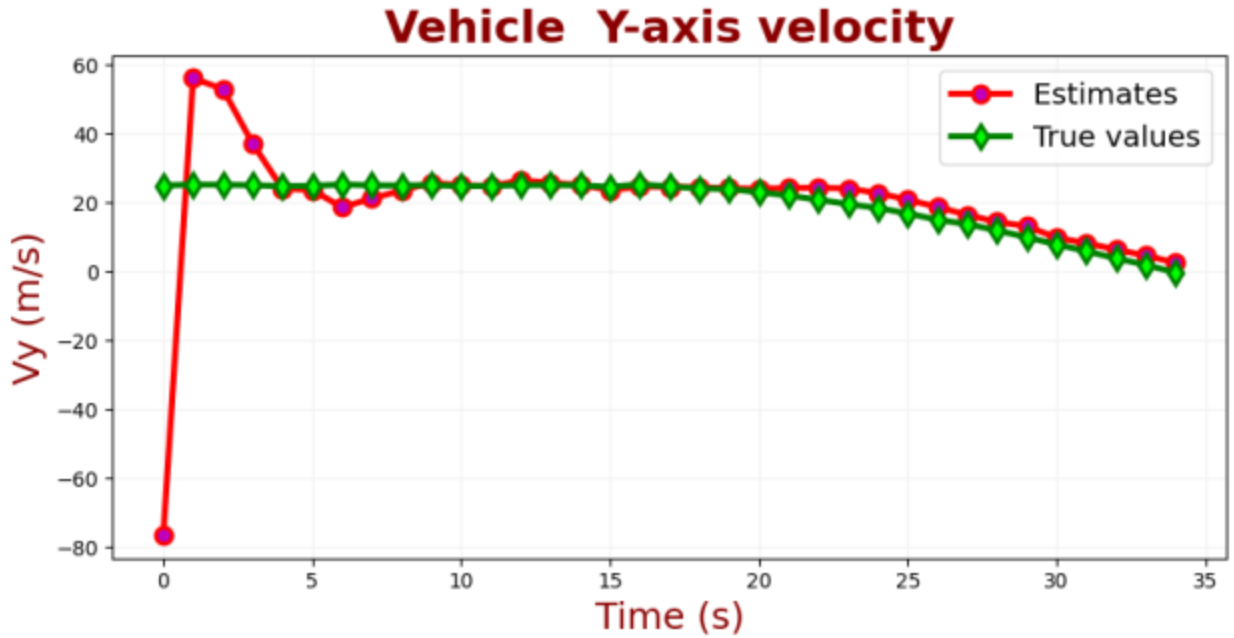
Step 3 - Predict

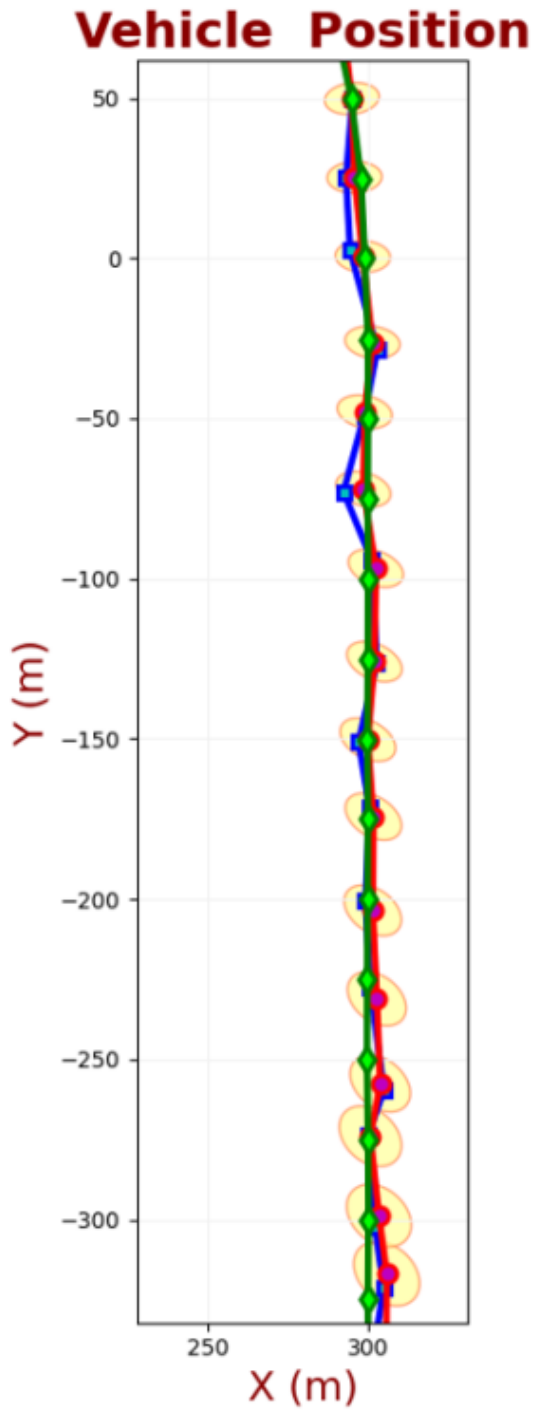
$$\hat{\boldsymbol{x}}_{2,1} = \boldsymbol{F}\hat{\boldsymbol{x}}_{1,1} = \begin{bmatrix} 252.42 \\ -73.8 \\ -18.45 \\ -504.13 \\ -102.06 \\ -25.52 \end{bmatrix}$$

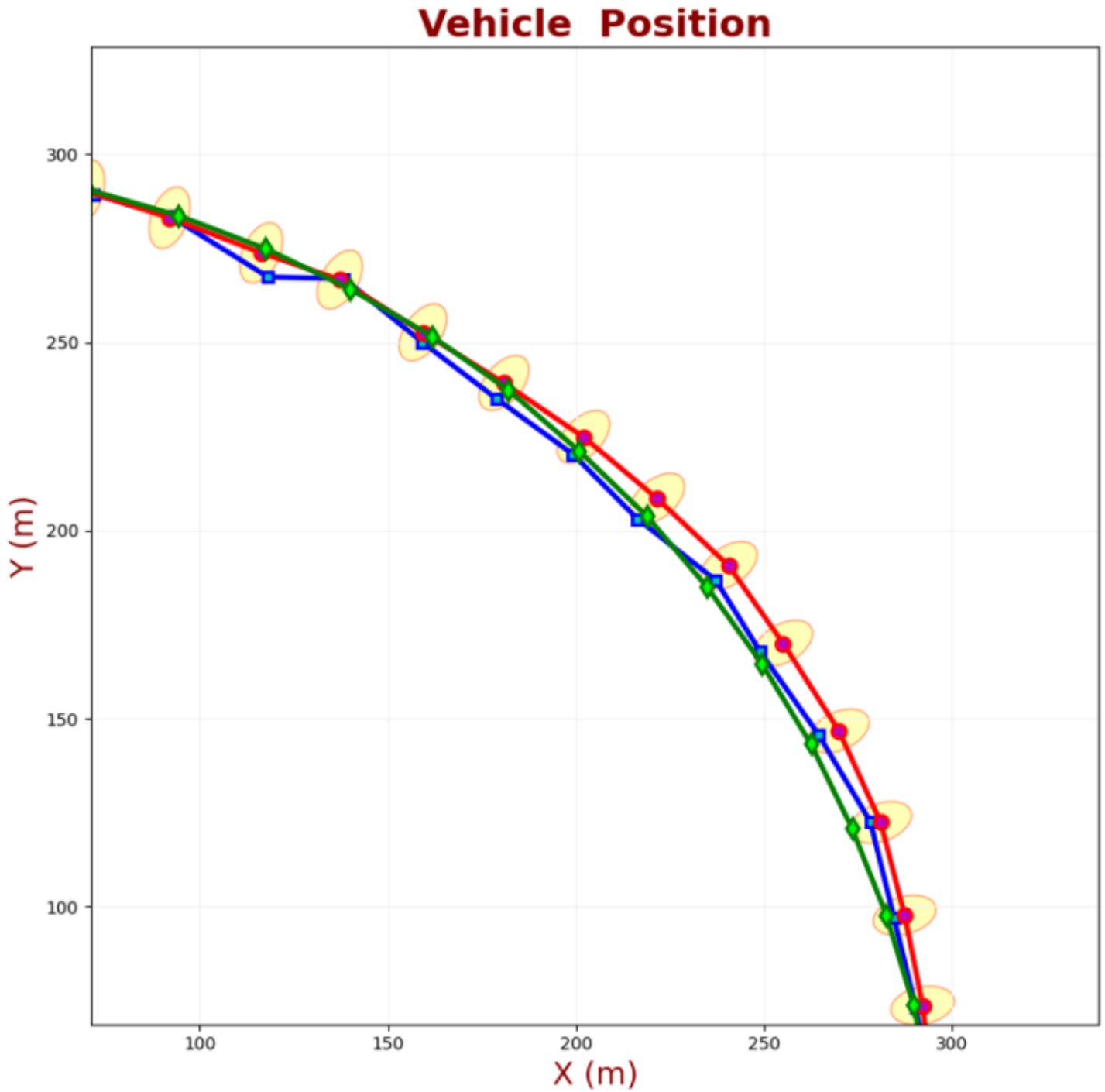
$$\boldsymbol{P}_{2,1} = \boldsymbol{F}\boldsymbol{P}_{1,1}\boldsymbol{F}^T + \boldsymbol{Q} = \begin{bmatrix} 1015.28 & 1257.7 & 564.46 & -8.7 & -4.35 & -1.09 \\ 1257.7 & 1628.94 & 782.3 & -4.35 & -2.18 & -0.54 \\ 564.46 & 782.3 & 445.62 & -1.09 & -0.54 & -0.14 \\ -8.7 & -4.35 & -1.09 & 1010.2 & 1255.16 & 563.83 \\ -4.35 & -2.18 & -0.54 & 1255.16 & 1627.67 & 781.98 \\ -1.09 & -0.54 & -0.14 & 563.83 & 781.98 & 445.54 \end{bmatrix}$$


Vehicle Position
Measurements
Estimates
True values
Y (m)
X (m)
300
200
100
0
−100
−200
−300
−400
50
100
150
200
250
300


Vehicle X-axis velocity
Estimates
True values
Vx (m/s)
Time (s)
10
0
−10
−20
−30
−40
−50
0
5
10
15
20
25
30
35


Vehicle Y-axis velocity
Estimates
True values
Vy (m/s)
Time (s)
60
40
20
0
−20
−40
−60
−80
0
5
10
15
20
25
30
35


Vehicle Position
50
0
−50
−100
−150
−200
−250
−300
Y (m)
250
300
X (m)


Vehicle Position
300
250
200
150
100
Y (m)
100
150
200
250
300
X (m)

Additional resources

- https://www.mathworks.com/videos/series/understanding-kalman-filters.html
- https://www.mathworks.com/help/vision/ug/using-kalman-filter-for-object-tracking.html
- https://www.mathworks.com/help/vision/ug/motion-based-multiple-object-tracking.html
- https://www.mathworks.com/help/predmaint/ug/Fault-Detection-Using-an-Extended-Kalman-Filter.html

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